Nonlinear Wave Equations

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1 Overview

The following notes are an introduction into the theory of nonlinear wave equations, and mathematical fluids that I've written in preparation for the mathematics PhD oral examination at Stony Brook. These notes cover a few fundamental results on the (local) well-posedness of (non)linear wave equations, energy estimates, Strichartz estimates, and related results in harmonic analysis and fluids. In some sense, these notes are meant to be an extensive log of everything I have studied, and therefore, does not contain many novel ideals. The content reflects my most current understanding of the key ideas, and were written alongside discussions with my advisor, Professor John Anderson. Moreover, these notes are strongly based on the corresponding notes of Professors Holzegel, Lawrie, and Luk. It goes without saying that any errors in the following notes are entirely my responsibility.

Part I
Functional and Harmonic Analysis

2 General Analytic Results

In this subsection, we will be introducing some general analytic results in functional analysis and harmonic analysis, that we will later use in our study of the (non)linear wave equations (and hyperbolic PDE, in general) [8].



2.1 Introductory Harmonic Analysis

In this subsection, we will see some applications of L^p spaces. In particular, we introduce some rudimentary objects in **harmonic analysis**. Roughly speaking, harmonic analysis had its origins in trying to "decompose" functions into basic building blocks composed of elementary waves. In this sense, harmonic analysis closely resembles the related field of *Fourier analysis*. In particular, the theory for Fourier analysis is especially well-developed in Hilbert spaces (e.g., like L^2), where basic results like orthogonality, Plancherel's theorem, and Parseval's identity make the analysis especially clear to study. But in general, studying only L^2 functions is incredibly restrictive, and might not necessarily illuminate the full range of properties of various operators that we study (e.g., the Fourier transform). Thus a natural question to ask is if the results that holds in L^2 (more broadly, Hilbert spaces) either hold, or have analogs in L^p for $p \neq 2$. Thus we will begin by trying to develop L^p-analogs to the foundational *Parseval's identity*, which will lead us to the **Riesz-Thorin interpolation theorem**. Next we will try to develop the L^p equivalent of *conjugate functions* and *allied series*, which will lead to the **Hilbert transform**, and then we will encounter a few spaces that are better than L^1, L^∞ spaces, which leads to the theory of **Hardy spaces**.

2.1.1 The Riesz-Thorin Interpolation Theorem

If $f \in L^2([0, 2\pi])$ is square integrable, then the L^2 Parseval relation tells us that

$$\sum_{n=-\infty}^{\infty} |a_n|^2 = \frac{1}{2\pi} \int_0^{2\pi} |f(\theta)|^2 \, d\theta, \tag{1}$$

where for each n,

$$a_n = \frac{1}{2\pi} \int_0^{2\pi} f(\theta) e^{-in\theta} \, d\theta$$

are the **Fourier coefficients** of f such that

$$f(\theta) \sim \sum_{n=-\infty}^{\infty} a_n e^{in\theta}.$$

Likewise, if $\{a_n\} \in \ell^2(\mathbb{Z})$, then there exists a unique $f \in L^2([0,2\pi])$ such that $f(\theta) \sim \sum_{n=-\infty}^{\infty} a_n e^{in\theta}$ and (1) holds. If we assume $f \in L^p([0,2\pi])$ or $\{a_n\} \in \ell^p(\mathbb{Z})$ for some $p \neq 2$, does the L^2 Parseval relation generalize to these cases? In general, one has the *Hausdorff-Young inequalities*,

$$\left(\sum |a_n|^q\right)^{1/q} \leq \left(\frac{1}{2\pi}\int_0^{2\pi} |f(\theta)|^p \, d\theta\right)^{1/p},$$

and

$$\left(\frac{1}{2\pi}\int_0^{2\pi} |f(\theta)|^q \, d\theta\right)^{1/q} \leq \left(\sum |a_n|^p\right)^{1/p}.$$

Thus we need a general way of proving these (and other) results. As we stated, the $p = 2$ case follows immediately from Parseval's identity. Likewise, the $p = 1$ case follows almost immediately. Thus we will now develop a general theorem that enables us to prove these types of results efficiently.

More broadly, when establishing various estimates for PDEs, it can be easier to establish corresponding inequalities for certain "endpoints", and to "interpolate" in the corresponding estimates for all intermediate values. I.e., suppose one has the estimates $A_0 \leq B_0$ and $A_1 \leq B_1$. A natural question to ask is if for any $\theta \in (0,1)$, $A_\theta \lesssim_\theta B_\theta$. In this subsubsection, we will focus on estimates for linear operators between L^p spaces, and we will give specific criteria under which interpolation in the above sense holds. In particular, we will answer the following question.

> Suppose (p_0, q_0) and (p_1, q_1) are two pairs of indices with $\leq p_i, q_j \leq \infty$. Assume also that T is a linear operator and that,
>
> $$\|Tf\|_{L^{q_0}} \leq M_0 \|f\|_{L^{p_0}} \qquad \|Tf\|_{L^{q_1}} \leq M_1 \|f\|_{L^{p_1}}.$$
>
> Does it follow necessarily that $\|Tf\|_{L^q} \leq M \|f\|_{L^p}$ for other pairs (p, q)?

We begin with the following technical lemma.

Lemma 2.1 (Three-Lines Lemma)**.** *Suppose $\Phi(z)$ is a holomorphic function in the strip $S = \{z \in \mathbb{C} : 0 < \mathrm{Re}(z) < 1\}$, that is also continuous and bounded on the closure of S. If*

$$M_0 = \sup_{y\in\mathbb{R}} |\Phi(iy)| \qquad \text{and} \qquad M_1 = \sup_{y\in\mathbb{R}} |\Phi(1+iy)|,$$

then

$$\sup_{y\in\mathbb{R}} |\Phi(t+iy)| \leq M_0^{1-t} M_1^t, \qquad \text{for all } 0 \leq t \leq 1.$$

Proof. Let Φ be a holomorphic function in the strip $S = \{z \in \mathbb{C} : 0 < \mathrm{Re}(z) < 1\}$ that is also continuous and bounded on $\bar{S}$. If Φ is constant, then the claim follows trivially. So assume Φ is nonconstant. First, we assume that $M_0, M_1 = 1$. We also assume that $\Phi(x+iy) \to 0$ as $|y| \to \infty$. By the hypotheses on Φ, there exists some $M > 0$ such that

$$\sup_{z\in S} |\Phi(z)| = M.$$

In particular, it follows that there exists some sequence $\{z_n\}_1^\infty \subset S$ such that $\Phi(z_n) \to M$ as $n \to \infty$. But by the hypotheses, the z_n cannot converge to ∞, and thus, must converge to some $z_0 \in S$. The maximum modulus principle and the fact that Φ is not constant forces $z_0 \in \partial S$. Thus by the hypotheses on M, one has that $M \leq 1$. Therefore, the proof concludes under these restrictive hypotheses.

Now assume that $M_0, M_1 = 1$, but that the decay rate is not necessarily satisfied. For each $\varepsilon > 0$, define

$$\Phi_\varepsilon(z) = \Phi(z) \exp\left(\varepsilon(z^2 - 1)\right).$$

Moreover, it follows that on the lines $\mathrm{Re}(z) = 0$ and $\mathrm{Re}(z) = 1$, $|\Phi_\varepsilon(z)| \leq 1$. Since Φ is bounded, it also follows that for each fixed $x \in [0,1]$,

$$\sup_{x\in[0,1]} |\Phi_\varepsilon(x+iy)| \xrightarrow{|y|\to\infty} 0.$$

Therefore, the first part of the proof shows that on $\bar{S}$, $|\Phi_\varepsilon| \le 1$. Taking $\varepsilon \to 0$ concludes the proof.

For the final part of the proof, assume the least possible number of hypotheses on Φ, and define $\widetilde{\Phi} = M_0^{z-1} M_1^{-z}$. Then $\widetilde{\Phi}$ satisfies the hypotheses of the previous case so that $|\widetilde{\Phi}| \le 1$ in the strip, concluding the proof. □

Remark 2.1. *The geometric intuition for this lemma is clear to see. We specify bounds for Φ on the two vertical lines $\operatorname{Re}\Phi = 0$ and $\operatorname{Re}(z) = 1$. Then under the right hypotheses, the above lemma gives bounds for any third vertical line $\operatorname{Re}(z) = t$ between $\operatorname{Re}(z) = 0$ and $\operatorname{Re}(z) = 1$.*

From this result, we now have the following interpolation theorem for L^p spaces.

Theorem 2.1 (Riesz-Thorin Interpolation Theorem)**.** *Suppose T is a linear mapping from $L^{p_0} + L^{p_1}$ to $L^{q_0} + L^{q_1}$. Assume that T is bounded from L^{p_0} to L^{q_0} and from L^{p_1} to L^{q_1}:*

$$\begin{cases} \|Tf\|_{L^{q_0}} \le M_0 \|f\|_{L^{p_0}}, \\ \|Tf\|_{L^{q_1}} \le M_1 \|f\|_{L^{p_1}}. \end{cases}$$

Then T is bounded from L^p to L^q: $\|Tf\|_{L^q} \le M \|f\|_{L^p}$, whenever the pair (p, q) can be written as

$$\frac{1}{p} = \frac{1-t}{p_0} + \frac{t}{p_1} \qquad \text{and} \qquad \frac{1}{q} = \frac{1-t}{q_0} + \frac{t}{q_1}$$

for some t with $0 \le t \le 1$. Moreover, the bound M satisfies $M \le M_0^{1-t} M_1^t$.

Proof. Recall that the space of simple functions is dense in L^p for all p. Thus, our strategy is as follows: (1) show that the result holds for all simple functions and for all $p < \infty$ and $q > 1$, (2) use the density of simple functions in L^p to show the result for all $f \in L^p$ $(1 \le p < \infty)$, (3) show the result holds for $p = 1$ and $q = \infty$, and (4) finally show the result holds for $p < \infty$ and $q = 1$. We will also use the fact that to show that $\|Tf\|_{L^q} \le M \|f\|_{L^p}$, it suffices to show that $|\int Tfg\, dx| \le M \|f\|_{L^p} \|g\|_{L^{q'}}$ for all simple functions g with $\|g\|_{L^{q'}} = 1$[1]

(1) First assume that $p < \infty$ and $q > 1$, and assume $f \in L^p$ is simple with $\|f\|_{L^p} = 1$. We wish to define some holomorphic function Φ that is bounded on $\operatorname{Re}(z) = 0, 1$ so that we can use the interpolation result for this function. Thus for $z \in \mathbb{C}$, define

$$f_z = |f|^{\gamma(z)} \frac{f}{|f|}, \qquad \gamma(z) = p\left(\frac{1-z}{p_0} + \frac{z}{p_1}\right),$$
$$g_z = |g|^{\delta(z)} \frac{g}{|g|}, \qquad \delta(z) = q'\left(\frac{1-z}{q_0'} + \frac{z}{q_1'}\right).$$

These definitions are useful for the following reasons. First, it is straightforward to see that for any $t \in [0,1]$, $f_t = f$ and $g_t = g$. Next, for $\operatorname{Re}(z) = 0$, we compute

$$\|f_z\|_{L^{p_0}}^{p_0} = \int \left||f|^{\gamma(z)}\right|^{p_0} d\nu = \int |f|^{p_0 \cdot \operatorname{Re}(\gamma(z))} = \int |f|^p = \|f\|_{L^p} = 1.$$

Likewise, for $\operatorname{Re}(z) = 1$, we compute

$$\|f(z)\|_{L^{p_1}}^{p_1} = \int \left||f|^{\gamma(z)}\right|^{p_1} d\nu = \int |f|^{p_1 \cdot \operatorname{Re}(\gamma(z))} = \int |f|^p = \|f\|_{L^p} = 1.$$

Likewise, one has

$$\|g_z\|_{L^{q_0'}} = 1 \qquad \text{and} \qquad \|g_z\|_{L^{q_1'}} = 1.$$

Thus define

$$\Phi(z) = \int (Tf_z) g_z \, d\nu.$$

[1]Note: by $'$, we shall denote the conjugate exponent of an exponent.

Since f, g are simple, assume

$$f_z = \sum |a_k|^{\gamma(z)} \frac{a_k}{|a_k|} \chi_{E_k} \qquad \text{and} \qquad g_z = \sum |b_j|^{\delta(z)} \frac{b_j}{|b_j|} \chi_{F_j},$$

where the E_k and F_j are sets of finite measure. This implies that

$$\Phi(z) = \sum_{j,k} |a_k|^{\gamma(z)} |b_j|^{\delta(z)} \frac{a_k}{|a_k|} \frac{b_j}{|b_j|} \left(\int T(\chi_{E_k}) \chi_{F_j} \, d\nu \right)$$

so that (a) Φ is holomorphic on the strip S, and bounded, continuous on $\bar{S}$. For $\mathrm{Re}(z) = 0$, one has

$$|\Phi(z)| \le \int |Tf_z||g_z| \, d\nu \le \|Tf_z\|_{L^{q_0}} \|g\|_{L^{q_0'}} = \|Tf_z\|_{L^{q_0}} \le M_0 \|f\|_{L^{p_0}} = M_0.$$

Likewise on $\mathrm{Re}(z) = 1$, one has

$$|\Phi(z)| \le \int |Tf_z||g_z| \, d\nu \le \|Tf_z\|_{L^{q_1}} \|g\|_{L^{q_1'}} = \|Tf_z\|_{L^{q_1}} \le M_1 \|f\|_{L^{p_1}} = M_1.$$

Therefore, by the three-lines lemma, one has that on the line $\mathrm{Re}(z) = t$ $(t \in [0,1])$, $|\Phi(z)| \le M_0^{1-t} M_1^t$. Thus this completes the proof.

(2) Now fix some $f \in L^p$ for $1 \le p < \infty$. Choose a sequence $\{f_n\}$ of simple functions in L^p so that $\|f_n - f\|_{L^p} \to 0$. Since $\|Tf_n\|_{L^q} \le M \|f_n\|_{L^p}$, Tf_n is a Cauchy sequence in L^q. Thus it suffices to show that $Tf_n \to Tf$ a.e. To prove this, let us define,

$$f^U(x) \coloneqq \begin{cases} f(x), & \{x : |f(x)| \ge 1\}, \\ 0, & \text{otherwise} \end{cases} \qquad \text{and} \qquad f^L(x) \coloneqq \begin{cases} f(x), & \{x : |f(x)| < 1\}, \\ 0, & \text{otherwise.} \end{cases}$$

Then clearly $f = f^U + f^L$. Likewise, let $f_n = f_n^U + f_n^L$ for all n. Now wlog assume that $p_0 \le p_1$ so that $p_0 \le p \le p_1$. Since $p_0 \le p$, and $1 \le p$,

$$\int |f^U|^{p_0} \le \int |f^U|^p \le \int |f|^p < \infty,$$

so that $f^U \in L^{p_0}$. Since $f_n \to f$ in L^p, it follows that $f_n^U \to f^U$ in L^{p_0}. Thus there exists a subsequence $\{Tf_{n_j}^U\}$ that converges to Tf^U pointwise a.e. Likewise, we have

$$\int |f^L|^{p_1} \le \int |f^L|^p \le \int |f|^p < \infty,$$

so that $f^L \in L^{p_1}$. Since $f_n \to f$ in L^p, it follows that $f_n^L \to f^L$ in L^{p_1}. Thus the sequence $f_{n_j}^L \to f^L$ in L^{p_1} which means that $\{Tf_{n_j}^L\}$ converges to Tf^L in L^{q_1}. This means there exists a subsubsequence $\{Tf_{n_{j_k}}^L\}$ that converges to Tf^L pointwise a.e.; by construction, $Tf_{n_{j_k}}^U$ converges to Tf^U pointwise a.e. Therefore, renaming this subsubsequence as $\{Tf_n\}$, we conclude that $Tf_n = Tf_n^U + Tf_n^L$ converges to Tf a.e. so that $\|Tf\|_{L^q} \le M \|f\|_{L^p}$.

(3) Now consider the case $q = 1$ and $p = \infty$. Suppose first that $p = \infty$. Necessarily, it follows that $p_0 = p_1 = \infty$. Then H older's inequality concludes the proof.

(4) Suppose $p < \infty$ and $q = 1$. Then $q_0 = q_1 = 1$. We can take $g_z = g$ for all z and argue as in the case for $q > 1$.

□

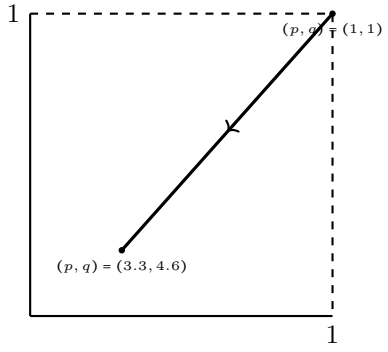
In particular, the following definition highlights the key essence of the Riesz-Thorin theorem.

Definition 2.1 (Operator of Type (p,q))**.** *Let $T : X \to Y$ be a linear operator that is initially defined on simple functions of X and maps such functions to functions on Y that are integrable on sets of finite measure. T is said to be an **operator of type** (p,q) if there exists $M > 0$ such that $\|Tf\|_{L^q} \le M \|f\|_{L^p}$ whenever f is simple.*

Since the space of simple functions is dense in every L^p space, it follows that if T is an operator of type (p,q) for some such pair, then the result extends easily to the entire L^p space for $p < \infty$ ($p = \infty$ if X has finite measure). We can denote the space of all such pairs (p,q) using a *Riesz diagram.*

Definition 2.2 (Riesz Diagram)**.** *Let T be a linear operator as defined above. Then the **Riesz diagram** of T is the set of all points $(1/p, 1/q)$ in the unit square $[0,1]^2$ such that T is of type (p,q). We define $M_{x,y}$ to be the least M for which $\|Tf\|_{L^q} \le M \|f\|_{L^p}$ among all (p,q) such that T is of type (p,q).*

For example, suppose T is a linear operator that satisfies the Riesz-Thorin Interpolation Theorem for $(p_0, q_0) = (1, 3.3)$, and $(q_0, q_1) = (1, 4.6)$. Then the Riesz diagram for T (as guaranteed by the Riesz-Thorin Interpolation Theorem) is the following,



We have the following corollary.

Corollary 2.1 (Riesz Convexity Theorem)**.** *With T as before,*

(a) The Riesz diagram of T is a convex set.
(b) $\log(M_{x,y})$ is a convex function on this set.

Now we are interested in applications of this theorem.

Corollary 2.2 (Hausdorff-Young Inequality)**.** *Let $X = [0, 2\pi]$ with the normalized Lebesgue measure $d\theta/(2\pi)$, and let $Y = \mathbb{Z}$ with its usual counting measure. Let T be the map defined by $Tf = \{a_n\}$, wher each a_n is the Fourier coefficient,*

$$a_n = \frac{1}{2\pi} \int_0^{2\pi} f(\theta) e^{-in\theta} \, d\theta.$$

If $1 \le p \le 2$ and $1/p + 1/q = 1$, then

$$\|Tf\|_{L^q(\mathbb{Z})} \le \|f\|_{L^p([0,2\pi])}.$$

Proof. This is a classic example of a proof using interpolation. Since $[0, 2\pi]$ has finite Lebesgue measure, one has that $L^2([0,2\pi]) \subset L^1([0,2\pi])$ and $L^2(\mathbb{Z}) \subset L^\infty(\mathbb{Z})$. Thus $L^2([0,2\pi]) + L^1([0,2\pi]) = L^1([0,2\pi])$, and $L^2(\mathbb{Z}) + L^\infty(\mathbb{Z}) = L^\infty(\mathbb{Z})$.

Now let $p_0, q_0 = 2$. Parseval's identity implies that

$$\|f\|_{L^{p_0}([0,2\pi])} = \left(\sum_{n=-\infty}^{\infty} |a_n|^2 \right)^{1/2} = \|Tf\|_{L^{q_0}(\mathbb{Z})}.$$

Thus the result follows immediately for this endpoint pair. Now suppose $p_1 = 1$ and $q_1 = \infty$. For any n, one has that

$$|a_n| \le \frac{1}{2\pi} \int_0^{2\pi} |f(\theta)| \, d\theta = \frac{1}{2\pi} \|f\|_{L^1([0,2\pi])} \le \|f\|_{L^1([0,2\pi])}.$$

Therefore, it follows immediately that $\|Tf\|_{L^\infty(\mathbb{Z})} \le \|f\|_{L^1([0,2\pi])}$. The Riesz-Thorin Interpolation Theorem then states that the result holds for all pairs (p,q) with

$$p = \left(\frac{1-t}{2} + t\right)^{-1} = \left(\frac{1-t+2t}{2t}\right)^{-1} = \frac{2t}{t+1}, \qquad \forall t \in [0,1],$$

and q the conjugate exponent of p; i.e., for all $1 \le p \le 2$ and q the conjugate exponent of p. Thus the proof concludes. □

Corollary 2.3 (The Fourier Transform)**.** *For simple functions f on $\mathbb{R}^d$, define the **Fourier transform** of f by*

$$T(f)(\xi) = \int_{\mathbb{R}^d} f(x) e^{-2\pi i x\cdot\xi}\, dx.$$

Then $\|Tf\|_{L^\infty(\mathbb{R}^d)} \le \|f\|_{L^1(\mathbb{R}^d)}$ so that T has a unique extension to $L^1(\mathbb{R}^d)$ for which this inequality continues to hold (i.e., by the Hahn-Banach Theorem). By Plancherel's Theorem, T also extends to $L^2(\mathbb{R}^d)$ as a unitary mapping. Thus, if $1 \le p \le 2$ and $1/p + 1/q = 1$, then the Fourier transform has a unique extension to a bounded map from L^p to L^q with $\|Tf\|_{L^q} \le \|f\|_{L^p}$.

Corollary 2.4 (Young's Inequality for Convolutions)**.** *Let $f \in L^p(\mathbb{R}^d)$ and $g \in L^r(\mathbb{R}^d)$. Then the convolution,*

$$(f * g)(x) \coloneqq \int_{\mathbb{R}^d} f(x-y) g(y)\, dy$$

is well-defined (i.e., $f(x-y)g(y)$ is integrable for almost every x), and

$$\|f * g\|_{L^q} \le \|f\|_{L^p} \|g\|_{L^r},$$

where $1/q = 1/p + 1/r - 1$ $(1 \le q \le \infty)$.

2.1.2 The Hilbert Transform

We begin with the following observations for L^2. Let $f \in L^2$, and assume that

$$f(\theta) \sim \sum_{n=-\infty}^{\infty} a_n e^{in\theta},$$

where the sum on the right is the Fourier series for f. We also define the *conjugate function* or *allied series* of f as follows.

Definition 2.3 (Allied Series)**.** *Let $f \in L^2$, and assume $f(\theta) \sim \sum_{-\infty}^{\infty} a_n e^{in\theta}$. Then we define the **conjugate function** or **allied series** of f by*

$$\widetilde{f}(\theta) \coloneqq \sum_{n=-\infty}^{\infty} \frac{\operatorname{sign}(n)}{i} a_n e^{in\theta}.$$

This specific definition is significant for the following reason:

$$\frac{1}{2}(f(\theta) + i\widetilde{f}(\theta) + a_0) = \sum_{n=0}^{\infty} a_n e^{in\theta} = F(z),$$

where F is the Cauchy integral (projection),

$$F(z) = \frac{1}{2\pi i} \int_0^{2\pi} \frac{f(\theta)}{e^{i\theta} - z} i e^{i\theta}\, d\theta.$$

In particular, it is clear that if f is real-valued (so that $\widetilde{f}$ is also real-valued), then $\widetilde{f}$ and $f + a_0$ represent, respectively, the real and imaginary parts of the boundary values of the analytic function $2F$ in the unit disc. Moreover, Parseval's identity then gives

$$\frac{1}{2\pi} \int_0^{2\pi} |\widetilde{f}(\theta)|^2\, d\theta + |a_0|^2 = \frac{1}{2\pi} \int_0^{2\pi} |f(\theta)|^2\, d\theta.$$

Now we would like to generalize this to other L^p spaces. This leads to the theory of the *Hilbert transform*. To make the transition clear, we begin with the L^2 formalism, and then use the Riesz-Thorin Interpolation theorem to generalize this result. We start with the following definition.

Definition 2.4 (Cauchy Integral)**.** *Let $f \in L^2(\mathbb{R})$. Then the **Cauchy integral** of f is defined to be*

$$F(z) = C(f)(z) = \frac{1}{2\pi i}\int_{-\infty}^{\infty} \frac{f(t)}{t-z}\,dt, \qquad \mathrm{Im}(z) > 0. \tag{2}$$

Remark 2.2. *The fact that we call $F(z)$ the Cauchy integral of f should look familiar. We recall that Cauchy's integral theorem tells us that if γ is a closed contour in $\mathbb{C}$, $f(z)$ is holomorphic inside the region enclosed by γ and on γ, and z_0 is a point contained inside the region enclosed by γ, then*

$$f(z_0) = \frac{1}{2\pi i}\oint_\gamma \frac{f(z)}{z-z_0}\,dz.$$

For our analysis, we replace the unit circle with the real line $\mathbb{R}$, and the unit disk with the upper half-plane $\mathbb{R}_+$. Thus for any $z \in \mathbb{R}_+$ (i.e., with $\mathrm{Im}(z) > 0$), it would be reasonable to consider the quantity

$$\frac{1}{2\pi i}\int_{-\infty}^{\infty} \frac{f(t)}{t-z}\,dt.$$

Thus we denote this quantity the Cauchy integral of f.

The first thing we must check is if $F(z)$ is well-defined; i.e., if $F(z)$ converges for arbitrary $f \in L^2(\mathbb{R})$. We compute,

$$|F(z)| \le \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{|f(t)|}{|t-z|}\,dt \le \|f\|_{L^2(\mathbb{R})}\left\|\frac{1}{\cdot - z}\right\|_{L^2(\mathbb{R})}.$$

If we can show that $(t-z)^{-1} \in L^2(\mathbb{R})$, then we are done. Taking $z = x + iy$ (where $y > 0$), we have

$$\begin{aligned}\int_{-\infty}^{\infty} \frac{1}{|t-x-iy|^2}\,dt &= \int_{-\infty}^{\infty} \frac{1}{(t-x)^2+y^2}\,dt \\ &= \int_{-\infty}^{\infty} \frac{1}{u^2+y^2}\,du = \frac{\pi}{y} < \infty.\end{aligned}$$

Therefore, it follows that $F(z)$ converges for all $x + iy$ with $y > 0$. But there is also a more clever way of writing the Cauchy integral of f,

$$F(z) = \int_0^{\infty} \hat{f}(\xi) e^{2\pi i z\xi}\,d\xi, \qquad \mathrm{Im}(z) > 0.$$

This notation allows us to demonstrate the following,

$$\begin{aligned}\|F(\cdot + iy) - P(f)(\cdot)\|_{L^2(\mathbb{R})} &= \left(\int_{-\infty}^{\infty}\left|\int_{-\infty}^{\infty} \hat{f}(\xi)\chi(\xi)e^{2\pi i x\xi}e^{-2\pi y\xi}\,d\xi - \int_{-\infty}^{\infty} \hat{f}(\xi)\chi(\xi)e^{2\pi i x\xi}\,d\xi\right|^2 dx\right)^{\frac{1}{2}} \\ &= \left(\int_{-\infty}^{\infty}\left|\int_{-\infty}^{\infty} \hat{f}(\xi)\chi(\xi)e^{2\pi i x\xi}(e^{-2\pi y\xi}-1)\,d\xi\right|^2 dx\right)^{1/2} \\ &= \left(\int_{-\infty}^{\infty}\left|\int_{-\infty}^{\infty} g(\xi)e^{2\pi i x\xi}\,d\xi\right|^2 dx\right)^{1/2} \\ &= \|\check{g}\|_{L^2(\mathbb{R})} = \|g\|_{L^2(\mathbb{R})}.\end{aligned}$$

Using the definition of g, we have

$$\|g\|_{L^2(\mathbb{R})}^2 = \int_{-\infty}^{\infty} |\hat{f}(\xi)|^2\chi(\xi)(e^{-2\pi y\xi}-1)^2\,d\xi$$

Pointwise, in the limit $y \to 0^+$, the integrand converges to 0. Thus by the Dominated Convergence Theorem, we conlude that $\|g\|_{L^2(\mathbb{R})} \to 0$ as $y \to 0^+$. Therefore, we conclude that $F(x+iy) \xrightarrow{y\to 0^+} P(f)(x)$ under the $L^2(\mathbb{R})$ norm, where we define

Definition 2.5 (Orthogonal Projection Operator)**.** *Let $P : L^2(\mathbb{R}) \to L^2(\mathbb{R})$ be the orthogonal projection operator that projects each $f \in L^2(\mathbb{R})$ to a function f' for which $\hat{f}'(\xi) = 0$ for almost every $\xi < 0$:*

$$P(f)(x) = \int_{-\infty}^{\infty} \hat{f}(\xi)\chi(\xi)e^{2\pi i x\xi}\, d\xi.$$

Now we define the *Hilbert transform*; a motivation for the Hilbert transform will follow later.

Definition 2.6 (Hilbert Transform)**.** *Let $f \in L^2(\mathbb{R})$. Then we define the **Hilbert transform** of f to be the function*

$$H(f)(x) = \int_{-\infty}^{\infty} \hat{f}(\xi)\frac{\operatorname{sign}(\xi)}{i}e^{2\pi i x\xi}\, d\xi. \tag{3}$$

From the definition, we make the following observations:

(i) $P = \frac{1}{2}(I + iH)$, where I is the identity operator.

(ii) H is unitary on L^2, and $H \circ H = H^2 = -I$; i.e., $\|H(f)\|_{L^2} = \|f\|_{L^2}$ and H is invertible with $H^{-1} = -H$.

Proof.

(i) Let $f \in L^2(\mathbb{R})$. Then

$$\begin{aligned}\frac{1}{2}(I + iH)(f) &= \frac{1}{2}\int_{-\infty}^{\infty} \hat{f}(\xi)e^{2\pi i x\xi}\, d\xi + \frac{1}{2}\int_{-\infty}^{\infty} \hat{f}(\xi)\operatorname{sign}(\xi)e^{2\pi i x\xi}\, d\xi \\ &= \int_{0}^{\infty} \hat{f}(\xi)e^{2\pi i x\xi}\, d\xi = \int_{-\infty}^{\infty} \hat{f}(\xi)\chi(\xi)e^{2\pi i x\xi}\, d\xi =: P(f)(x),\end{aligned}$$

where the integral over all $\xi \in (-\infty, 0]$ vanishes because of the $\operatorname{sign}(\xi)$ in the second integral.

(ii) Fix $f \in L^2(\mathbb{R})$. Then we compute,

$$\begin{aligned}\hat{H}(f)(\xi) &= \int_{-\infty}^{\infty} H(f)(x)e^{-2\pi i x\cdot\xi}\, dx \\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \hat{f}(\eta)\frac{\operatorname{sign}(\eta)}{i}e^{2\pi i x(\eta-\xi)}\, d\eta dx. \\ \|H(f)\|^2_{L^2(\mathbb{R})} &= \int_{-\infty}^{\infty}\left|\int_{-\infty}^{\infty} \hat{f}(\xi)\frac{\operatorname{sign}(\xi)}{i}e^{2\pi i x\xi}\, d\xi\right|^2 dx \\ &= \int_{-\infty}^{\infty}\left|\int_{-\infty}^{\infty} g(\xi)e^{2\pi i x\xi}\, d\xi\right|^2 dx \\ &= \|\hat{g}\|^2_{L^2(\mathbb{R})} = \|\hat{g}\|^2_{L^2(\mathbb{R})} = \left\|\hat{f}\right\|^2_{L^2(\mathbb{R})} = \|f\|^2_{L^2(\mathbb{R})}\end{aligned}$$

□

If we wish to extend the Hilbert transform to other L^p spaces, then it is desirable for H to be at least bounded (if not unitary). To work towards this goal, we have the following proposition.

Proposition 2.1 (Hilbert Transform as a Singular Integral)**.** *If $f \in L^2(\mathbb{R})$, then*

$$H(f)(x) = \lim_{\varepsilon\to 0}\frac{1}{\pi}\int_{|t|\geq\varepsilon} f(x-t)\frac{dt}{t}. \tag{4}$$

I.e., if for each $\varepsilon > 0$, we denote

$$H_\varepsilon(f)(x) := \frac{1}{\pi}\int_{|t|\geq\varepsilon} f(x-t)\frac{dt}{t},$$

then $H_\varepsilon(f) \in L^2(\mathbb{R})$ and the convergence is accordingly in the L^2 norm.

Remark 2.3. *The proof needs to be incorporated. But the main idea is that for each $\varepsilon > 0$, $\|H_\varepsilon(f)\|_{L^2(\mathbb{R})} \leq A\|f\|_{L^2(\mathbb{R})}$, where A is independent of ε and f.*

In particular, the remark is suggestive of the following theorem of Riesz.

Theorem 2.2 (Riesz's Theorem for Hilbert Transform)**.** *Suppose $1 < p < \infty$. Then the Hilbert transform H, initially defined on $L^2 \cap L^p(\mathbb{R})$ by either (3) or (4), satisfies the estimate*

$$\|H(f)\|_{L^p(\mathbb{R})} \leq A_p \|f\|_{L^p(\mathbb{R})}, \qquad \forall f \in L^2 \cap L^p(\mathbb{R}), \tag{5}$$

where A_p is independent of f. Thus the Hilbert transform has a unique extension to all of L^p satisfying the same bound.

Proof. [!! Complete Later !!] □

2.1.3 Hardy Spaces

Now we turn to the theory of the *Hardy spaces*, $\mathbf{H}^1_r(\mathbb{R}^d)$. Recall that many of the results we've been studying thus far have been based on L^p space theory. Broadly speaking, we have been interested in one of the following three spaces: L^1, L^2, and L^∞. But restricting ourselves to just one of these three spaces are indeed too restrictive. For instance, if $f \in L^1(\mathbb{R})$, then the only results we can say about their Fourier transforms is that they are bounded, and vanish at infinity (the *Riemann-Lebesgue lemma*). But these transforms are not guaranteed to be integrable, nor do we have any other important constraints on their "size". But if $f \in \mathbf{H}^1_r$, then we actually have the following estimate (cf. exercises)

$$\int_{\mathbb{R}^d} |\hat{f}(\xi)| \frac{d\xi}{|\xi|^d} \leq A \|f\|_{\mathbf{H}^1_r}.$$

Thus the main advantage of working with $\mathbf{H}^1_r$ is that it is a Banach space that is, in some sense, "close" to L^1, while being more helpful in establishing estimates. Therefore, we will develop this theory now.

We begin with the notion of an *atom*.

Definition 2.7 (Atom)**.** *$\mathfrak{a}$ is an atom associated to a ball $B \subset \mathbb{R}^d$ if $\mathfrak{a}$ is a bounded measurable function on $\mathbb{R}^d$ such that*

(i) $\mathfrak{a}$ is supported in B, with $|\mathfrak{a}(x)| \leq 1/m(B)$ for all x, and
(ii) $\int_{\mathbb{R}^d} \mathfrak{a}(x)\, dx = 0$.

Then, we define

Definition 2.8 (Hardy Space)**.** *The space $\mathbf{H}^1_r(\mathbb{R}^d)$ consists of all functions $f \in L^1(\mathbb{R}^d)$ that can be written as*

$$f = \sum_{k=1}^{\infty} \lambda_k \mathfrak{a}_k,$$

where the $\mathfrak{a}_k$ are atoms, and the λ_k are scalars with

$$\sum_{k=1}^{\infty} |\lambda_k| < \infty.$$

Of course, the first thing we must check is if the above sum for f actually converges. By definition of atoms, it follows that for each k, $\|\mathfrak{a}\|_{L^1(\mathbb{R})} = 1$. Thus one has that the sum converges in the L^1 norm. We can turn $\mathbf{H}^1_r$ into a normed space as follows:

Definition 2.9 (Hardy Norm)**.** *Let $f \in \mathbf{H}^1_r(\mathbb{R}^d)$. Then the $\mathbf{H}^1_r(\mathbb{R}^d)$ norm of f, is defined to be the quantity*

$$\|f\|_{\mathbf{H}^1_r(\mathbb{R}^d)} \coloneqq \inf \left\{ \sum |\lambda_k| : f = \sum_1^{\infty} \lambda_k \mathfrak{a}_k \right\}.$$

We make the following observation.

(i) The space $\mathbf{H}^1_r$ is complete under the above norm, and thus is a Banach space. If $f \in \mathbf{H}^1_r$, then $f \in L^1$, and $\|f\|_{L^1(\mathbb{R}^d)} \le \|f\|_{\mathbf{H}^1_r}$. Moreover, $\int f(x)\,dx = 0$.

We also have the following theorem.

Proposition 2.2 (L^p, Bounded Support Functions and Hardy Spaces)**.** *Suppose $f \in L^p(\mathbb{R}^d)$, $p > 1$, and f has bounded support. Then f belongs to $\mathbf{H}^1_r(\mathbb{R}^d)$ if and only if $\int_{\mathbb{R}^d} f(x)\,dx = 0$.*

Now we will generalize the notion of atoms to include *p-atoms*.

Definition 2.10 (p-Atoms)**.** *For any p with $p > 1$, define a p-**atom**, associated to a ball B to be a measurable function $\mathfrak{a}$ which satisfies:*

(i) $\mathfrak{a}$ is supported in B, and $\|\mathfrak{a}\|_{L^p} \le m(B)^{-1+1/p}$.
(ii) $\int_{\mathbb{R}^d} \mathfrak{a}(x)\,dx = 0$.

Thus in defining $\mathbf{H}^1_r$, we no longer need to restrict ourselves to just working with atoms, and can instead also consider p-atoms, and obtain equivalent norms. Now, we arrive at another reason why the Hardy space is more desirable than L^1. Recall that the Hilbert transform is not bounded on L^1 (indeed, the transform is bounded on L^p only for $1 < p < \infty$). However, we have the following boundedness result for Hardy spaces.

Theorem 2.3 (Hilbert Transform and Boundedness)**.** *Let $f \in \mathbf{H}^1_r(\mathbb{R})$. Then $H_\varepsilon(f) \in L^1(\mathbb{R})$ for every $\varepsilon > 0$. Moreover, $H_\varepsilon(f)$ converges in the L^1 norm, as $\varepsilon \to 0$. Its limit, defined as $H(f)$, satisfies*

$$\|H(f)\|_{L^1(\mathbb{R})} \le A\,\|f\|_{\mathbf{H}^1_r(\mathbb{R})}.$$

2.1.4 Maximal Functions

In this subsubsection, we introduce the notion of *maximal functions*, which will become more useful when we investigate oscillatory integrals. We start with the following "rough" definition.

Definition 2.11 (Maximal Function)**.** *For appropriate functions f given on $\mathbb{R}^d$, define the **maximal function** f^* by*

$$f^*(x) \coloneqq \sup_{x \in B} \frac{1}{m(B)} \int_B |f(y)|\,dy, \tag{6}$$

where the supremum is taken over all balls B containing x, and m, dy denote the Lebesgue measure on $\mathbb{R}^d$.

Then we have the following L^p inequality.

Theorem 2.4 (L^p Inequality for the Maximal Function)**.** *Suppose $f \in L^p(\mathbb{R}^d)$ with $1 < p \le \infty$. Then $f^* \in L^p(\mathbb{R}^d)$, and*

$$\|f^*\|_{L^p} \le A_p\,\|f\|_{L^p},$$

where A_p depends on p, but is independent of f.

Remark 2.4. *The proof for the case $p = \infty$ follows immediately:*

$$|f^*(x)| = \sup_{x \in B} \frac{1}{m(B)} \int_B |f(y)|\,dy \le \|f\|_{L^\infty} \cdot \sup_{x \in B} \frac{1}{m(B)}\,dy = \|f\|_{L^\infty}.$$

Since this is true for any $x \in \mathbb{R}^d$, $\|f^*\|_{L^\infty} \le \|f\|_{L^\infty}$. What about for the case $p = 1$? Let f be the characteristic function of the unit ball B so that $f \in L^1(\mathbb{R}^d)$. Let f^* be the maximal function for f. Fix $x \in B$, and define B' to be the ball centered at the origin of radius $1 + |x|$. Then

$$\frac{1}{m(B')} \int_{B'} |f(y)|\,dy = \frac{m(B \cap B')}{m(B')} = \frac{1}{(1+|x|)^d}.$$

Thus by definition of the maximal function, $|f^*(x)| \ge 1/(1+|x|^d)$. But the right side of the estimate is not integrable at infinity. However, we do have a class of weak-type inequalities.

Proposition 2.3 (Weak-Type Inequalities)**.** *Let $f \in L^1(\mathbb{R}^d)$. Then there is a bound A (independent of f) such that*

$$m(\{x : f^*(x) > \alpha\}) < \frac{A}{\alpha} \|f\|_{L^1(\mathbb{R}^d)}.$$

The proof follows using the Vitali covering lemma.

Now we will turn to the application of Hardy space theory to maximal functions. The maximal function we defined above is not immediately applicable to here, since it requires the absolute value of f, and does not adequately exploit the properties of being contained in $\mathbf{H}^1_r$. Moreover, the characteristic functions of balls is not smooth enough for our purposes. Thus we begin with approximations of identity.

Lemma 2.2 (Approximation to Identity)**.** *Fix Φ be a bounded function with compact support such that $\int \Phi = 1$, and for every $\varepsilon > 0$, set $\Phi_\varepsilon(x) = \varepsilon^{-d}\Phi(x/\varepsilon)$. Then for any $f \in L^1$,*

$$(f * \Phi_\varepsilon(x)) \to f(x), \qquad \text{as } \varepsilon \to 0 \text{ for a.e. } x.$$

Thus we define the *maximal function* in the following new (albeit related) way.

Definition 2.12 (Maximal Function)**.** *Having fixed Φ as above, and $f \in L^1$, we define the **maximal function** of f by*

$$M(f)(x) = \sup_{\varepsilon > 0} |(f * \Phi_\varepsilon)(x)|.$$

Thus we have the following theorem.

Theorem 2.5 (Maximal Function is a Bounded Operator)**.** *Suppose Φ is a C^1 function with compact support on $\mathbb{R}^d$. With M defined as above, $M(f) \in L^1(\mathbb{R}^d)$, whenever $f \in \mathbf{H}^1_r(\mathbb{R}^d)$. Moreover,*

$$\|M(f)\|_{L^1(\mathbb{R}^d)} \le A \|f\|_{\mathbf{H}^1_r(\mathbb{R}^d)}.$$

Proof. Suppose $f \in \mathbf{H}^1_r(\mathbb{R}^d)$ and $f = \sum \lambda_k \mathfrak{a}_k$ is an atomic decomposition. Then

$$\begin{aligned} |(f * \Phi_\varepsilon)(x)| &= \left| \left(\left(\sum_1^\infty \lambda_k \mathfrak{a}_k \right) * \Phi_\varepsilon \right)(x) \right| \\ &= \left| \sum_1^\infty (\lambda_k (\mathfrak{a}_k * \Phi_\varepsilon)(x)) \right| \\ &\le \sum_1^\infty |\lambda_k| |(\mathfrak{a}_k * \Phi_\varepsilon)(x)|. \end{aligned}$$

Thus taking the suprema of both sides over all $\varepsilon > 0$, we conclude that $M(f) \le \sum_1^\infty |\lambda_k| M(\mathfrak{a}_k)$. Thus it suffices to prove the bound when f is an atom $\mathfrak{a}$. Now we observe the following property of atoms: for $r > 0$, if $\mathfrak{a}_r(x) = r^d \mathfrak{a}(rx)$, then $(\mathfrak{a}_r * \Phi_\varepsilon)(x) = r^d(\mathfrak{a} * \Phi_{\varepsilon r})(rx)$ so that $M(\mathfrak{a}_r)(x) = r^d M(\mathfrak{a})(rx)$. Moreover, the mapping $\mathfrak{a} \mapsto M(\mathfrak{a})$ commutes with translations. Thus, it suffices to assume that the atom $\mathfrak{a}$ is associated to the unit ball centered at the origin. Now we consider the following two steps.

1. ($|x| \le 2$) Since $M(\mathfrak{a})(x) \le c$, $\int_{|x| \le 2} M(\mathfrak{a})(x)\, dx \le c'$.
2. ($|x| > 2$) We start by writing

$$\begin{aligned} (a * \Phi_\varepsilon)(x) &= \varepsilon^{-d} \int_{\mathbb{R}^d} \mathfrak{a}(y) \Phi\left(\frac{x-y}{\varepsilon}\right) dy \\ &= \varepsilon^{-d} \int_{\mathbb{R}^d} \mathfrak{a}(y) \left[\Phi\left(\frac{x-y}{\varepsilon}\right) - \Phi\left(\frac{x}{\varepsilon}\right) \right] dy, \end{aligned}$$

where we used the fact that $\int_{\mathbb{R}^d} \mathfrak{a}(y)\, dy = 0$. However, since $|x| \ge 2$, and $|y| \le 1$, $|x-y| \ge |x|/2$. Moreover, since $\Phi \in C^1$ we have that

$$\left| \Phi\left(\frac{x-y}{\varepsilon}\right) - \Phi\left(\frac{x}{\varepsilon}\right) \right| \le \frac{c|y|}{\varepsilon} \le \frac{c}{\varepsilon}.$$

Since Φ has compact support, $(\mathfrak{a} * \Phi_\varepsilon)(x)$ vanishes unless $|(x-y)/\varepsilon| \le A$ for some bound A. This then implies $\varepsilon > |x|/(2A)$.

Thus this concludes the proof. □

The Space BMO In this section, we introduce the space *BMO*, which we will show is a natural substitute for the space $L^\infty(\mathbb{R}^d)$. We begin with the following definition of what it means for a function to be of *bounded mean oscillation*.

Definition 2.13 (BMO Function)**.** *A locally integrable function f on $\mathbb{R}^d$ is said to be of* ***bounded mean oscillation (BMO)*** *if*

$$\|f\|_{\mathrm{BMO}} = \sup_B \frac{1}{m(B)} \int_B |f(x) - f_B|\, dx < \infty, \tag{7}$$

where

$$f_B \coloneqq \fint_B f(x)\, dx.$$

$\|\cdot\|_{\mathrm{BMO}}$ *defines a norm on the space of all BMO functions.*

I.e., the space BMO consists of all locally integrable functions with the property that on any ball B, the average difference between f and its average on the ball is bounded. We make the following observations about this space.

- The null elements of the norm are the constant functions. To see this, assume $\|f\|_{\mathrm{BMO}} = 0$. Then on any ball B,
$$\int_B |f(x) - f_B|\, dx = 0,$$
which means $|f(x) - f_B| = 0$ a.e. But this means that $f = f_B$ a.e. so that f is constant a.e.
- L^∞ functions are contained in BMO. To see this, assume that $f \in L^\infty(\mathbb{R}^d)$ be locally integrable. Then fix an arbitrary ball B, and let f_B be the average of f over B. Then
$$\frac{1}{m(B)} \int_B |f(x) - f_B|\, dx \le \frac{1}{m(B)} \int_B \|f\|_{L^\infty(\mathbb{R}^d)} + |f_B|\, dx = \|f\|_{L^\infty(\mathbb{R}^d)} + |f_B|. \tag{8}$$
On the other hand, one has that
$$|f_B| \le \frac{1}{m(B)} \int_B |f(x)|\, dx \le \|f\|_{L^\infty(\mathbb{R}^d)}.$$
Thus, since the upper bound for (8) is independent of B, taking the supremum over all balls, we conclude that $\|f\|_{\mathrm{BMO}} \le 2\|f\|_{L^\infty(\mathbb{R}^d)} < \infty$.

The spaces BMO and $\mathbf{H}^1_r$ are actually more connected than it would originally seem. In particular, we will soon show that BMO is the dual space of $\mathbf{H}^1_r$.

Theorem 2.6 (BMO as the Dual of $\mathbf{H}^1_r$)**.** *Suppose $g \in \mathrm{BMO}$. Then the linear functional ℓ defined by*

$$\ell(f) = \int_{\mathbb{R}^d} f(x)g(x)\, dx,$$

initially considered for $f \in H^1_0$ (the dense subspace of $\mathbf{H}^1_r$ consisting of finite linear combinations of atoms), has a unique extension to $\mathbf{H}^1_r$ that satisfies

$$\|\ell\| \le c\|g\|_{\mathrm{BMO}}.$$

Conversely, every bounded linear functional ℓ on $\mathbf{H}^1_r$ can be written in the form of the linear functional above with $g \in BMO$ and

$$\|g\|_{BMO} \le c'\|\ell\| \coloneqq \|\ell\|_{(\mathbf{H}^1_r)^*}.$$

2.2 Distributions and Generalized Functions

In this subsection, we introduce the notion of *distributions*, which will become essential in our future study of nonlinear wave equations. Classically, we are used to the idea of a function f (on $\mathbb{R}^n$) as assigning a definite value $f(x)$ for every $x \in \mathbb{R}^n$. But we also saw in measure theory that it suffices to define a function up to a set of measure zero (i.e., we can allow f to remain undefined on points contained in a set of measure zero).[2] Now, we will see that it suffices to relax this definition even more in certain cases, leading to the idea of **distributions** or **generalized functions**. Rather than assigning a definite value at each point in its domain, we define a distribution by its averages taken with respect to a certain class of test functions (e.g., $C_c^\infty(\mathbb{R}^n)$). As we will see in Section §.3, one might not have classical solutions of desired regularity (in the sense of ordinary functions) for various partial differential equations. However, there could exist generalized functions, that although do not necessarily have the regularity to satisfy the original differential equations, that satisfy other properties that a normal classical solution would have. For example, consider the one-dimensional wave equation,

$$\frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} = 0.$$

Multiplying both sides by an arbitrary smooth function, $\varphi(x,t)$ with compact support, and integrating one has,

$$\int_{\mathbb{R}}\int_{\mathbb{R}} \frac{\partial u}{\partial t}\varphi(x,t)\, dtdx + \int_{\mathbb{R}}\int_{\mathbb{R}} \frac{\partial u}{\partial x}\varphi(x,t)\, dtdx.$$

Using Fubini's theorem, and integration by parts, one obtains

$$\int_{\mathbb{R}}\int_{\mathbb{R}} u(t,x)\frac{\partial \varphi}{\partial t}\, dtdx + \int_{\mathbb{R}}\int_{\mathbb{R}} u(t,x)\frac{\partial \varphi}{\partial x}\, dtdx = 0.$$

Thus any solution to the wave equation satisfies the above equation as long as it is at least C^1. However, it is possible that there exist functions u that satify the above equation for all such φ but is possibly not differentiable (so that u does not satisfy the wave equation, itself). Hence, as we will see, distributions enable us to make sense of such u (which we shall call *weak solutions* to a differential equation).

Thus we have the following definitions.

Definition 2.14 (Test Functions on Ω)**.** *Fix an open set Ω in $\mathbb{R}^d$. The **test functions** are those functions $\varphi \in C_c^\infty(\Omega)$ (or equivalently, $C_0^\infty(\Omega)$) - the set of all smooth functions with compact support in Ω. We denote the space of all test functions on Ω by the symbol $\mathcal{D}(\Omega)$.*

Definition 2.15 (Convergence in $\mathcal{D}(\Omega)$)**.** *Let $\{\varphi_n\} \subset \mathcal{D}(\Omega)$ and $\varphi \in \mathcal{D}$. φ_n is said to converge to φ if the supports of φ_n are contained in a common compact set, and for every multiindex α, one has that $\partial_x^\alpha \varphi_n \to \partial_x^\alpha \varphi$ uniformly in x as $n \to \infty$.*

Definition 2.16 (Distributions on Ω)**.** *A **distribution** F on Ω is a complex-valued linear functional $\varphi \mapsto F(\varphi)$, defined for all $\varphi \in \mathcal{D}$, that is continuous in the sense that $F(\varphi_n) \to F(\varphi)$ whenever $\varphi_n \to \varphi$ in $\mathcal{D}$. We shall denote the vector space of all distributions on Ω by $\mathcal{D}^*(\Omega)$.*

For example, all signed Radon measures on an open subset Ω, and all locally integrable function on Ω are examples of distributions on Ω. Our ultimate goal is to "calculus" using distributions. For smooth functions f, one of the most important results we have (as we shall see in the Section §.3) is **integration by parts**: for any $f \in C^\infty(\Omega)$ and $\varphi \in \mathcal{D}(\Omega)$, a simple inductive argument yields,

$$\int_\Omega (\partial_x^\alpha f)\varphi(x)\, dx = (-1)^{|\alpha|}\int_\Omega f(\partial_x^\alpha \varphi)\, dx,$$

where the boundary terms are identically zero because φ has compact support. Thus, generalizing this definition,

[2] which is to say that a function is really just an equivalence class of functions that agree pointwise a.e.

Definition 2.17 (Derivative of a Distribution)**.** *If $F \in \mathcal{D}^*(\Omega)$, for any multiindex α, one defines the* ***distributional derivative of*** *F, $(\partial_x^\alpha F)$ as follows: for any $\varphi \in \mathcal{D}(\Omega)$,*

$$(\partial_x^\alpha F)(\varphi) = (-1)^{|\alpha|} F(\partial_x^\alpha \varphi). \tag{9}$$

Thus while an arbitrary locally integrable function f may not have sufficient regularity to talk about its classical derivatives, we can certainly talk about its partial derivatives as distributions (which exist).

We can generalize some basic properties of functions to distributions.

Definition 2.18 (Some Basic Operations on Distributions)**.** *Let Ω be an open subset of $\mathbb{R}^d$, and let $F \in \mathcal{D}^*$. Then we define the following.*

1. *If $\psi \in C^\infty(\Omega)$, then the product $\psi \cdot F$ is defined by $(\psi \cdot F)(\varphi) \coloneqq F(\psi\varphi)$ for every $\varphi \in \mathcal{D}$. It stands that $\psi \cdot F \in \mathcal{D}^*$.*
2. *Let $\tau_{\cdot}$ denote the translation operator: $\tau_h f(x) \coloneqq f(x-h)$ for any $h \in \mathbb{R}^d$. Then the corresponding definition for distributions is,*
$$\tau_h(F)(\varphi) = F(\tau_{-h}\varphi), \qquad \forall \varphi \in \mathcal{D}.$$
3. *If L is a nonsingular linear transformation on $\mathbb{R}^d$, then if $f_L(x) \coloneqq f(Lx)$ for functions, on distributions, one has*
$$F_L(\varphi) = |\det L|^{-1} \cdot F(\varphi_{L^{-1}}), \qquad \forall \varphi \in \mathcal{D}.$$

Another property we would like to generalize is convolution of a distribution with a test function. We make the following definitions:

Definition 2.19 (Convolution of a Distribution with Test Functions)**.** *Let $F \in \mathcal{D}(\mathbb{R}^d)$, and $\psi \in C_c^\infty(\mathbb{R}^d)$. Then we make the definitions,*

1. *(As a function): $(F * \psi)(x) \coloneqq F(\widetilde{\psi_x})$, where $\widetilde{\psi_x} \coloneqq \psi(x-y)$,*
2. *(As a distribution): $(F * \psi)(\varphi) \coloneqq F(\widetilde{\psi} * \varphi)$, for all $\varphi \in \mathcal{D}(\mathbb{R}^d)$, with $\widetilde{\psi} \equiv \widetilde{\psi_0}$.*

To see where these definitions could come from, assume first that F is a locally integrable function. Then one has that

$$(F * \psi)(x) \coloneqq \int_{\mathbb{R}^d} F(x-y)\psi(y)\, dy = \int_{\mathbb{R}^d} F(y)\psi(x-y)\, dy.$$

This leads naturally to the first definition. On the other hand, for any $\varphi \in \mathcal{D}(\mathbb{R}^d)$, if F is a locally integrable function, then

$$\begin{aligned}
(F * \psi)(\varphi) &= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} F(y)\psi(x-y)\varphi(x)\, dydx \\
&= \int_{\mathbb{R}^d} \int_{\mathbb{R}^d} F(y)\psi(x-y)\varphi(x)\, dxdy \qquad \text{(by Fubini's Theorem)} \\
&= \int_{\mathbb{R}^d} F(y)(\widetilde{\psi} * \varphi)(y)\, dy \\
&= F(\widetilde{\psi} * \varphi).
\end{aligned}$$

This leads naturally to the second definition.

Proposition 2.4 (Equivalence of Definitions of Convolution)**.** *Suppose F is a distribution and $\psi \in \mathcal{D}$. Then*

1. *The two definitions of $F * \psi$ given above coincide.*
2. *The distribution $F * \psi$ is a C^∞ function.*

We can also talk about distributions converging in $\mathcal{D}^*$.

Definition 2.20 (Weak Convergence of Distributions)**.** *A sequence of distributions $\{F_n\} \subset \mathcal{D}^*$ converges to a distribution F in the* ***weak sense*** *(or in the* ***sense of distributions****) if $F_n(\varphi) \to F(\varphi)$ for all $\varphi \in \mathcal{D}$.*

In particular, the following result follows.

Proposition 2.5 (Approximating Distibutions with Smooth Functions)**.** *Suppose F is a distribution on $\mathbb{R}^d$. Then there exists a sequence $\{f_n\}$, with $f_n \in C^\infty$, and $f_n \to F$ in the weak sense.*

However, the space of test functions $C_c^\infty(\mathbb{R}^d)$ could potentially be too restrictive at points. Thus we generalize instead to the Schwartz space, $\mathcal{S} = \mathcal{S}(\mathbb{R}^d)$ of **Schwartz functions**, which are smooth functions that rapidly decrease at infinity with all of their derivatives. I.e., we define

$$\mathcal{S}(\mathbb{R}^d) \coloneqq \{\varphi \in C_c^\infty(\mathbb{R}^d) : \sup_{x\in\mathbb{R}^d} |x^\beta(\partial_x^\alpha \varphi)(x)| < \infty, \quad \forall |\alpha|, |\beta| \le N \text{ and } \forall N \in \mathbb{N}\}.$$

We denote,

$$\|\varphi\|_N = \sup_{x\in\mathbb{R}^d, |\alpha|, |\beta|\le N} |x^\beta(\partial_x^\alpha \varphi)(x)|, \qquad \forall N \in \mathbb{N};$$

we say that $\varphi_k \to \varphi$ in $\mathcal{S}$ iff $\|\varphi_k - \varphi\|_N \to 0$ for every $N \ge 1$. Accordingly, we make the following definition.

Definition 2.21 (Tempered Distributions)**.** *F is said to be a **tempered distribution** if it is a linear functional on $\mathcal{S}$ which is continuous in the sense that $F(\varphi_k) \to F(\varphi)$ whenever $\varphi_k \to \varphi$ in $\mathcal{S}$. By $\mathcal{S}^*$, we shall denote the space of all tempered distributions on a domain.*

Tempered distributions satisfy the following bounding criterion.

Proposition 2.6 (Bounding for Distributions)**.** *Suppose F is a tempered distribution. Then there is a positive integer N and a constant $c > 0$, so that*

$$|F(\varphi)| \le c\|\varphi\|_N, \qquad \forall \varphi \in \mathcal{S}.$$

Proof. Assume that this bound is not satisfied. Then for every positive integer n, there exists $\psi_n \in \mathcal{S}$ with $\|\psi_n\|_n = 1$, and $|F(\psi_n)| \ge n$. For each n, set $\varphi_n = \psi_n/n^{1/2}$. This means $\|\varphi_n\|_N \le \|\varphi_n\|_n$ for all $n \ge N$. In particular, $\|\varphi_n\|_N \le n^{-1/2} \to 0$ as $n \to \infty$. On the other hand, $|F(\varphi_n)| \ge n^{1/2} \to \infty$, which contradicts the continuity of F. □

Now we recall that the Fourier transform of Schwartz functions are Schwartz. As a result, it follows that the Fourier transform maps the class of tempered distributions onto itself, so that $\mathcal{S}^*$ is closed under the Fourier transform. But we have not yet described what it means to take the Fourier transform of a distribution.

Recall that for all $\varphi, \psi \in \mathcal{S}$, one has

$$\begin{aligned}\int_{\mathbb{R}^d} \hat\varphi(x)\psi(x)\,dx &= \int_{\mathbb{R}^d}\left(\int_{\mathbb{R}^d} \varphi(y)\psi(x)e^{-2\pi i y\cdot x}\,dy\right)dx \\ &= \int_{\mathbb{R}^d}\left(\int_{\mathbb{R}^d} \varphi(y)\psi(x)e^{-2\pi i y\cdot x}\,dx\right)dy \qquad \text{(by Fubini's Theorem)} \\ &= \int_{\mathbb{R}^d} \varphi(y)\hat\psi(y)\,dy =: \int_{\mathbb{R}^d} \varphi(x)\hat\psi(x)\,dx.\end{aligned}$$

Thus, one seeks to generalize this to distributions as follows:

Definition 2.22 (Fourier Transform for Tempered Distributions)**.** *Let $F \in \mathcal{S}^*$ be a tempered distribution. Then the **Fourier distribution** of F, denoted by $F^\wedge$, is defined by*

$$F^\wedge(\varphi) = F(\varphi^\wedge), \qquad \forall \varphi \in \mathcal{S}.$$

In particular, the Fourier transform of tempered distributions satisfy the same properties as the Fourier transform for functions.

2.2.1 Important Examples of Distributions

Example 2.1 (Principle Value)**.** *Consider the function $f(x) = 1/x$ for x real and nonzero. Since f is not integrable in any neighborhood of the origin, it follows that $f(x)$ does not represent a distribution on $\mathbb{R}$. However, consider instead the linear functional,*

$$\varphi \in \mathcal{S}(\mathbb{R}^d) \mapsto \lim_{\varepsilon \to 0} \int_{|x| \geq \varepsilon} \varphi(x)\, \frac{dx}{x}.$$

First, we must check that the limit actually exists. We may assume $\varepsilon \leq 1$. Then we write

$$\int_{|x| \geq \varepsilon} \varphi(x)\, \frac{dx}{x} = \int_{1 \geq |x| \geq \varepsilon} \varphi(x) \frac{dx}{x} + \int_{|x|>1} \varphi(x)\, \frac{dx}{x}.$$

Since φ is Schwartz, clearly the rightmost integral converges. On the other hand, since

$$A \int_{1 \geq |x| \geq \varepsilon} \frac{dx}{x} = 0,$$

for any constant A since $1/x$ is an odd function, we can write the first integral on the right side as

$$\int_{1 \geq |x| \geq \varepsilon} \varphi(x)\, \frac{dx}{x} = \int_{1 \geq |x| \geq \varepsilon} \frac{\varphi(x) - \varphi(0)}{x}\, dx.$$

Then

$$\left| \int_{1 \geq |x| \geq \varepsilon} \frac{\varphi(x) - \varphi(0)}{x}\, dx \right| \leq \int_{1 \geq |x| \geq \varepsilon} \frac{|\varphi(x) - \varphi(0)|}{|x|}\, dx \leq c \int_{1 \geq |x| \geq \varepsilon} \frac{|x|}{|x|}\, dx < \infty,$$

where $c = \sup |\varphi'(x)|$. Thus the limit converges; we denote the limit by

$$\operatorname{pv} \int_{\mathbb{R}} \varphi(x)\, \frac{dx}{x} \coloneqq \lim_{\varepsilon \to 0} \int_{|x| \geq \varepsilon} \varphi(x)\, \frac{dx}{x}.$$

In particular, one has that the principal value is bounded above by (some constant multiple of) $\|\varphi\|_1$. Therefore, the above linear functional is a tempered distribution, which we shall henceforth denote by $\operatorname{pv}\left(\frac{1}{x}\right)$.

Example 2.2 (Principal Value and Hilbert Transform)**.** *Recall that we also defined the Hilbert transform in the form (4); namely,*

$$H(f) = \lim_{\varepsilon \to 0} \frac{1}{\pi} \int_{|y| \geq \varepsilon} f(x-y)\, \frac{dy}{y} =: \frac{1}{\pi} \operatorname{pv}\left(\frac{1}{x}\right) * f, \qquad \forall f \in \mathcal{S}.$$

Fundamental Solutions Perhaps in these notes, the most striking application of distribution theory is to the notion of *fundamental solutions* to partial differential equations. We begin by presenting a few key ideas of PDEs in this subsection. We start by defining **fundamental solutions**.

Definition 2.23 (Fundamental Solutions)**.** *Suppose L is a the partial differential operator*

$$L = \sum_{|\alpha| \leq m} a_\alpha \partial_x^\alpha \qquad \text{on } \mathbb{R}^d,$$

*with a_α complex constants. A **fundamental solution** of L is a distribution F so that $L(F) = \delta$, where δ is the Dirac delta function. Note that a fundamental solution is not necessarily unique since we can always add a solution to the homogeneous problem $Lu = 0$ to F.*

The following calculations show why using the fundamental solution is a "good" idea. Let $T : \mathcal{D} \to C^\infty$ be the operator defined by $T : f \mapsto F * f$, where F is the fundamental solution to f. Then we claim that $LT = TL = 1$. Indeed, since $\partial_x^\alpha (F * f) = (\partial_x^\alpha F) * f = F * (\partial_x^\alpha F)$ and $\delta * f = f$, we have

$$LTf = \sum_{|\alpha| \leq m} a_\alpha \partial_x^\alpha (F * f) = \sum_{|\alpha| \leq m} F * a_\alpha \partial_x^\alpha f = F * Lf = TLf.$$

Thus the fundamental solution can be thought of as an "inverse" to a partial differential operator.

Now, we are interested in actually obtaining the fundamental solution to various PDE explicitly. Thus we start with the following definition.

Definition 2.24 (Characteristic Polynomial)**.** *Let L be the partial differential operator,*

$$L = \sum_{|\alpha| \leq m} a_\alpha \partial_x^\alpha.$$

*Then the **characteristic polynomial** of the operator L is the polynomial,*

$$P(\xi) = \sum_{|\alpha| \leq m} a_\alpha (2\pi i \xi)^\alpha.$$

The significance is as follows: we know that if a fundamental solution F exists for L, then $LF = \delta$. On the other hand, for any $f \in \mathcal{S}$, one has that $\widehat{(Lf)} = P \cdot \hat{f}$. This would imply that $1 = \widehat{(LF)} = P \cdot \hat{F}$. Thus it would be desirable to just proclaim the fundamental solution F to be given by $\hat{F}(\xi) = 1/P(\xi)$, or

$$F = \int_{\mathbb{R}^d} \frac{1}{P(\xi)} e^{2\pi i x \cdot \xi} \, d\xi.$$

However, the issues with this definition is that we must account for the zeros of P (and the difficulty with defining $1/P$ as a distribution arising from the zeros). There are a few cases where this definition follows naturally. Consider, for example, the case of Laplace's equation.

Theorem 2.7. *Fundamental Solution to Laplace's Equation ($d \geq 3$) For $d \geq 3$, the locally integrable function F defined by $F(x) = C_d |x|^{-d+2}$ is a fundamental solution for the operator Δ, with $C_d = -\frac{\Gamma(\frac{d}{2}-1)}{4\pi^{\frac{d}{2}}}$.*

Theorem 2.8 (Fundamental Solution to Laplace's Equation ($d = 2$))**.** *When $d = 2$, the locally integrable function $\frac{1}{2\pi}\log(|x|)$ is a fundamental solution of Δ.*

2.2.2 Calderón-Zygmund Distributions

In this subsubsection, we cover some rudimentary principles of *Calderón-Zygmund theory*. For motivation, consider Poisson's equation $\Delta\phi = F$ under sufficient regularity assumptions on ϕ and F. Suppose that we wish to estimate the partial derivatives of ϕ (e.g., assume we wish to estimate the second derivatives of ϕ). Taking the Fourier transform of the Poisson equation, we obtain

$$-4\pi^2 |\xi|^2 \hat{\phi} = \hat{F} \iff \hat{\phi} = -\frac{1}{4\pi^2|\xi|^2} \hat{F}.$$

Then to study the L^p norms of $\partial_{ij}^2 \phi$ for $1 < p < \infty$, it suffices to study $\mathcal{F}^{-1}(-4\pi^2 \xi_i \xi_j \hat{\phi})$, which we can write as

$$\mathcal{F}^{-1}\left(-4\pi^2 \xi_i \xi_j \hat{\phi}\right) = \mathcal{F}^{-1}\left(\frac{\xi_i \xi_j}{|\xi|^2} \hat{F}\right).$$

We note that the kernel, $\hat{K} = (\xi_i \xi_j)^{-1}/|\xi|^2$ is locally integrable, smooth away from the origin, is bounded, tempered, and satisfies the bound $|\partial_\xi^\alpha \hat{K}| \leq c'_\alpha |\xi|^{-\alpha}$ for all multiindices. Thus if $K_{ij} = \mathcal{F}^{-1}(\hat{K})$, by the convolution theorem, we have

$$\mathcal{F}^{-1}\left(\frac{\xi_i \xi_j}{|\xi|^2} \hat{F}\right) = K * F.$$

Therefore, if we can show that the operator T defined by $TF = K * F$, initially just defined on $\mathcal{S}(\mathbb{R}^d)$, then we will have shown that T extends to a bounded operator on all of $L^p(\mathbb{R}^d)$ (for all $1 < p < \infty$), from which we will have deduced the corresponding L^p estimates on the second partial derivatives of ϕ.

The theory developed in this subsubsection will identify the specific hypotheses on K that enables us to make such estimates. Therefore, the material in this subsubsection is an extension of our discussion of the Hilbert transform. We begin with the following regularity assumption that we will be using throughout this subsubsection.

Definition 2.25 (Regularity Assumption)**.** *Let K be a distribution on $\mathbb{R}^d$. K is said to be **regular** in the Calderón-Zygmund sense if there exists a function k that is smooth away from the origin, $K = k$ away from the origin, and $|\partial_x^\alpha k(x)| \le C_\alpha |x|^{-d-\alpha}$ for all α (in particular, when $\alpha = 0$, one has that K is a tempered distribution).*

Moreover, we require the following definitions.

Definition 2.26 ($C^{(n)}$-Normalized Bump Functions)**.** *Fix an integer n. Then φ is said to be a $C^{(n)}$-**normalized bump function** if φ is supported in the unit ball $B_1(0)$, and for all multiindices $|\alpha| \le n$, $\sup_x |\partial_x^\alpha \varphi(x)| \le 1$.*

Definition 2.27 (Cancellation Condition)**.** *For any $C^{(n)}$-normalized bump function φ, define $\varphi_r(x) = \varphi(rx)$ for all $r > 0$. Then K is said to satisfy the **cancellation condition** in the sense of Calderón-Zygmund if*

$$\sup_{0<r} |K(\varphi_r)| \le A, \tag{10}$$

for all such functions φ.

Then we define *Calderón-Zygmund distributions* as the following.

Definition 2.28 (Calderón-Zygmund Distribution)**.** *A distribution K on $\mathbb{R}^d$ is said to be a **Calderón-Zygmund distribution** if it is regular and satisfies the cancellation condition outlined above.*

Proposition 2.7 (Equivalent Formulations of Calderón-Zygmund Distributions)**.** *A distribution K is a Calderón-Zygmund distribution on $\mathbb{R}^d$ if and only if it is a tempered distribution such that $\hat{K} \in C^\infty(\mathbb{R}^d \setminus \{0\}) \cap L^\infty(\mathbb{R}^d \setminus \{0\})$, and $|\partial_\xi^\alpha \hat{K}(\xi)| \le c'_\alpha |\xi|^{-\alpha}$.*

Then one immediately obtains the following result.

Theorem 2.9 (Calderón-Zygmund Decomposition Lemma)**.** *Roughly speaking, every integrable function can be split into a "good" and a "bad" term. I.e., for very $f \in L^1(\mathbb{R}^d)$ and $\alpha > 0$, we can find an open set E_α and a decomposition $f = g + b$ so that:*

(a) $m(E_\alpha) \le \dfrac{c}{\alpha} \|f\|_{L^1(\mathbb{R}^d)}$.

(b) $|g(x)| \le c\alpha$, *for all x.*

(c) E_α *is a union $\bigcup Q_k$ of cubes Q_k whose interiors are disjoint. Moreover $b = \sum_k b_k$, with each function* $\operatorname{supp}(b_k) \subseteq Q_k$ *and*

$$\int |b_k(x)|\, dx \le c\alpha m(k) \qquad \text{and} \qquad \int_{Q_k} b_k(x)\, dx = 0.$$

I.e., b is supported in E_α, so that $g(x) = f(x)$ if $x \notin E_\alpha$. Moreover, each b_k is of the form $c\alpha m(Q_k)\mathfrak{a}_k$, where $\mathfrak{a}_k$ is a 1-atom.

I.e., the theorem tells us that every integrable function can be exressed as the sum of a good function g, which is bounded, and a bad function $b = \sum b_k$, where each b_k is supported in a small cube so that the support of the bad function b is contained in a measurable set E_α which is bounded in size.

Theorem 2.10 (Calderón-Zygmund Distributions and L^p Estimates)**.** *Let T be the linear operator on $L^p(\mathbb{R}^d)$, initially only defined on $\mathcal{S}(\mathbb{R}^d)$, defined by $Tf = f * K$, where K is a Calderón-Zygmund distribution. Then T extends to a bounded operator on $L^p(\mathbb{R}^d)$ for $1 < p < \infty$; i.e., for $1 < p < \infty$, there exists $A_p = A_p(p)$ such that for all $f \in L^p(\mathbb{R}^d)$,*

$$\|Tf\|_{L^p(\mathbb{R}^d)} \le A_p \|f\|_{L^p(\mathbb{R}^d)}.$$

Let us consider some examples.

Example 2.3 (Hilbert Transform)**.** *Recall the following definition of the Hilbert transform,*

$$H(f)(x) \coloneqq \int_{\mathbb{R}^d} -i\hat{f}(\xi)\,\mathrm{sign}(\xi)e^{2\pi i x\cdot\xi}\,d\xi = \mathcal{F}^{-1}(-i\,\mathrm{sign}(\xi)\hat{f}(\xi)).$$

Consider the distribution K such that $\hat{K} = -i\,\mathrm{sign}(\xi)$. It follows that $\hat{K} \in C^\infty(\mathbb{R}^d \setminus \{0\}) \cap L^\infty(\mathbb{R}^d \setminus \{0\})$. To show that K is a tempered distribution, it suffices to show that $\hat{K}$ is a tempered distribution. Suppose $\{\varphi_n\}_{n=1}^\infty, \varphi \in \mathcal{S}(\mathbb{R}^d)$ such that $\varphi_n \to \varphi$ in $\mathcal{S}(\mathbb{R}^d)$. Then

$$\left|\int_{\mathbb{R}^d} \hat{K}(\xi)\varphi(\xi) - \hat{K}(\xi)\varphi_n(\xi)\,d\xi\right| \le \int_{\mathbb{R}^d} |\hat{K}(\xi)\|\varphi(\xi) - \varphi_n(\xi)|\,d\xi = \|\varphi - \varphi_n\|_{L^1(\mathbb{R}^d)} \xrightarrow{n\to\infty} 0.$$

Thus, $\hat{K}(\varphi_n) \to \hat{K}(\varphi)$ so that $\hat{K}$ (and therefore, K) is tempered. This means that K is a Calderón-Sygmund distribution. Therefore, by the convolution theorem, it follows that for all $f \in L^p(\mathbb{R}^d)$ $(1 < p < \infty)$,

$$H(f)(x) \coloneqq (f * K)(x).$$

Thus by the preceding theorem, we conclude that the Hilbert transform is a bounded operator on $L^p(\mathbb{R}^d)$ for all $1 < p < \infty$, which was precisely the conclusion of Theorem 2.2.

Example 2.4 (Estimates for Poisson's Equation)**.** *Consider again the Poisson equation, $\Delta\phi = F$, where ϕ and F are assumed to be sufficiently regular and contained in the appropriate function spaces (e.g., assume $F \in L^p(\mathbb{R}^d)$ for some $1 < p < \infty$). As we saw above, one has that*

$$\hat{\phi} = -\frac{1}{4\pi^2|\xi|^2}\hat{F} \qquad \text{and} \qquad \mathcal{F}^{-1}(-4\pi^2\xi_i\xi_j\hat{\phi}) = \mathcal{F}^{-1}\left(\frac{\xi_i\xi_j}{|\xi|^2}\hat{F}\right),$$

for all $1 \le i, j \le d$. Consider the distribution K_{ij} on $\mathbb{R}^d$ so that $\hat{K}_{ij} = \frac{\xi_i\xi_j}{|\xi|^2}$ is locally integrable. Since for any $1 \le i, j \le d$, $|\xi_i| \le |\xi|$, it follows that $\hat{K}_{ij} \in C^\infty(\mathbb{R}^d \setminus \{0\}) \cap L^\infty(\mathbb{R}^d \setminus \{0\})$. It is also straightforward to check that the derivatives of $\hat{K}_{ij}$ satisfy the differential inequality stated in the definition of Calderón-Zygmund distributions. Thus it remains to be seen that $\hat{K}_{ij}$ is tempered. Let $\{\varphi_n\}, \varphi \in \mathcal{S}(\mathbb{R}^d)$ so that $\varphi_n \to \varphi$ in $\mathcal{S}(\mathbb{R}^d)$. Then we compute

$$\left|\int_{\mathbb{R}^d} \frac{\xi_i\xi_j}{|\xi|^2}(\varphi - \varphi_n)\,d\xi\right| \le \int_{\mathbb{R}^d} \left|\frac{\xi_i\xi_j}{|\xi|^2}\right| |\varphi - \varphi_n|\,d\xi \le \left\|\hat{K}_{ij}\right\|_{L^\infty(\mathbb{R}^d)} \|\varphi - \varphi_n\|_{L^1(\mathbb{R}^d)} \xrightarrow{n\to\infty} 0.$$

Thus this establishes that $\hat{K}_{ij}$ (and therefore, K_{ij}) is tempered; K_{ij} is a Calderón-Zygmund distribution. From the theorem above, it follows that for any $1 < p < \infty$ and for any $1 \le i, j \le d$,

$$\left\|\partial_{ij}^2\phi\right\|_{L^p(\mathbb{R}^d)} \le A_p \|F\|_{L^p(\mathbb{R}^d)}.$$

This gives us a $W^{2,p}(\mathbb{R}^d)$ regularity estimate for the Poisson equation. We can use this result to establish $W^{k+2,p}(\mathbb{R}^d)$ regularity estimates for the Poisson equation. Start by assuming $F \in W^{k,p}(\mathbb{R}^d)$ for some fixed $k \ge 0$. Let α be a multiindex with $|\alpha| = k + 2$. Then if we let $\beta = \alpha - e_i - e_j$ (so that $|\beta| = k$), one has that

$$\|\partial_x^\alpha\phi\|_{L^p(\mathbb{R}^d)} = \left\|\partial_{ij}^2(\partial_x^\beta\phi)\right\|_{L^p(\mathbb{R}^d)} \le A_p \left\|\partial_x^\beta F\right\|_{L^p(\mathbb{R}^d)}.$$

Then, by definition of the $W^{k+2,p}(\mathbb{R}^d)$ norm, one has that

$$\begin{aligned}
\|\phi\|_{W^{k+2,p}(\mathbb{R}^d)} = \left(\sum_{|\alpha|\le k+2} \|\partial_x^\alpha\phi\|_{L^p(\mathbb{R}^d)}^p\right)^{1/p} &\le \left(\sum_{|\alpha|<2} \|\partial_x^\alpha\phi\|_{L^p(\mathbb{R}^d)}^p + \sum_{0\le|\beta|\le k} C_p^p \left\|\partial_x^\beta F\right\|_{L^p(\mathbb{R}^d)}^p\right)^{1/p} \\
&\le \left(\sum_{|\alpha|<2} \|\partial_x^\alpha\phi\|_{L^p(\mathbb{R}^d)}^p\right)^{1/p} + C_p\left(\sum_{0\le|\beta|\le k} \left\|\partial_x^\beta F\right\|_{L^p(\mathbb{R}^d)}^p\right)^{1/p} \\
&= \|\phi\|_{W^{1,p}(\mathbb{R}^d)} + C_p \|F\|_{W^{k,p}(\mathbb{R}^d)}.
\end{aligned}$$

Thus, we obtain the $W^{k+2,p}$ regularity estimate,

$$\|\phi\|_{W^{k+2,p}(\mathbb{R}^d)} \le \|\phi\|_{W^{1,p}(\mathbb{R}^d)} + C_p \|F\|_{W^{k,p}(\mathbb{R}^d)}.$$

2.3 Oscillatory Integrals in Fourier Analysis

In this subsection, we consider *oscillatory integrals* and their applications in Fourier analysis. As we will see, the idea of oscillatory integrals are indispensable for PDEs, in particular, for dispersion equations. For example, consider the Klein-Gordon equation $\Box\phi - \phi = 0$, with compactly supported smooth initial data $(\phi, \partial_t\phi)\restriction_{\{t=0\}}$, on $\mathbb{R}^{1+d}$. Then one can show that the solution ϕ satisfies the dispersion estimate,

$$\sup_x |\phi|(t,x) \lesssim_{(\varphi_0,\varphi_1)} \frac{1}{(1+t)^{\frac{d}{2}}}.$$

Likewise, for the linear wave equation, $\Box\phi = 0$ on $\mathbb{R}^{1+d}$, one has the dispersion rate

$$\sup_x |\phi|(t,x) \lesssim_{(\phi_0,\phi_1)} \frac{1}{(1+t)^{\frac{d-1}{2}}}.$$

Proving these estimates requires us to study the asymptotics of oscillatory integrals. Thus, we turn to the theory of oscillatory integrals in this subsection.

2.3.1 Oscillatory Integrals

We begin with the following definition.

Definition 2.29 (Oscillatory Integrals)**.** *By **oscillatory integrals**, we mean integrals of the form*

$$I(\lambda) = \int_{\mathbb{R}^d} e^{i\lambda\Phi(x)}\psi(x)\,dx.$$

The types of questions we are interested in studying is the behavior of $I(\lambda)$ for large λ. Here, we assume that both the parameter λ and the **phase** Φ are real-valued, but in general, ψ may be complex-valued. The main idea behind attaining decay estimates for these oscillatory integrals is the idea of **stationary phase**. Essentially, if the gradient of the phase is nonvanishing, then the integral is rapidly decreasing in λ (and thus negligible). Therefore, the only meaningful contributions are from the critical points of Φ (i.e., when the gradient vanishes). We will see this procedure in detail over the course of this section. But we begin with the following opening estimate.

Proposition 2.8 (First Estimate for Oscillatory Integrals)**.** *Let $I(\lambda)$ be the oscillatory integral from above. Suppose $|\nabla\Phi(x)| \geq c > 0$ for all x in the support of ψ. Then for every $N \geq 0$,*

$$|I(\lambda)| \leq C_N\lambda^{-N}, \qquad \text{whenever } \lambda > 0.$$

Proof. We start by considering the vector field

$$L = \frac{1}{i\lambda}\sum_{k=1}^{d} a_k\frac{\partial}{\partial x_k} = \frac{1}{i\lambda}(a\cdot\nabla),$$

where $a = (a_1,\dots,a_n) = \nabla\Phi/|\nabla\Phi|^2$. This is because we then observe that

$$L(e^{i\lambda\Phi(x)}) = \frac{a}{i\lambda}\cdot\nabla(e^{i\lambda\Phi(x)}) = \frac{a}{i\lambda}\cdot\left(i\lambda e^{i\lambda\Phi(x)}\nabla\Phi\right) = e^{i\lambda\Phi(x)}.$$

Thus for any positive integer N, $L^N(e^{i\lambda\Phi}) = e^{i\lambda\Phi}$. We then compute the transpose $L^\top$ of L to be

$$L^\top(f) = -\frac{1}{i\lambda}\sum_{k=1}^{d}\frac{\partial}{\partial x_k}(a_k f) = -\frac{1}{i\lambda}\nabla\cdot(af).$$

Since we assume that $|\nabla\Phi(x)| \geq c > 0$ for all x in the support of ψ, the a_j and all of their partial derivatives are each bounded on the support of ψ. Thus, we compute,

$$|I(\lambda)| \leq \int_{\mathbb{R}^d}\left|L^N(e^{i\lambda\Phi})\psi\right|\,dx = \int_{\mathbb{R}^d}|e^{i\lambda\Phi}||(L^\top)^N(\psi)|\,dx = \int_{\mathbb{R}^d}|(L^\top)^N(\psi)|\,dx$$

Thus by smoothness and compact support of ψ, we conclude that $|I(\lambda)| \le c_N \lambda^{-N}$, where the prefactor C follows from bounds on the derivatives of a and ψ. □

In dimension one, we can actually get more precise estimates at the cost of simpler hypotheses.

Proposition 2.9 (Oscillatory Estimate in One Dimension)**.** *Let $I_1(\lambda)$ be the following integral,*

$$I_1(\lambda) = \int_a^b e^{i\lambda\Phi(x)}\, dx,$$

where a, b are any real numbers, $\Phi \in C^2$, and $\Phi'(x)$ is monotonic (either increasing or decreasing) with $|\Phi'(x)| \ge 1$ on $[a,b]$. Then

$$|I_1(\lambda)| \le c\lambda^{-1},$$

for all $\lambda > 0$, with $c = 3$.

Proof. We use an operator L similar to the operator used in the proof above. Without loss of generality, we may assume that $\Phi'(0) > 0$ on $[a,b]$ since the case for $\Phi' < 0$ follows by taking the complex conjugates. Then in this case, $L = (-i/\lambda)(1/\Phi'(x))\frac{d}{dx}$, and likewise, $L^\top(f) = \frac{i}{\lambda}\frac{d}{dx}\left(\frac{f}{\Phi'}\right)$. Then, we compute

$$I_1(\lambda) = \int_a^b L(e^{i\lambda\Phi})\, dx = \int_a^b e^{i\lambda\Phi} L^t(1)\, dx + \left[e^{i\lambda\phi}\frac{1}{i\lambda\Phi'}\right]_a^b,$$

where the boundary terms appear since the amplitude does not identically vanish at the endpoints. Focus first on the boundary terms. Since $|\Phi'| \ge 1$,

$$\left|\left[e^{i\lambda\Phi}\frac{1}{i\lambda\Phi'}\right]_a^b\right| \le \frac{1}{\lambda|\Phi'(b)|} + \frac{1}{\lambda|\Phi'(a)|} \le \frac{2}{\lambda}.$$

On the other hand, the first term can be majorized as follows:

$$\left|\int_a^b L^\top(1)\, dx\right| \le \frac{1}{\lambda}\int_a^b \left|\frac{d}{dx}\left(\frac{1}{\Phi'}\right)\right| dx = \frac{1}{\lambda}\left|\int_a^b \frac{d}{dx}\left(\frac{1}{\Phi'}\right) dx\right| = \frac{1}{\lambda}\cdot\left|\frac{1}{\Phi'(b)} - \frac{1}{\Phi'(a)}\right|,$$

where the first equality follows from the fact that Φ' is monotonic and continuous on the interval $[a,b]$ and $|\Phi'(x)| \ge 1$ so that the derivative does not change signs on the interval. Thus this concludes the proof. □

We have a similar result when Φ has a *nondegenerate critical point* at x_0.

Proposition 2.10 (van der Corput's Estimate)**.** *Let $I_1(\lambda)$ be the oscillatory integral defined above. Assume $\Phi \in C^2([a,b])$, with nondegenerate critical point x_0, and $|\Phi''(x)| \ge 1$. Then for all $\lambda > 0$ and with $c' = 8$, the following estimate applies*

$$|I_1(\lambda)| \le c'\lambda^{-1/2}.$$

Corollary 2.5. *Under the same hypotheses as in van der Corput's estimate, one has the estimate*

$$\left|\int_a^b e^{i\lambda\Phi(x)}\psi(x)\, dx\right| \le c_\psi\lambda^{-1/2},$$

where

$$c_\psi = 8\left(\int_a^b |\psi'(x)|\, dx + |\psi(b)|\right).$$

While there are no direct analogs of the above strict estimates for $\mathbb{R}^d$ $(d > 1)$, we have the following generalization.

Proposition 2.11 (Generalization of Decay in Higher Dimensions)**.** *Suppose Φ and ψ are smooth, ψ has compact support, and $\det\{\nabla^2\Phi\} \ne 0$ on $\operatorname{supp}(\Psi)$. Then*

$$I(\lambda) = \int_{\mathbb{R}^d} e^{i\lambda\Phi(x)}\psi(x)\, dx = \mathcal{O}(\lambda^{-d/2}), \qquad \text{as } \lambda \to \infty.$$

Fourier Transform of Surface-Carried Measures

Part II
Nonlinear Wave Equations

3 Linear Wave Equation

3.1 Introduction

Through these notes, our goal is to study the properties of the nonlinear wave equation, which can either be a "simple" scalar equation or a system of equations [5]:

$$\frac{1}{\sqrt{-\det(g)}}\partial_\mu\left(\sqrt{-\det g}\,g^{\mu\nu}\partial_\nu\phi\right) = F(\phi,\partial\phi), \tag{11}$$

where g is a Lorentzian metric on $I\times\mathbb{R}^n$, and $\phi : I\times\mathbb{R}^n\to\mathbb{R}$ is an unknown function of some regularity. For example, if $n=1$, and g is the flat Minkowski metric, $\mathrm{diag}(1,-1)$, then the nonlinear wave equation is merely

$$\partial_\mu(\partial^{\mu\nu}\partial_\nu\phi) = F(\phi,\partial\phi) \iff \partial_t^2 - (\partial_x\phi)^2 = F(\phi,\partial\phi).$$

Note that we denote the coordinate of I by t (to designate the *time* coordinate), and the coordinates of $\mathbb{R}^n$ by $x=(x^1,\dots,x^n)$. An equation of the form (11) is said to be

(i) **linear**: g is independent of ϕ and F is a linear function of both of its arguments;
(ii) **semilinear**: g is independent of ϕ and F is a nonlinear function;
(iii) **quasilinear**: g is independent of ϕ and/or $\partial\phi$.

Before we build up to the nonlinear wave equation, we will first examine the linear wave equation. Generally speaking, we are interested in the following properties:

(i) **Existence and Uniqueness**: Does a solution to (11) exist, and if so, is it the only solution?
(ii) **Dispersion**: What physical properties do solutions to (11) have?
(iii) **Well-Posedness**: If we consider the *Cauchy wave problem* consisting of the equation (11) along with initial data $(\phi,\partial_t\phi)$ on the hypersurface $\{t=0\}$, where $\phi,\partial_t\phi$, does there exist a unique solution to the problem that satisfy the initial conditions and is continuously dependent on the initial data?

We will gradually work through each of these results, sometimes developing new techniques and machinery. We first begin by developing some general analytic results, and idea of energy estimates. Then we will apply these to the study of linear wave equations. This will then give a foundation for the study of nonlinear wave equations.

3.2 Energy Estimates

In this section, we introduce the notion of the energy of a solution to the wave equation. We begin by considering the homogeneous linear Cauchy problem,

$$\begin{cases} \Box\,\phi = 0, \\ (\phi, \partial_t\phi)\restriction_{\{t=0\}} = (\phi_0, \phi_1), \end{cases} \tag{12}$$

where we initially assume $\phi_0, \phi_1 \in C_c^\infty \subset \mathcal{S}(\mathbb{R}^n)$, and $\Box$ denotes the **wave operator**. As of now, we do not yet have an existence (or uniqueness) result for this Cauchy wave problem. Thus we simply assume that such a unique (sufficiently regular) solution exists for now. Then we define the **energy** of a solution to be the following L^2-type norm,

Definition 3.1 (Energy)**.** *For $(\phi_0, \phi_1) \in C_c^\infty(\mathbb{R}^n) \times C_c(\mathbb{R}^n)$ and ϕ the solution to the Cauchy problem (12), the **energy** of ϕ at the spacetime point (t, x) is defined to be the L^2-type norm,*

$$E(t) = \frac{1}{2}\int_{\mathbb{R}^n} \left((\partial_t\phi)^2 + \sum_1^n (\partial_j\phi)^2 \right)(t,x)\, dx. \tag{13}$$

Intuitively, the definition for the energy is as we expected: consider an oscillating 1-dimensional string, and let $\phi(t,x)$ denote the amplitude of the wave at a point x at time t. Then the quantity $(1/2)(\partial_t\phi)^2$ resembles the kinetic energy of an infinitesimal point particle at the spacetime location (t,x), while the quantity $\sum_1^n (\partial_j\phi)^2$ resembles the (spring) potential energy at (t,x). Thus under these observations, $E(t)$ resembles the total energy of the system at some time t. Since the Cauchy wave problem (12) has no source term (i.e., is homogeneous), one expects the following result [5]:

Proposition 3.1 (Conservation of Energy)**.** *Let ϕ be a (sufficiently regular) solution to (12). Define $E(t)$ as given in Definition 3.1. Then E is constant in time.*

Remark 3.1. *We will give two proofs for this result; one relies on straightforward differentiation of $E(t)$ and follows the steps outlined in [5], while the other introduces the concept of conservation laws and follows the steps outlined in [4].*

Proof # 1. Let ϕ be a sufficiently regular solution to (2), and define the energy $E(t)$ of ϕ to be the function,

$$E(t) = \frac{1}{2}\int_{\mathbb{R}^n} \left((\partial_t\phi)^2 + \sum_1^n (\partial_j\phi)^2 \right)(t,x)\, dx,$$

where we assume that the initial data ϕ_0, ϕ_1 are both smooth with compact support. As we will see in a later section, $\phi \in C^\infty(I, \mathcal{S}(\mathbb{R}^n))$. Thus by the Dominated Convergence Theorem allows us to pass a time derivative inside the integral sign. We then obtain,

$$\begin{aligned}
\frac{d}{dt}E(t) &= \frac{1}{2}\int_{\mathbb{R}^n} (\partial_t\phi)(\partial_{tt}\phi) + \sum_1^n \partial_t(\partial_j\phi)^2\, dx \\
&= \frac{1}{2}\int_{\mathbb{R}^n} (\partial_t\phi)\cdot\left(\sum_1^n \partial_j^2\phi\right) + \sum_1^n \partial_t\phi\cdot(\partial_j\phi)^2\, dx \\
&= \frac{1}{2}\int_{\mathbb{R}^n} (\partial_t\phi)\cdot\left(\sum_1^n \partial_j^2\phi\right) + \frac{1}{2}\sum_1^n \int_{\mathbb{R}^n} \partial_j(\partial_t\phi)(\partial_j\phi)\, dx \\
&= \frac{1}{2}\int_{\mathbb{R}^n} (\partial_t\phi)\cdot\left(\sum_1^n \partial_j^2\phi\right) + \frac{1}{2}\sum_1^n \left[(\partial_t\phi)(\partial_j\phi)\Big|_{|x|=\infty} - \int_{\mathbb{R}^n} (\partial_t\phi)(\partial_j^2\phi)\, dx \right] \\
&= \frac{1}{2}\int_{\mathbb{R}^n} (\partial_t\phi)\cdot\left(\sum_1^n \partial_j^2\phi\right) dx - \frac{1}{2}\int_{\mathbb{R}^n} (\partial_t\phi)\cdot\left(\sum_1^n \partial_j^2\phi\right) dx = 0,
\end{aligned}$$

where the boundary term is identically zero since ϕ is Schwartz. Thus, the energy is conserved. □

Proof # 2. We start by claiming that solutions to (12) are the stationary points of a variational integral. Define the linear functional $\mathcal{L} : C^\infty(\mathbb{R}; \mathcal{S}(\mathbb{R}^n)) \to \mathbb{R}$ by

$$\mathcal{L}(\phi) = \frac{1}{2} \int_{\mathbb{R}\times\mathbb{R}^n} (\partial_t \phi)^2 - \sum_1^n (\partial_j \phi)^2 \, dtdx;$$

to obtain this functional, we can use the Euler-Lagrange equations for a physical system. Now let $\psi \in C_c^\infty(\mathbb{R} \times \mathbb{R}^n)$. We recall that $\langle \mathcal{L}'(\phi), \psi \rangle$, where $\mathcal{L}'(\phi)$ is the Fréchet derivative is defined as the directional derivative along ψ; i.e.,

$$\begin{aligned}
\langle \mathcal{L}'(\phi), \psi \rangle &= \frac{d}{d\varepsilon} \mathcal{L}(\phi + \varepsilon\psi)\Big|_{\varepsilon=0} \\
&= \frac{d}{d\varepsilon}\Big|_{\varepsilon=0} \left(\mathcal{L}(\phi) + \varepsilon \int_{\mathbb{R}\times\mathbb{R}^n} (-\nabla\phi \cdot \nabla\psi + (\partial_t\phi)(\partial_t\psi)) \, dtdx + \mathcal{O}(\varepsilon^2) \right) \\
&= \int_{\mathbb{R}\times\mathbb{R}^n} -(\nabla\phi)\cdot(\nabla\psi) + (\partial_t\phi)(\partial_t\psi) \, dtdx.
\end{aligned}$$

Since ψ is smooth with compact support, and ϕ is Schwartz, using Fubini's theorem and integration by parts, and recognizing that the boundary terms are identically zero, one obtains

$$\langle \mathcal{L}'(\phi), \psi \rangle = -\int_{\mathbb{R}\times\mathbb{R}^n} (\Box\phi)\psi \, dtdx = 0.$$

Thus since $\psi \in C_c^\infty$ was arbitrary, we conclude that ϕ is stationary for $\mathcal{L}$ iff $\Box\phi = 0$. Now, we recall Nöther's theorem, which states that if $\mathcal{L}$ is invariant with respect to a 1-parameter family of diffeomorphisms, then there exists a conservation law for an extreme value of $\mathcal{L}$. We note that $\mathcal{L}$ is time-translation invariant: if $G_\varepsilon : \phi(t,x) \mapsto \phi(t+\varepsilon, x)$, then one observes that $\mathcal{L}(\phi) = \mathcal{L}(G_\varepsilon\phi)$. Therefore, since the vector field ∂_t generates the flow corresponding to time translations, one has

$$\begin{aligned}
0 &= \phi_t \, \Box \, \phi \\
&= \phi_{tt}\phi_t - \Delta\phi\phi_t \\
&= \partial_t \left(\frac{(\partial_t\phi)^2}{2} - \frac{(\nabla\phi)^2}{2} \right) - \operatorname{div}(\nabla\phi\phi_t).
\end{aligned}$$

Then, integrating over $\mathbb{R}^n$ yields,

$$0 = \frac{d}{dt} \left(\frac{1}{2} \int_{\mathbb{R}^n} (\partial_t\phi)^2 - (\nabla\phi)^2 \, dx \right) - \int_{\mathbb{R}^n} \operatorname{div}(\nabla\phi\phi_t) \, dx.$$

The last term vanishes by using integration by parts and recalling that ϕ is Schwartz. Thus, one has that the energy is independent of time. □

Remark 3.2. *As the second proof shows, because of the time translation invariance, to show that the energy is conserved, it suffices to integrate the quantity $(\partial_t\phi)(\Box\phi)$ by parts.*

Likewise, we have the following conservation law in Sobolev norms [3].

Corollary 3.1 (Sobolev Conservation Law I)**.** *Given initial data $(\phi, \partial_t\phi) \restriction_{\{t=0\}} = (\phi_0, \phi_1) \in \mathcal{S}(\mathbb{R}^n) \times \mathcal{S}(\mathbb{R}^n)$, and assuming the existence of a solution $\phi \in C^\infty(\mathbb{R}; \mathcal{S}(\mathbb{R}^n))$ to the Cauchy wave problem (12), for all $s \in \mathbb{R}$,*

$$\|\partial_t\phi\|_{H^{s-1}(\mathbb{R}^n)} + \|\nabla\phi\|_{H^{s-1}(\mathbb{R}^n)} = \|\phi_0\|_{H^{s-1}(\mathbb{R}^n)} + \|\nabla\phi_1\|_{H^{s-1}(\mathbb{R}^n)}. \tag{14}$$

Proof. Assume that $\phi \in C^\infty(\mathbb{R}; \mathcal{S}(\mathbb{R}^n))$ is a solution to (12) with initial data $(\phi, \partial_t\phi) \restriction_{\{t=0\}} = (\phi_0, \phi_1) \in \mathcal{S}(\mathbb{R}^n) \times \mathcal{S}(\mathbb{R}^n)$. If $\hat{\cdot}$ denotes the Fourier transform of $\cdot$, then one has the equation

$$(\partial_t\phi)^2 + 4\pi^2|\xi|^2(\hat{\phi})^2 = (\hat{\phi}_0)^2 + 4\pi^2|\xi|^2(\hat{\phi}_1)^2.$$

Note, this equation becomes clearer once we establish a representation formula for ϕ in a later section. Multiplying through by $(1 + |\xi|^2)^{s-1}$ (for arbitrary $s \in \mathbb{R}$), and integrating, yields the desired conservation law. □

One immediate advantage of having conservation laws is that if the Cauchy problem (12) has a solution, then these laws immediately imply uniqueness.

Corollary 3.2 (Uniqueness I)**.** *Given initial data* $(\phi, \partial_t\phi) \upharpoonright_{\{t=0\}} = (\phi_0, \phi_1) \in H^1(\mathbb{R}^n) \times L^2(\mathbb{R}^n)$*, if we have two solutions* $\phi^{(1)}$ *and* $\phi^{(2)}$ *to the linear wave equation (12) with the given data such that both of them satisfy* $(\phi^{(i)}, \partial_t\phi^{(i)}) \in L^\infty([0,T]; H^1(\mathbb{R}^n)) \times L^\infty([0,T]; L^2(\mathbb{R}^n))$ *for* $i = 1, 2$*, then* $\phi^{(1)} = \phi^{(2)}$ *almost everywhere in* $[0,T] \times \mathbb{R}^n$*.*

Proof. The immediate difficulty with trying to use conservation of energy to prove uniqueness of solutions under the regularity assumptions made above is that we do not have enough regularity in the time variable to justify passing the time derivative inside the integral when differentiating $E(t)$. Thus we must use a mollification argument. First, let us assume there exist two solutions $\phi^{(1)}, \phi^{(2)} \in L^\infty([0,T]; H^1(\mathbb{R}^n))$ to the linear wave equation, where both solution satisfy the initial data (ϕ_0, ϕ_1). Define $\phi = \phi^{(1)} - \phi^{(2)}$, which solves the linear wave equation for initial data that are identically zero. For now, we provide a heuristic argument, with a rigorous statement to be filled in afterwards. The idea is that we can consider mollifying ϕ with temporal mollifiers φ_ε for each $\varepsilon > 0$. Then for each $\varepsilon > 0$, one has that $\phi_\varepsilon = \phi * \varphi_\varepsilon$ is smooth and the energy conservation applies. Thus passing to the limit $\varepsilon \to 0$ would enable us to conclude that ϕ is identically zero, yielding uniqueness. □

To extract more useful information, we can localize the energy identity, thus obtaining the following result.

Proposition 3.2 (Localized Energy Identity)**.** *Define, for* $0 \le t \le R$*,*

$$E(t; x, R) \coloneqq \frac{1}{2} \int_{B(x,R-t)} \left((\partial_t\phi)^2 + \sum_1^n (\partial_j\phi)^2 \right)(t, y)\, dy. \tag{15}$$

Then $E(t; x, R)$ *is a nonincreasing function of* t*.*

Remark 3.3. *This result tells us that the energy at time* t *is less than or equal to the energy at time* $s > t$ *in the backwards light cone with vertex at* x*.*

Proof. As suggested in the second proof of Proposition 3.1, we will start with the expression $(\partial_t\phi)\Box\phi = 0$ and integrate by parts in the region to the future of $\{t_1\} \times B(x, R-t_1)$ and to the past of $(\{t_2\} \times B(x, R-t_2)) \cup \left(\bigcup_{s\in[t_1,t_2]} (\{s\} \times \partial B(x, R-s))\right)$. The idea behind this choice is that the boundary terms conveniently give us $E(t_1; x, R)$ and $E(t_2; x, R)$. The boundary term on $\bigcup_{s\in[t_1,t_2]} (\{s\} \times \partial B(x, R-s))$ then has a definite sign, which then allows us to compare the final and initial energy. Indeed, one has the following computations,

$$\begin{aligned}
0 &= \int_{t_1}^{t_2} \int_{B(x,R-t)} \left((\partial_t\phi) \left(-\partial_t^2\phi + \sum_1^n \partial_j^2\phi \right) \right)(t, y)\, dydt \\
&= -\int_{t_1}^{t_2} \int_{B(x,R-t)} \partial_t \left(\frac{1}{2}(\partial_t\phi)^2 + \frac{1}{2}\sum_1^n (\partial_j\phi)^2 \right) dydt + \sum_1^n \int_{t_1}^{t_2} \int_{B(x,R-t)} \partial_j((\partial_t\phi)\cdot(\partial_j\phi))\, dydt \\
&= -\int_{t_1}^{t_2} \left[\frac{1}{2}\frac{d}{dt} \int_{B(x,R-t)} (\partial_t\phi)^2 + \Delta\phi \right] dt - \frac{1}{2}\int_{t_1}^{t_2} \int_{\partial B(x,R-t)} \left((\partial_t\phi)^2 + \sum_1^n (\partial_j\phi)^2 \right) \\
&\qquad + \int_{t_1}^{t_2} \int_{\partial B(x,R-t)} (\partial_t\phi) \cdot \sum_1^n (\partial_j\phi) \cdot \frac{(y-x)_j}{|y-x|}\, d\sigma(y)dt \\
&= -E(t_2; x, R) + E(t_1; x, R) - \frac{1}{2}\int_{t_1}^{t_2} \int_{\partial B(x,R-t)} \left((\partial_t\phi)^2 + \sum_1^n (\partial_j\phi)^2 - 2(\partial_t\phi)\cdot\sum_1^n \frac{(y-x)_j}{|y-x|}(\partial_j\phi) \right) d\sigma(y)dt.
\end{aligned}$$

where the second equality follows from Theorem C.4.6 of [1]. Now we estimate the final term of the last

equation. Cauchy-Schwarz tells us the following (CS arises in the second line),

$$\begin{aligned} 2(\partial_t\phi)\cdot\sum_1^n \frac{(y-x)_j}{|y-x|}(\partial_j\phi) &\le (\partial_t\phi)^2 + \left(\sum_1^n \frac{(y-x)_j}{|y-x|}(\partial_j\phi)\right)^2 \\ &\le (\partial_t\phi)^2 + \left(\sum_1^n \frac{(y-x)_j}{|y-x|}\right)^2 \cdot \left(\sum_1^n (\partial_j\phi)^2\right) \\ &= (\partial_t\phi)^2 + \sum_1^n (\partial_j\phi)^2. \end{aligned}$$

Thus this implies that the final term is negative. So then, one must have $-E(t_2;x,R) + E(t_1;x,R) \ge 0$, whence $E(t_2;x,R) \le E(t_1;x,R)$. □

While we are at it, we also consider the following linear, non-constant coefficient wave equation,

$$\begin{cases} \partial_\alpha(a^{\alpha\beta}\partial_\beta\phi) = F, \\ (\phi, \partial_t\phi)\restriction_{\{t=0\}} = (\phi_0, \phi_1), \end{cases} \tag{16}$$

where a is assumed to be a symmetric $(n+1)\times(n+1)$ matrix on $I\times\mathbb{R}^n$ satisfying

$$\sum_{\alpha,\beta} |a^{\alpha\beta} - m^{\alpha\beta}| < \frac{1}{10}.$$

While the energy is not conserved, in general, due to the source term F, the growth of the energy is controlled in the following sense.

Theorem 3.1 (Growth Bound on Energy)**.** *Assume ϕ is a solution to (16) of some sufficient regularity; let $E(t)$ be the energy of ϕ in the sense of Proposition (3.1). Then there exists a positive constant $C = C(n) > 0$ such that the following energy estimate holds*

$$\left\|E^{1/2}\right\|_{C([0,T])} =\le C\left(\|(\phi_0,\phi_1)\|_{\dot H^1(\mathbb{R}^n)\times L^2(\mathbb{R}^n)} + \int_0^T \|F\|_{L^2(\mathbb{R}^n)}(t)\,dt\right)\exp\left(C\int_0^T \|\partial a\|_{L^\infty(\mathbb{R}^n)}(t)\,dt\right), \tag{17}$$

where by $\partial\cdot$, we denote $(\partial_t\cdot)^2 + \Delta\cdot$.

To prove this result, we first require **Grönwall's lemma**.

Lemma 3.1 (Grönwall's Lemma)**.** *Let $f:\mathbb{R}\to\mathbb{R}$ be a nonnegative continuous function and $g:\mathbb{R}\to\mathbb{R}$ be a nonnegative integrable function such that*

$$f(t) \le A + \int_0^t f(s)g(s)\,ds$$

holds for some $A \ge 0$, and every $t\in[0,T]$. Then

$$f(t) \le A\exp\left(\int_0^t g(s)\,ds\right)$$

for every $t\in[0,T]$.

Proof.

(1) (Standard Proof): The proof follows by showing that the result holds for any $A > 0$, and then passing to the limit $A\to 0$. First, we compute

$$\frac{d}{dt}\left(A + \int_0^t f(s)g(s)\,ds\right) = f(t)g(t) \le g(t)\left(A + \int_0^t f(s)g(s)\,ds\right).$$

In particular, this implies that

$$\frac{d}{dr}\log\left[A+\int_0^t f(s)g(s)\,ds\right]\leq g(r),$$

so that integrating,

$$\log\left[A+\int_0^t f(s)g(s)\,ds\right]\leq\int_0^t g(r)\,dr+\log A.$$

Thus, we conclude that

$$f(t)\leq A+\int_0^t f(s)g(s)\,ds\leq A\exp\left(\int_0^t g(r)\,dr\right).$$

(2) (Bootstrap Method): Now we will prove the same result again, using the **bootstrap** method. For each $\varepsilon>0$, consider the following condition:

$$f(t)\leq(1+\varepsilon)A\exp\left((1+\varepsilon)\int_0^t g(s)\,ds\right),\tag{$\star$}$$

and define the subset $B\subseteq[0,T]$ by

$$B\coloneqq\{t\in[0,T]:(\star)\text{ holds for all }s\in[0,t]\}.$$

It follows immediately that B is nonempty as it at least contains the point $0\in B$. Likewise, by continuity of f, it follows that B is closed. By continuity of f, if we can show that f satisfies a *better* estimate for $t\in B$, then it must follow that f satisfies the above bound in some small neighborhood of t, which would show that B is open in $[0,T]$. Indeed, let $t\in B$. Then combining the hypothesis, one has that

$$\begin{aligned}f(t)&\leq A+\int_0^t g(s)\,ds\leq A+(1+\varepsilon)A\int_0^t g(s)\exp\left((1+\varepsilon)\int_0^s g(r)\,dr\right)ds\\&\leq A\left(1+\left(\exp\left((1+\varepsilon)\int_0^t g(s)\,ds\right)-1\right)\right)\qquad\text{(by integration by parts)}\\&=A\exp\left((1+\varepsilon)\int_0^t g(s)\,ds\right).\end{aligned}$$

Thus, this shows that B must be open. Since $[0,T]$ is connected, one must have $B=[0,T]$. Hence the modified condition holds for all $t\in[0,T]$ and $\varepsilon>0$ so that passing to the limit $\varepsilon\to0$ yields the desired estimate.

□

Now we will use Grönwall's lemma to prove the energy estimate from above.

Proof of Theorem 3.1. For the sake of this proof, we will be assuming that ϕ is compactly supported so that we can ignore the spatial boundary terms in the integration. The fundamental idea behind the techniques in this proof is to bound the energy of the solution on the time interval $[0,T]$ by looking at the total energy at some fixed time, and then summing these contributions. In other words, we wish to integrate the first term of the quantity, $\partial_t\phi(\partial_\alpha(a^{\alpha\beta}\partial_\beta\phi)-F)$ by parts. Before doing so, we make the following observations.

(1) Suppose first that $\alpha=\beta=0$. Then one has that

$$\begin{aligned}\partial_t\phi(\partial_\alpha(a^{\alpha\beta}\partial_\beta\phi))&=(\partial_t\phi)(\partial_t(a^{tt}\partial_t\phi))\\&=(\partial_t\phi)((\partial_t a^{tt})(\partial_t\phi))+a^{tt}(\partial_t\phi)(\partial_{tt}\phi)\\&=(\partial_t a^{tt})(\partial_t\phi)^2+\frac{1}{2}a^{tt}\partial_t(\partial_t\phi)^2,\end{aligned}$$

so that

$$\begin{aligned}\int_0^T \int_{\mathbb{R}^n} (\partial_t\phi)(\partial_t(a^{tt}\partial_t\phi))\,dxdt &= \int_0^T \int_{\mathbb{R}^n} (\partial_t a^{tt})(\partial_t\phi)^2 + \frac{1}{2}a^{tt}\partial_t(\partial_t\phi)^2\,dxdt \\ &= \frac{1}{2}\int_{\{T\}\times\mathbb{R}^n} a^{tt}(\partial_t\phi)^2\,dx - \frac{1}{2}\int_{\{0\}\times\mathbb{R}^n} a^{tt}(\partial_t\phi)^2\,dx \\ &\quad + \frac{1}{2}\int_0^T \int_{\mathbb{R}^n} (\partial_t a^{tt})(\partial_t\phi)^2\,dxdt,\end{aligned}$$

where the second line follows from Fubini/Tonelli, and the fundamental theorem of calculus.

(2) Now assume that one of the indices is t, while the other index runs from $j = 1,\dots,n$. Then one has that

$$\begin{aligned}\partial_t\phi(\partial_\alpha(a^{\alpha\beta}\partial_\beta\phi)) &= (\partial_t\phi)\cdot\left[\partial_j(a^{jt}\partial_t\phi) + \partial_t(a^{tj}\partial_j\phi)\right] \\ &= (\partial_t\phi)\cdot\left[(\partial_j a^{jt})(\partial_t\phi) + a^{jt}\partial_j(\partial_t\phi) + \partial_t(a^{jt})(\partial_j\phi) + a^{jt}\partial_j(\partial_t\phi)\right] \\ &= (\partial_j a^{jt})(\partial_t\phi)^2 + a^{jt}\partial_j(\partial_t\phi)^2 + (\partial_t a^{jt})(\partial_t\phi)(\partial_j\phi).\end{aligned}$$

Observe that $a^{jt} = a^{tj}$ by symmetry of a. We first consider the integral of the middle term. Integration by parts tells us that

$$\begin{aligned}\int_0^T \int_{\mathbb{R}^n} a^{jt}\partial_j(\partial_t\phi)^2\,dxdt &= \int_0^T \left[a^{jt}(\partial_t\phi)^2|_{\text{bdr.}} - \int_{\mathbb{R}^n}(\partial_j a^{jt})(\partial_t\phi)^2\,dx\right]dt \\ &= -\int_0^T \int_{\mathbb{R}^n}(\partial_j a^{jt})(\partial_t\phi)^2\,dxdt,\end{aligned}$$

where the boundary terms vanish because of our assumption that ϕ is compactly supported. Thus in fact,

$$\int_0^T \int_{\mathbb{R}^n} \partial_t\phi(\partial_\alpha(a^{\alpha\beta}\partial_\beta\phi))\,dxdt = \int_0^T \int_{\mathbb{R}^n}(\partial_j a^{jt})(\partial_t\phi)^2 - (\partial_j a^{jt})(\partial_t\phi)^2 + (\partial_t a^{jt})(\partial_t\phi)(\partial_j\phi)\,dxdt.$$

(3) Now assume that both indices run over $i, j = 1,\dots,n$. Then one has that

$$\begin{aligned}\int_0^T \int_{\mathbb{R}^n} \partial_t\phi(\partial_i(a^{ij}\partial_j\phi))\,dxdt &= (\partial_t\phi)(a^{ij}\partial_j\phi)\Big|_{\text{bdr.}} - \int_0^T \int_{\mathbb{R}^n} \partial_t\partial_i\phi a^{ij}\partial_j\phi\,dxdt \\ &= -\frac{1}{2}\int_0^T \int_{\mathbb{R}^n} a^{ij}\partial_t(\partial_i\phi\partial_j\phi)\,dxdt \\ &= -\frac{1}{2}\int_{\{T\}\times\mathbb{R}^n} a^{ij}\partial_i\phi\partial_j\phi\,dx + \frac{1}{2}\int_{\{0\}\times\mathbb{R}^n} a^{ij}\partial_i\phi\partial_j\phi\,dx \\ &\quad + \frac{1}{2}\int_0^T \int_{\mathbb{R}^n}(\partial_t a^{ij}(\partial_i\phi)(\partial_j\phi)\,dxdt,\end{aligned}$$

where the final line follows from the fact that

$$\partial_t(a^{ij}\partial_i\phi\partial_j\phi) - (\partial_t a^{ij})(\partial_i\phi)(\partial_j\phi) = a^{ij}\partial_t(\partial_i\phi\partial_j\phi),$$

we used the fundamental theorem of calculus after using the above fact, and the boundary terms in the first line vanish since ϕ has compact support.

Now we obtain the following result:

$$\begin{aligned}&\int_0^T \int_{\mathbb{R}^n} (\partial_t\phi)(\partial_\alpha(a^{\alpha\beta}\partial_\beta\phi)) = \int_0^T \int_{\mathbb{R}^n}(\partial_t\phi)F \\ \implies &\frac{1}{2}\int_{\{T\}\times\mathbb{R}^n} a^{tt}(\partial_t\phi)^2\,dx - \frac{1}{2}\int_{\{T\}\times\mathbb{R}^n} a^{ij}\partial_i\phi\partial_j\phi\,dx = \frac{1}{2}\int_{\{0\}\times\mathbb{R}^n} a^{tt}(\partial_t\phi)^2\,dx - \frac{1}{2}\int_{\{0\}\times\mathbb{R}^n} a^{ij}\partial_i\phi\partial_j\phi\,dx \\ &\quad + \int_0^T \int_{\mathbb{R}^n}(\partial_t\phi)F - \frac{1}{2}(\partial_t a^{tt})(\partial_t\phi)^2 - (\partial_j a^{jt})(\partial_t\phi)^2 \\ &\quad + (\partial_j a^{jt})(\partial_t\phi)^2 - (\partial_t a^{jt})(\partial_t\phi)(\partial_j\phi)\,dxdt\end{aligned}$$

Now, one has that

$$\left|\int_0^T \int_{\mathbb{R}^n} (\partial_t\phi)F - \frac{1}{2}(\partial_t a^{tt})(\partial_t\phi)^2 - (\partial_j a^{jt})(\partial_t\phi)^2 + (\partial_j a^{jt})(\partial_t\phi)^2 - (\partial_t a^{jt})(\partial_t\phi)(\partial_j\phi)\,dxdt\right|$$
$$\leq C'\int_0^T \|\partial_t\phi\|_{L^2(\mathbb{R}^n)}\|F\|_{L^2(\mathbb{R}^n)}(t) + \|\partial a\|_{L^\infty(\mathbb{R}^n)}(t)\|\partial\phi\|^2_{L^2(\mathbb{R}^n)}(t)\,dt,$$

where we used Hölder's inequality to obtain the first term, and assumed $\partial a \in L^\infty(\mathbb{R}^n)$. On the other hand, by our hypothesis on $a^{\alpha\beta}$, there exists some constant $K > 0$ large enough so that (taking $C = \min\{C', K\}$), Explain how?

$$\|\partial\phi\|_{L^2(\mathbb{R}^n)}(T) \leq C\left|\frac{1}{2}\int_{\{T\}\times\mathbb{R}^n} a^{tt}(\partial_t\phi)^2\,dx - \frac{1}{2}\int_{\{T\}\times\mathbb{R}^n} a^{ij}\partial_i\phi\partial_j\phi\,dx\right|.$$

Thus this implies that

$$\|\partial\phi\|_{L^2(\mathbb{R}^n)}(T) \leq C\|\partial\phi\|^2_{L^2(\mathbb{R}^n)}(0) + C\int_0^T \|\partial_t\phi\|_{L^2(\mathbb{R}^n)}\|F\|_{L^2(\mathbb{R}^n)}(t) + \|\partial a\|_{L^\infty(\mathbb{R}^n)}(t)\|\partial\phi\|^2_{L^2(\mathbb{R}^n)}(t)\,dt.$$

Since one can repeat this argument for any smaller interval $[0, T']$, where $0 < T' < T$, one obtains

$$\sup_{t\in[0,T]}\|\partial\phi\|_{L^2(\mathbb{R}^n)}(t) \leq C\|\partial\phi\|^2_{L^2(\mathbb{R}^n)}(0) + C\int_0^T \|\partial_t\phi\|_{L^2(\mathbb{R}^n)}\|F\|_{L^2(\mathbb{R}^n)}(t) + \|\partial a\|_{L^\infty(\mathbb{R}^n)}(t)\|\partial\phi\|^2_{L^2(\mathbb{R}^n)}(t)\,dt.$$

For the first term inside the integral, one has that

$$\int_0^T \|\partial_t\phi\|_{L^2(\mathbb{R}^n)}\|F\|_{L^2(\mathbb{R}^n)}(t)\,dt \leq \sup_{t\in[0,T]}\|\partial_t\phi\|_{L^2(\mathbb{R}^n)}\cdot\left(\int_0^T \|F\|_{L^2(\mathbb{R}^n)}(t)\,dt\right)$$
$$\leq \delta\|\partial_t\phi\|^2_{L^2(\mathbb{R}^n)} + \frac{1}{4\delta}\left(\int_0^T \|F\|_{L^2(\mathbb{R}^n)}(t)\,dt\right)^2.$$

where the second line follows from Young's inequality with ε[3]. If we fix $\delta > 0$ small enough so that the first term can be combined with the left side of the above inequality (and absorbing δ into the constant C), one has that

$$\sup_{t\in[0,T]}\|\partial\phi\|_{L^2(\mathbb{R}^n)}(t) \leq C\|\partial\phi\|^2_{L^2(\mathbb{R}^n)}(0) + C\left(\int_0^T \|F\|^2_{L^2(\mathbb{R}^n)}(t)\,dt\right)^2 + C\int_0^T \|\partial a\|_{L^\infty(\mathbb{R}^n)}(t)\|\partial\phi\|^2_{L^2(\mathbb{R}^n)}(t)\,dt.$$

Thus using Grönwall's lemma concludes the proof. □

We can, of course, say more. Using the above estimate, we can control the size of the higher-order derivatives of ϕ via the following result.

Proposition 3.3 (Higher Order Energy Estimate I)**.** *Assume ϕ is a smooth solution to the Cauchy wave problem 16, and let k be a positive integer. Then there exists some constant $C = C(n,k) > 0$, the following energy estimates holds:*

$$\sup_{t\in[0,T]}\|\phi\|_{H^k(\mathbb{R}^n)}(t) + \sup_{t\in[0,T]}\|\partial_t\phi\|_{H^{k-1}(\mathbb{R}^n)}(t)$$
$$\leq C(1+T)\exp\left(C\int_0^T \|\partial a\|_{L^\infty(\mathbb{R}^n)}(t)\,dt\right)\left(\|f\|_{H^k(\mathbb{R}^n)} + \|g\|_{H^{k-1}(\mathbb{R}^n)} + \int_0^T \|F\|_{H^{k-1}(\mathbb{R}^n)}(t)\,dt\right.$$
$$\left. + C\int_0^T \sum_{|\alpha|+|\beta|\leq k-1}\left\|\partial\partial_x^\alpha a\partial\partial_x^\beta\phi\right\|_{L^2(\mathbb{R}^n)}(t) + \sum_{|\alpha|+|\beta|\leq k-1}\left\|\partial_x^\alpha a\partial\partial_x^\beta\phi\right\|_{L^2(\mathbb{R}^n)}(t)\,dt\right). \quad (18)$$

Proof. [!! Complete Later !!] □

[3] For any $a, b \geq 0$ and any $\varepsilon > 0$,

$$ab \leq \frac{\varepsilon^p a^p}{p} + \frac{1}{e^q q}b^q$$

, where p, q are conjugate exponents.

3.3 Solutions to the Linear Wave Equation (Constant Coefficients)

In the previous subsection, we outlined various simple energy estimates solutions to the (non-)constant coefficient linear wave equations would satisfy if they existed, and also used these estimates to establish uniqueness results. But we have not yet shown that such solutions exist, nor what their physical properties. We dedicate this subsection to answering these questions.

Consider again the Cauchy wave problem (12),

$$\begin{cases} \Box \phi = 0, \\ (\phi, \partial_t \phi) \restriction_{\{t=0\}} = (\phi_0, \phi_1), \end{cases}$$

where we initially assume $\phi_0, \phi_1 \in C_c^\infty \subset \mathcal{S}(\mathbb{R}^n)$, and $\Box$ denotes the **wave operator**. Then one has the following result.

Theorem 3.2 (Existence and Uniqueness for (12))**.** *Given initial data $(\phi, \partial_t \phi) \restriction_{\{t=0\}} = (\phi_0, \phi_1) \in C_c^\infty(\mathbb{R}^n) \times C_c^\infty(\mathbb{R}^n) \subset \mathcal{S}(\mathbb{R}^n) \times \mathcal{S}(\mathbb{R}^n)$, there exists a unique solution $\phi \in C^\infty(\mathbb{R}; \mathcal{S}(\mathbb{R}^n))$ that solves (12).*

Proof. Assuming that ϕ is an appropriate space of functions, taking the Fourier transformation of the wave equation yields,

$$\partial_t^2 \hat{\phi} + 4\pi^2 |\xi|^2 \hat{\phi} = 0 \implies \partial_t^2 \hat{\phi} = -4\pi^2 |\xi|^2 \hat{\phi},$$

which is a second-order ordinary differential equation. Thus solving this system, one obtains the general solution

$$\hat{\phi}(t, \xi) = A \cos(2\pi t |\xi|) + B \sin(2\pi t \xi).$$

Thus matching the initial conditions, it follows that

$$\hat{\phi}(t, \xi) = \hat{\phi}_0 \cos(2\pi t \xi) + \frac{\hat{\phi}_1}{2\pi \xi} \sin(2\pi t \xi).$$

But to complete the proof of existence, it remains to be seen that we can compute the inverse Fourier transform of $\hat{\phi}$. It suffices to show, at least, that $\hat{\phi}(t, \cdot) \in L^1(\mathbb{R}^n)$. Thus, we compute,

$$\int_{\mathbb{R}^n} |\hat{\phi}(t, \xi)| \, d\xi \leq \int_{\mathbb{R}^n} |\hat{\phi}_0(\xi)| \, d\xi + \int_{\mathbb{R}^n} \frac{|\hat{\phi}_1(\xi)|}{2\pi |\xi|} |\sin(2\pi t |\xi|)| \, d\xi.$$

Since $\hat{\phi}_0(\xi)$ is Schwartz, it is integrable, and thus the first term is finite. $\hat{\phi}_1(\xi)$ is also Schwartz, and thus bounded above; i.e., there exists some $M > 0$ such that for all ξ, $|\hat{\phi}_1(\xi)| \leq M$. We note that $\sin(2\pi t \xi)/\xi \to t$ as $\xi \to 0$. This means there exists some $\varepsilon > 0$ such that for all $\xi \in B_\varepsilon(0)$, $|\sin(2\pi t \xi)/\xi - t| \lesssim_t 1$. This means that

$$\begin{aligned} \int_{\mathbb{R}^n} |\hat{\phi}(t, \xi)| \, d\xi &\leq \int_{\mathbb{R}^n} |\hat{\phi}_0(\xi)| \, d\xi + \int_{\mathbb{R}^n} \frac{|\hat{\phi}_1(\xi)|}{2\pi |\xi|} |\sin(2\pi t |\xi|)| \, d\xi \\ &\lesssim \int_{\mathbb{R}^n} |\hat{\phi}_0| \, d\xi + \int_{B_\varepsilon(0)} M \, d\xi + \int_{\mathbb{R}^n \setminus B_\varepsilon(0)} |\hat{\phi}_1(\xi)| \, d\xi < \infty. \end{aligned}$$

Therefore, it follows that $\hat{\phi}(t, \cdot) \in L^1(\mathbb{R}^n)$. The Fourier Inversion Theorem then tells us that

$$\phi(t, x) = \int_{\mathbb{R}^n} \hat{\phi}(t, \xi) e^{2\pi i x \cdot \xi} \, d\xi$$

The final piece of the proof of existence is that $\phi(t, x)$ is a solution to (12). We first compute,

$$\phi(0, x) = \int_{\mathbb{R}^n} \hat{\phi}(0, \xi) e^{2\pi i x \cdot \xi} \, d\xi = \int_{\mathbb{R}^n} \hat{\phi}_0 e^{2\pi i x \cdot \xi} \, d\xi = \phi_0(x).$$

Then using integrability and the dominated convergence theorem, one obtains:

$$\partial_t \phi(t,x) = \int_{\mathbb{R}^n} \partial_t \hat{\phi}(t,\xi) e^{2\pi i x\cdot\xi}\, d\xi,$$
$$\partial_t^2 \phi(t,x) = \int_{\mathbb{R}^n} \partial_t^2 \hat{\phi}(t,\xi) e^{2\pi i x\cdot\xi}\, d\xi,$$
$$\Delta\phi(t,x) = -\int_{\mathbb{R}^n} 4\pi^2|\xi|^2 \hat{\phi}(t,\xi) e^{2\pi i x\cdot\xi}\, d\xi,$$

so that

$$\partial_t \phi(0,x) = \int_{\mathbb{R}^n} \hat{\phi}_1(\xi) e^{2\pi i x\cdot\xi}\, d\xi = \phi_1(x),$$

and

$$\Box\phi = \partial_t^2\phi(t,x) - \Delta\phi(t,x) = \int_{\mathbb{R}^n} \partial_t^2\hat{\phi}(t,\xi) + 4\pi^2|\xi|^2\hat{\phi}(t,\xi)\, d\xi = 0.$$

Thus, an integrable solution exists for (12) under the hypothesis assumed above. In fact, since $\hat{\phi}(t,x) \in \mathcal{S}(\mathbb{R}^n)$, one actually has that $\phi(t,\cdot) \in \mathcal{S}(\mathbb{R}^n)$. To prove uniqueness, assume $\phi^{(1)}$ and $\phi^{(2)}$ are two solutions to (12). Linearity of $\Box$ then implies that $\phi := \phi^{(1)} - \phi^{(2)}$ also solves the linear wave equation with zero initial data. Thus, if $E(t) = \|\partial\phi\|^2_{L^2(\mathbb{R}^n)}$ denotes the energy of ϕ, then $E(0) = 0$. Thus conservation of energy implies that $E(t) = 0$ for all $t \geq 0$, whence $\phi \equiv 0$. Therefore, $\phi^{(1)} \equiv \phi^{(2)}$. □

In particular, one does not completely require the initial data to smooth with compact support. We have the following existence and uniqueness result.

Theorem 3.3 (Existence and Uniqueness for (12))**.** *Given initial data* $(\phi, \partial_t\phi)\restriction_{\{t=0\}} = (\phi_0, \phi_1) \in H^1(\mathbb{R}^n) \times L^2(\mathbb{R}^n)$*, there exists a unique solution* $\phi \in L^\infty([0,T]; H^1(\mathbb{R}^n)) \times L^\infty([0,T]; L^2(\mathbb{R}^n))$ *that solves (12).*

Proof. The following proof is due to [3]. [!! Complete Later !!] □

3.3.1 Fundamental Solutions

Now we will present another approach to solving (12). In particular, we consider instead the *fundamental solution* using distributions (see Section 2.2 for a review of distributions). We immediately have the following definition.

Definition 3.2 (Forward Fundamental Solution)**.** *The distribution* E_+ *is said to be a* ***forward fundamental solution*** *to the linear wave equation on* $\mathbb{R}^{n+1}$ *if*

1. $\Box E_+ = \delta_0$ *(where* δ_0 *is defined as* $\delta_0(\varphi) = \varphi(t=0, x=0)$*),*
2. $\operatorname{supp}(E_+) \subseteq \{(t,x) \in \mathbb{R}^{1+n} \mid 0 \leq |x| \leq t\}$.

In other words, the forward fundamental solution to the linear wave equation is the solution to linear wave equation on the *forward light cone*. Immediately, one has the following uniqueness result.

Proposition 3.4 (Uniqueness of Forward Fundamental Solution)**.** *A forward fundamental solution on* $\mathbb{R}^{n+1}$*, if it exists, is unique.*

Proof. Let E_1, E_2 be two forward fundamental solutions. Then one has,

$$E_1 = E_1 * \delta_0 = E_1 * (\Box E_2) = E_1 * (\Box\delta_0 * E_2) = (\Box\delta_0 * E_1) * E_2 = \Box E_1 * E_2 = \delta_0 * E_2 = E_2,$$

where we use the support properties of E_1 and E_2 to justify well-definedness of the convolution of the distributions and to justify swapping the wave operator. □

Now, we can use the forward fundamental solution to craft a solution to (12).

Proposition 3.5 (Forward Fundamental Solutions and Representation Formulas)**.** *Let E_+ be a forward fundamental solution to the linear wave equation on $\mathbb{R}^{1+n}$. Then if $(\phi_0, \phi_1) \in C_c^\infty(\mathbb{R}^n) \times C_c^\infty(\mathbb{R}^n)$, the unique solution to (12) is given by,*

$$\phi(t,x) = -E_+ * (\phi_1 \delta_{\{t=0\}}) - (\partial_t E_+) * (\phi_0 \delta_{\{t=0\}}).$$

More generally, for $\Box\phi = F$, where $F \in C^\infty(\mathbb{R}^n)$, ϕ is given for $t \geq 0$ by the following formula

$$\phi(t,x) = -E_+ * (\phi_1 \delta_{\{t=0\}}) - (\partial_t E_+) * (\phi_0 \delta_{\{t=0\}}) + (F\chi_{\{t\geq 0\}}) * E_+,$$

where

$$\chi_{\{t\geq 0\}}(t,x) \coloneqq \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

is the characteristic function for the set $\{t \geq 0\}$.

Proposition 3.6 (Unique Fundamental Solution)**.** *The unique forward fundamental solution is given by*

$$E_+(t,x) = -\frac{\pi^{\frac{1-n}{2}}}{2} \chi_{\{t\geq 0\}} \chi_+^{-\frac{n-1}{2}}(t^2 - |x|^2), \text{??} \tag{19}$$

where

$$\chi_+^a \coloneqq \frac{x^a \chi_{\{x\geq 0\}}}{\Gamma(a+1)}.$$

From the fundamental solution, one has the following result.

Corollary 3.3 (Propagation of Waves)**.** *Solutions to the wave equation $\Box\phi = F$ propagate in time; i.e., $\frac{d}{dt}\operatorname{supp}(\phi)(t) > 0$.*

Now we can derive some of the explicit representation formulas given in the Appendix.

Proposition 3.7 (d'Alembert's Formula)**.** *Let $(\phi_0, \phi_1) \in C_c^\infty(\mathbb{R}^n) \times C_c^\infty(\mathbb{R}^n)$ be initial data to (12) in $\mathbb{R}^{1+1}$. Then the unique solution is given by*

$$\phi(t,x) = \frac{1}{2}(\phi_0(x+t) - \phi_0(x-t)) + \frac{1}{2}\int_{x-t}^{x+t} \phi_1(y)\, dy.$$

Proof. First, we have

$$\chi_+^0(t^2 - |x|^2) = \chi_{\{t^2 - |x|^2 \geq 0\}},$$

so that

$$(\chi_{\{t\geq 0\}}\chi_+^0(t^2-|x|^2)) * (\phi_1\delta_{\{t=0\}}) = \int_{-\infty}^{\infty} \chi_{\{t\geq 0\}}\chi_{\{t^2-|x-y|^2\}}\phi_1(y)\, dy = \int_{x-t}^{x+t} \phi_1(y)\, dy.$$
$$(\partial_t(\chi_{\{t\geq 0\}}\chi_+^0(t^2-|x|^2))) * (\phi_0\delta_{t=0}) = \partial_t \int_{x-t}^{x+t} \phi_0(y)\, dy = \phi_0(x+t) - \phi_0(x-t).$$

Thus using the formula for the forward fundamental solution, we conclude that

$$\phi(t,x) = \frac{1}{2}(\phi_0(x+t) - \phi_0(x-t)) + \frac{1}{2}\int_{x-t}^{x+t} \phi_1(y)\, dy.$$

□

The support of the forward fundamental solution also yields the following results.

Corollary 3.4 (Finite Speed of Propagation)**.** *Suppose $\phi \in C^\infty(\mathbb{R}^{n+1})$ is a solution to the linear wave equation with data (ϕ_0, ϕ_1). If $(\phi_0, \phi_1) = (0,0)$ in $\{y \in \mathbb{R}^n : |x-y| \leq t\}$, then $\phi(t,x) = 0$.*

Corollary 3.5 (Strong Huygens Principle)**.** *Let $n \geq 3$ be an odd integer. Suppose $\phi \in C^\infty(\mathbb{R}^{n+1})$ is a solution to the linear wave equation with data (ϕ_0, ϕ_1). If $(\phi_0, \phi_1) = (0,0)$ in $\{y \in \mathbb{R}^n : |x-y| = t\}$, then $\phi(t,x) = 0$.*

Proof. The proof just follows by looking at the forward fundamental solution,

$$E_+(t,x) = -\frac{\pi^{\frac{1-n}{2}}}{2}\chi_{\{t\geq 0\}}\frac{1}{\Gamma(\frac{3-n}{2})}\frac{1}{(t^2-|x|^2)^{\frac{n-1}{2}}}\chi_{\{|x|\leq t\}}.$$

Consider $n \geq 3$ odd. Then $1/\Gamma((3-n)/2)$ is identically zero since $3-n$ is a nonpositive integer, and the Γ function has poles at these values. Since for $|x| < t$, $1/(t^2-|x|^2)^{(n-1)/2}$ is bounded, it follows that the forward fundamental solution is identically zero for all $|x| < t$. On the other hand, one has that on $|x| = t$, the $1/(t^2-|x|^2)^{(n-1)/2}$ is unbounded, while $1/\Gamma((3-n)/2)$ is zero. Thus the support focuses on the spherical shell $|x| = t$. On the other hand, for $n \geq 2$ even, $\Gamma((3-n)/2)$ is bounded as is $1/(t^2-|x|^2)^{(n-1)/2}$ for $|x| < t$. Thus the forward fundamental solution is not identically zero in the interior of the balls. □

3.3.2 Decay Estimates

Corollary 3.3 shows that as t increases, the support of the unique solution to the (non)homogeneous wave equation increases (e.g., consider ripples on the surface of a lake). For $n \geq 2$, the measure of the support increases in time, while for $n = 1$, the measure remains constant. However, we showed that the energy of a solution to (12) is conserved. This must imply that the amplitude of the wave $|\phi(t,x)|$ decays as $t \to \infty$. We have the following decay estimates.

Proposition 3.8 (Pointwise Decay Rate)**.** *Fix any $R > 0$. For $(\phi_0, \phi_1) \in C_c^\infty \times C_c^\infty(\mathbb{R}^n)$ and ϕ the unique solution to the linear Cauchy wave problem (12),*

$$\sup_{|x|\leq R}|\phi(t,x)| \lesssim_{(\phi_0,\phi_1,R)} \frac{1}{(1+t)^{n-1}}. \tag{20}$$

Proof. The fundamental ideas behind the proof are as follows: we wish to consider the expression

$$\phi(t,x) = \int_{\mathbb{R}^n} e^{2\pi i(t|\xi|+x\cdot\xi)}\left(\frac{\hat{\phi}_0}{2} + \frac{\hat{\phi}_1}{2\pi i|\xi|}\right) + e^{-2\pi i t|\xi|}\left(\frac{\hat{\phi}_0}{2} - \frac{\hat{\phi}_1}{2\pi i|\xi|}\right) d\xi,$$

and then bound each of the terms in the integral. For sake of demonstrating the main ideas, we shall assume $n = 3$. The proof follows trivially for all $|t| \leq 1$: let

$$I_1 = \int_{\mathbb{R}^3} e^{2\pi i(t|\xi|+x\cdot\xi)}\frac{\hat{\phi}_0}{2}\,d\xi, \qquad I_2 = \int_{\mathbb{R}^3} e^{-2\pi i(t|\xi|+x\cdot\xi)}\frac{\hat{\phi}_0}{2}\,d\xi.$$

$$I_3 = \int_{\mathbb{R}^3} e^{2\pi i(t|\xi|+x\cdot\xi)}\frac{\hat{\phi}_1}{2\pi i|\xi|}\,d\xi \qquad I_4 = -\int_{\mathbb{R}^3} e^{-2\pi i(t|\xi|+x\cdot\xi)}\frac{\hat{\phi}_1}{2\pi i|\xi|}\,d\xi.$$

Since $\hat{\phi}_0$ is Schwartz, and thus integrable, there exist $M_1, M_2 > 0$ such that for all $|t| \leq 1$, $|I_i| \leq M_i$ for $i = 1,2$. Now for I_3 (and analogously for I_4), we can write

$$\begin{aligned}
|I_3| &\leq \int_{\mathbb{R}^3}\frac{|\hat{\phi}_1(\xi)|}{2\pi|\xi|}\,d\xi = \int_{B_1(0)}\frac{|\hat{\phi}_1(\xi)|}{2\pi|\xi|}\,d\xi + \int_{\mathbb{R}^3\setminus B_1(0)}\frac{|\hat{\phi}_1(\xi)|}{2\pi|\xi|}\,d\xi \\
&\leq \int_{B_1(0)}\frac{K_1}{2\pi|\xi|}\,d\xi + \int_{\mathbb{R}^3\setminus B_1(0)}\frac{|\hat{\phi}_1(\xi)|}{2\pi|\xi|}\,d\xi \\
&= \int_{B_1(0)}\frac{K_1}{2\pi|\xi|}\,d\xi + \int_{\mathbb{R}^3\setminus B_1(0)}\frac{|\hat{\phi}_1(\xi)|}{2\pi|\xi|}\,d\xi < \infty,
\end{aligned}$$

where the second line follows from the fact that since $\hat{\phi}_1 \in \mathcal{S}(\mathbb{R}^n)$, it is continuous on the compact set $\overline{B}_1(0)$ and thus bounded above by some positive constant K_1, and the final line follows from the fact that $1/|\xi|$ is integrable over $B_1(0)$ for all $n \geq 2$, and the second term in the final line is controlled above by Hölder's inequality; I_4 is controlled in the same way. Thus, one has that for every $x \in \mathbb{R}^n$, and all $|t| \leq 1$, $|\phi(t,x)| \leq M$ for some constant M. Taking $C > 0$ large enough so that $C \geq M \cdot 2^{n-1}$, one has that

$$\sup_{|x| \leq R} |\phi(t,x)| \leq \frac{M}{(1+t)^2};$$

note that since M_1, M_2, K_1, K_2 are dependent on ϕ_0, ϕ_1, the proof concludes. So now we focus on the case with $t > 1$. Now introduce polar coordinates on $\mathbb{R}^3$ as follows: let $(|\xi|, \xi_\theta, \xi_\phi)$ be the polar coordinates on $\mathbb{R}^3_\xi$ with $\xi_1 = |\xi| \sin\xi_\theta \cos\xi_\phi$, $\xi_2 = |\xi| \sin\xi_\theta \sin\xi_\phi$, and $\xi_3 = |\xi| \cos\xi_\theta$. The corresponding volume form is,

$$d\xi = |\xi|^2 \sin\xi_\theta \, d|\xi| d\xi_\theta d\xi_\phi.$$

Furthermore, one has that for any $N \in \mathbb{N}_0$,

$$\left(\frac{1}{2\pi i t} \frac{\partial}{\partial |\xi|}\right)^N e^{2\pi i t|\xi|} = e^{2\pi i t|\xi|}.$$

Now using integration by parts,

$$\begin{aligned} I &= \int_0^{2\pi} \int_0^{\pi} \int_0^{\infty} e^{2\pi i t|\xi|} e^{2\pi i (x\cdot\xi)} \frac{\hat{\phi}_1(\xi)}{2\pi i} |\xi| \sin(\xi_\theta) \, d|\xi| d\xi_\theta d\xi_\varphi \\ &= \left[\frac{1}{(2\pi i)^2 t} \int_0^{\pi} \int_0^{2\pi} e^{2\pi i t|\xi|} e^{2\pi i (x\cdot\xi)} \hat{\phi}_1(\xi) |\xi| \sin(\xi_\theta) \, d\xi_\varphi d\xi_\theta \right]_{|\xi|=0}^{R\to\infty} \\ &\quad - \int_0^{2\pi} \int_0^{\pi} \int_0^{\infty} \frac{1}{2\pi i t} e^{2\pi i t|\xi|} \frac{\partial}{\partial |\xi|} \left(e^{2\pi i x\cdot\xi} \frac{\hat{\phi}_1(\xi)}{2\pi i} |\xi| \right) \sin(\xi_\theta) \, d|\xi| d\xi_\theta d\xi_\varphi \end{aligned}$$

We claim that the first term is identically zero. To prove this, we first note that

$$\lim_{|\xi|\to\infty} \left| 2\pi i t|\xi| e^{2\pi i (x\cdot\xi)} \hat{\phi}_1(\xi) |\xi| \sin(\xi_\theta) \right| = \lim_{|\xi|\to\infty} |\hat{\phi}_1(\xi)||\xi| = 0,$$

since $\hat{\phi}_1(\xi) \in \mathcal{S}(\mathbb{R}^3)$. On the other hand, the integrand vanishes identically at $|\xi| = 0$. Now, we see that one application of integration by parts gave us a power of t. To get another power of t, we will use integration by parts on the second term. This time, after dealing with a boundary term at $|\xi| = 0$, we get

$$I = -\int_{\mathbb{R}^3} e^{2\pi i t|\xi|} \frac{1}{8\pi^3 i t^2} (\cdots) \sin(\xi_\theta) \, d|\xi| d\xi_\theta d\theta_\varphi + \frac{1}{2\pi^2 i t^2} \hat{\phi}_1(\xi = 0).$$

The boundary term does not allow us to use integration by parts again. Thus bounding each of the term and using the fact that $\hat{\phi}_1 \in \mathcal{S}$, one obtains

$$|I| \leq \frac{C(1+R^2)}{t^2}.$$

□

Note that this decay bound is not uniform for all $x \in \mathbb{R}^n$. We can actually get a uniform decay bound for all $x \in \mathbb{R}^n$, but at the cost of the strength.

Theorem 3.4 (Uniform Decay Rate)**.** *For $(\phi_0, \phi_1) \in C_c^\infty(\mathbb{R}^n) \times C_c^\infty(\mathbb{R}^n)$, and ϕ the unique solution to the homogeneous initial value problem, there exists $C = C(\phi_0, \phi_1) > 0$ so that*

$$\sup_{x\in\mathbb{R}^n} |\phi(t,x)| \leq \frac{C}{(1+t)^{\frac{n-1}{2}}}$$

for $t \geq 0$.

Proof. As before, we only present the proof for $n = 3$ and for the term

$$I = \left(\int_{\mathbb{R}^2} e^{2\pi i(t|\xi|+x\cdot\xi)} \frac{\hat{\phi}_1(\xi)}{2\pi i|\xi|} \, d\xi \right).$$

It is always possible to find a coordinate system in which $x = (0, 0, \alpha)$ for some $\alpha \in \mathbb{R}$; thus, we will work in these coordinates. Let $\rho = \sqrt{\xi_1^2 + \xi_2^2}$, and let us use the coordinate system $(\xi_3, \rho, \xi_\theta)$, where ξ_θ is given by $\rho\cos(\xi_\theta) = \xi_1$. First note that

$$\frac{\partial|\xi|}{\partial\rho} = \frac{\partial}{\partial\rho}\sqrt{\rho^2 + \xi_3^2} = \frac{\rho}{|\xi|}.$$

Moreover, $\partial_\rho e^{2\pi i x\cdot\xi} = 0$. Thus, since for any nonnegative integer N,

$$\left(\frac{|\xi|}{2\pi i\rho t} \frac{\partial}{\partial\rho} e^{2\pi i t|\xi|} \right)^N = e^{2\pi i t|\xi|},$$

integration by parts gets us,

$$\begin{aligned} I &= \int_{\mathbb{R}^3} \frac{|\xi|}{2\pi i\rho t} \frac{\partial}{\partial\rho}(e^{2\pi i t|\xi|}) e^{2\pi i x\cdot\xi} \frac{\hat{\phi}_1(\xi)}{2\pi i|\xi|} \, \rho d\rho d\xi_\theta d\xi_3 = \int_{\mathbb{R}^3} \frac{1}{2\pi i t} \frac{\partial}{\partial\rho}(e^{2\pi i t|\xi|}) e^{2\pi i x\cdot\xi} \frac{\hat{\phi}_1(\xi)}{2\pi i} \, d\rho d\xi_\theta d\xi_3 \\ &= -\int \frac{1}{4\pi^2 t} e^{2\pi i(t|\xi|+x\cdot\xi)} \hat{\phi}_1(\xi)\Big|_{\rho=0}^{\infty} d\xi_\theta d\xi_3 + \int_{\mathbb{R}^3} \frac{1}{4\pi^2 t} e^{2\pi i(t|\xi|+x\cdot\xi)} \frac{\partial}{\partial\rho}(\hat{\phi}_1(\xi)) \, d\rho d\xi_\theta d\xi_3 \\ &= \int \frac{1}{2\pi t} e^{2\pi i(t|\xi_3|+\alpha\xi_3)} \hat{\phi}_1(\rho = 0, \xi = \xi_3) \, d\xi_3 + \int_{\mathbb{R}^3} \frac{1}{4\pi^2 t} e^{2\pi i(t|\xi|+x\cdot\xi)} \frac{\partial}{\partial\rho}(\hat{\phi}_1(\xi)) \, d\rho d\xi_\theta d\xi_3. \end{aligned}$$

Since $\hat{\phi}_1(\xi)$ and $\partial_\rho\hat{\phi}_1(\xi)$ are Schwartz, it follows that

$$|I| \le \frac{\|\hat{\phi}_1\|_{L^1}}{2\pi t} + \frac{\|\partial_\rho\hat{\phi}_1\|_{L^1}}{4\pi^2 t} =: \frac{M}{t}.$$

Now we split into two cases. First assume $t \in [0, 1]$. Then $(1 + t)^{-1} \ge 2^{-1}$ so that $2/(1 + t) \ge 1$. Directly bounding $|I(t, x)|$, one has that $|I(t, x)| \le M_0$ for some $M_0 > 0$, and all $t \in [0, 1]$, $x \in \mathbb{R}^3$. Thus in this case, $|I| \le M_0 \le \frac{2M_0}{1+t}$. Now for the $t > 1$ case, $2t \ge 1 + t$ so that $2/(1 + t) \ge \frac{1}{t}$. Therefore, $|I| \le \frac{2M}{1+t}$. Take $C = \max\{2M_0, 2M\}$ so that $|I| \le C/(1 + t)$ for all $t \in [0, 1]$ and $x \in \mathbb{R}^3$. The other terms of $\phi(t, x)$ can be controlled the same way. □

The key result is the following: for all $n \ge 2$, solutions to (12) undergo *geometric dispersion*, while no dispersion occurs for $n = 1$. Even more strikedly, we notice that the decay rate inside a bounded ball is *faster* than the globally uniform decay rate. Computationally, this is because we were able to undertake a second integration by parts for the bounded ball case, with each integration by parts adding a power of t (in the case of $n = 3$). We can try to understand why this would make sense intuitively. Looking at the fundamental solution, (**??**), we notice that the decay is primarily "saturated" along the wavefront. Consider first the case of $n = 3$. Fix $R > 0$ large enough, and thus fix the ball $B_R(0)$ of radius R around the origin. Suppose that at $t = 0$, a light is flashed at the origin. The emitted light waves propagate in a spherical shell at the speed of light ($c = 1$). Because of conservation of energy, we know that the signal weakens as the wavefront approaches $\partial B_R(0)$. The moment the wavefront passes $\partial B_R(0)$

3.4 Solutions to the Linear Wave Equation (Variable Coefficients)

In this subsection, we continue studying the (non)homogeneous linear wave equation with *variable* coefficients [5]. Consider again the equation (16),

$$\begin{cases} \partial_\alpha(a^{\alpha\beta}\partial_\beta\phi) = F \\ (\phi, \partial_t\phi) \restriction_{\{t=0\}} = (\phi_0, \phi_1) \in H^k(\mathbb{R}^n) \times H^{k-1}(\mathbb{R}^n), \end{cases}$$

where $k \in \mathbb{N}$, $\phi : I \times \mathbb{R}^n \to \mathbb{R}$, and $a^{\alpha\beta}$ is assumed to be a symmetric matrix such that[4].

$$\sum_{\alpha,\beta} |a^{\alpha\beta} - m^{\alpha\beta}| < \frac{1}{10}.$$

Furthermore, we shall assume that a and all its derivatives (of all orders) are bounded in $[0,T] \times \mathbb{R}^n$, and that $F : [0,T] \times \mathbb{R}^n \to \mathbb{R}^n \in L^1([0,T]; H^{k-1}(\mathbb{R}^n))$. As in the case of the linear wave equation with constant coefficients, we wish to proceed with an existence and uniqueness result. But this requires us to first present the following result.

Lemma 3.2 (Bound on Sobolev Norms)**.** *Let L denote the partial differential operator $L\phi = \partial_\alpha(a^{\alpha\beta}\partial_\beta\phi)$. Let L^* be defined by $L^*\psi = \partial_\alpha(a^{\alpha\beta}\partial_\beta\psi)$ be the (formal) adjoint of L. Suppose $\psi \in C_c^\infty((-\infty,T)\times\mathbb{R}^n)$. Then for every $m \in \mathbb{Z}$, there exists $C = C(m,T,a,b) > 0$ such that*

Explain?

$$\|\psi\|_{H^m(\mathbb{R}^n)}(t) \le C \cdot \int_t^T \|L^*\psi\|_{H^{m-1}(\mathbb{R}^n)}(s)\, ds. \tag{21}$$

Proof. [!! Complete Later !!] □

Theorem 3.5 (Existence of Solutions to the Linear Equation)**.** *Let $k \in \mathbb{N}$. Given $F \in L^1([0,T]; H^{k-1}(\mathbb{R}^n))$, there exists a solution*

$$(\phi, \partial_t\phi) \in L^\infty([0,T]; H^k(\mathbb{R}^n)) \times L^\infty([0,T]; H^{k-1}(\mathbb{R}^n))$$

solving (16).

Proof. Unlike for the linear case with constant coefficients where we explicitly derived a representation formula, the proof of this result requires a bit of functional analysis (in particular the *Hahn-Banach theorem*). The proof will be carried out over two steps.

1. First assume that $(\phi_0, \phi_1) = (0,0)$. Let L be the differential operator $\partial_\alpha(a^{\alpha\beta}\partial_\beta\bullet)$, and let $L^* : C_c^\infty((-\infty,T)\times\mathbb{R}^n) \to C_c^\infty((-\infty,T)\times\mathbb{R}^n)$ be the formal adjoint of L[5]. For every $\psi \in C_c^\infty((-\infty,T)\times\mathbb{R}^n)$, define a map

$$\Phi : L^*\psi \in L^\infty(C_c^\infty((-\infty,T)\times\mathbb{R}^n)) \mapsto \int_0^T \int_{\mathbb{R}^n} \psi F \, dxdt =: \langle F, \psi \rangle .$$

We claim that this functional Φ is bounded. Indeed, one has

$$\begin{aligned}
|\Phi(L^*\psi)| &= \left| \int_0^T \int_{\mathbb{R}^n} \psi F \, dxdt \right| \le \int_0^T \int_{\mathbb{R}^n} |\psi F| \, dxdt \\
&\le \int_0^T \|F\|_{H^{k-1}(\mathbb{R}^n)} \|\psi\|_{H^{1-k}(\mathbb{R}^n)} \, dt \\
&\le C \cdot \left(\int_0^T \|F\|_{H^{k-1}(\mathbb{R}^n)} \, dt \right) \cdot \left(\sup_{t\in[0,T]} \|\psi\|_{H^{1-k}(\mathbb{R}^n)} \right) \\
&\le C' \cdot \int_0^T \|L^*\psi\|_{H^{-k}(\mathbb{R}^n)}(s)\, ds,
\end{aligned}$$

where (1) the second line follows from the generalized Hölder's inequality[6], (2) the final line follows from incorporating $\int_0^T \|F\|_{H^{k-1}(\mathbb{R}^n)} \, dt$ into C, and using the previous lemma. It is also straightforward to see that the Φ is linear. Thus, by the Hahn-Banach theorem, there exists $\phi \in (L^1((-\infty,T); H^{-k}(\mathbb{R}^n)))^* = L^\infty((-\infty,T); H^k(\mathbb{R}^n))$ with $\phi = 0$ for $t < 0$ that extends Φ: $\langle F, \psi \rangle = \langle \phi, L^*\psi \rangle$ for every $\psi \in C_c^\infty((-\infty,T)\times\mathbb{R}^n)$. Thus ϕ is a *weak solution* (see e.g., Section 2.2) to the Cauchy wave problem with trivial data.

[4] to ensure hyperbolicity; [!! Complete Later !!]
[5] In the sense that for every $\phi, \psi \in C_c^\infty((-\infty,T)\times\mathbb{R}^n)$, $\langle L\phi, \psi \rangle = \langle \phi, L^*\psi \rangle$.
[6] $\|uv\|_{L^1} \le \|u\|_{W^{(-s,p)}} \|v\|_{W^{(-s,q)}}$, where p, q are conjugate exponents

2. Now turn to the general case with $(\phi_0, \phi_1) \in H^k(\mathbb{R}^n) \times H^{k-1}(\mathbb{R}^n)$, and let u be a function in $[0,T] \times \mathbb{R}^n$ such that
$$(u, \partial_t u) \restriction_{\{t=0\}} = (\phi_0, \phi_1).$$
Then the system $L\eta = F - Lu$ has trivial initial data. Then the work in (1) gives us a weak solution η to the problem. Setting $\phi = \eta + u$ gives the desired solution[7].

□

Remark 3.4. *It is important to note that the above theorem only gives the existence of weak solutions. If the initial data is sufficiently regular, we can upgrade to a classical solution via the Sobolev Embedding Theorem.*

Theorem 3.6 (Sobolev Embedding Theorem)**.** *For every $s > \frac{n}{2}$, and for all $\phi \in H^s(\mathbb{R}^n)$,*
$$\|\phi\|_{L^\infty(\mathbb{R}^n)} \lesssim_{(n,s)} \|\phi\|_{H^s(\mathbb{R}^n)}.$$
I.e., for every $s > n/2$, $H^s(\mathbb{R}^n) \hookrightarrow C_0(\mathbb{R}^n)$.

3.5 More on Linear Hyperbolic Systems

Here, we consider additional properties of the linear wave equation not studied above, and also consider the more general class of *hyperbolic systems* [7]. As we shall see, this leads naturally to the notion of *propagation*. Some of these results are presented, perhaps with less exposition, in the Appendix.

Method of Characteristics We start with a simple setting: consider the homogeneous linear wave equation in $\mathbb{R}^{1+1}$, given by
$$\partial_t \phi + c(t,x)\, \partial_x \phi = 0,$$
where $c(t,x)$ is assumed to be a smooth real-valued function. We also consider the ordinary differential equation, $\dot{x}(t) = c(t, x(t))$. For a smooth function ϕ, one has that along the integral curves $(1, x(t))$,
$$\frac{d}{dt}\phi(t, x(t)) = (\partial_t \phi + c\, \partial_x \phi)|_{(t,x(t))}.$$
Thus it follows that the solutions of the homogeneous 1D linear wave equation are precisely those solutions ϕ that are constant along the integral curves $(t, x(t))$. We call these curves the **characteristic curves** of the partial differential operator $L = \partial_t + c(t,x)\partial_x$.

The **method of characteristics** then boils down to considering a differential equation in one spatial dimension, reducing the equation to an ODE on integral curves, and constructing an overall solution from the solutions on each integral curve (e.g., see Chapter 1 of [2] for an in-depth coverage of the subtleties and generalizations of the method of characteristics). It then follows that certain existence and uniqueness theorems follow easily from this method. E.g.,

Proposition 3.9 (Existence and Uniqueness for 1D Wave Equation)**.** *If $c \in \mathbb{R}$ is constant, then $\phi \in C^\infty([0,T] \times \mathbb{R})$ satisfies*
$$\partial_t \phi + c\partial_x \phi = 0$$
if and only if there is an $f \in C^\infty(\mathbb{R})$ such that $\phi = f(x - ct)$. The function f is uniquely determined.

Proof. By the method of characteristics, for constant c, the characteristics along which u is constant are the lines $(t, x + ct)$. Thus, $u(t,x) = u(0, x - ct)$. If one sets $f(x) = u(0,x)$, then the forward direction is satisfied. Conversely, if there exists $f \in C^\infty$ such that $\phi(t,x) = f(x - ct)$, then
$$\partial_t \phi = -cf'(x - ct), \qquad c\partial_x \phi = cf'(x - ct).$$
Thus, $\partial_t \phi + c\partial_x \phi = 0$. □

[7] Complete proof that $(\phi, \partial_t \phi)$ belong to the appropriate spaces

I.e., the Cauchy problem above is uniquely solvable by a C^∞ function, and the solutions are *propagating* waves translating rigidly with velocity c. We may also consider the same differential equation, but where $c(t,x)$ is *not* bounded.

Example 3.1 (Nonuniqueness for Unbounded Speed)**.** *Consider the differential equation,*

$$\partial_t \phi + c(t,x)\,\partial_x \phi = 0,$$

where $c(t,x) = x^2$ is not bounded. Fix a point $(0,x_0)$ in spacetime. The characteristic through this point is the solution of the differential equation,

$$x' = x^2, \qquad x(0) = x_0,$$

whence

$$x(t) = \frac{x_0}{1 - x_0 t}.$$

Inverting this, one obtains

$$x_0 = \frac{x}{1 + xt}.$$

Thus for each point (t,x) with $t \geq 0$, there is a unique characteristic tracing back to $t = 0$. Therefore, given initial data $\phi(0,x) = g(x)$, the solution $\phi(t,x)$ is uniquely determined in $t \geq 0$ by requiring $t \geq 0$ by requiring ϕ to be constant on the characteristics. By plotting the integral curves, we observe the following: for $|x_0| \geq 1$, the corresponding integral curves all diverge before $t = 1$, for $x_0 = 1$, the integral curve passing through $(0,x_0)$ diverges at $t = 1$, and for $x_0 \in (0,1)$, the integral curves blow up at finite time $t = 1/x_0 > 1$. The backward characteristic through $(0,-1)$ diverges to $-\infty$ at $t = -1$. These observations have the following consequences.

Since information about the data on $[1,\infty)$ blows up before $t = 1$, the data for $|x| \geq 1$ does not influence the solution in $t \geq 1$. Consequently, there has been a loss of information so that the initial values do not uniquely determine a solution in $t < 0$.

Thus, going forward, we shall make the following hypotheses:

> **Hypotheses 3.1** *Suppose that for all $T > 0$, $\partial^\alpha_{t,x} c \in L^\infty([0,T]\times\mathbb{R})$, and that the coefficient $d(t,x)$ satisfies analogous bounds.*

One can also generalize to *systems* of hyperbolic equations. For instance, the method of characteristics applies to systems of the form

$$u_t + \operatorname{diag}(c_1(t,x),\cdots,c_N(t,x))u = 0,$$

where $u = (u_1(t,x),\cdots,u_N(t,x))^\intercal$, and each u_j is constant on characteristics with speed $c_j(t,x)$ that satisfy Hypotheses 3.1. More generally, the method of characteristics extends to systems given by the partial differential operator,

$$L = \partial_t + A(t,x)\partial_x + B(t,x),$$

where A, B are smooth matrix-valued functions such that for all $T \in \mathbb{R}$ and for all multiindices α, $\partial^\alpha_{t,x} A, \partial^\alpha_{t,x} B \in L^\infty([-T,T]\times\mathbb{R})$ when the following hypothesis is satisfied,

> **Hypothesis 3.2** There is smooth matrix-valued function $M(t,x)$ so that for all $T \in \mathbb{R}$, and all multiindices α,
>
> $$\partial^\alpha_{t,x}(M), \partial^\alpha_{t,x}(M^{-1}) \in L^\infty([0,T]\times\mathbb{R}),$$
>
> and MAM^{-1} is diagonal and real.

Here, real diagonalizability of A just ensures that the linear operator remains hyperbolic-type, and the bounds on M and A just ensure that the characteristic speeds do not diverge (thus preventing the loss of information and/or uniqueness of solutions).

Definition 3.3 (Strictly Hyperbolic Systems)**.** *If for each (t,x), the matrix A has N distinct real eigenvalues $c_1(t,x) < c_2(t,x) < \cdots < c_N(t,x)$, the system is said to be **strictly hyperbolic**. To ensure that the estimates on M and M^{-1} are uniform as $|x| \to \infty$, we also make the additional assumption*

$$\inf_{(t,x)\in[0,T]\times\mathbb{R}} \min_{2\le j\le N} (c_j(t,x) - c_{j-1}(t,x)) > 0.$$

Explain?

A strictly hyperbolic system satisfies Hypothesis 3.2.

In particular, if the hypothesis holds, then since one has that $A = MDM^{-1}$ with $D = \operatorname{diag}(c_1,\dots,c_N)$, we can write

$$L = \partial_t + MDM^{-1}\partial_x + B(x,t)$$

[!! Complete Later !!]

3.5.1 Energy Estimates

Here, we return again to energy estimates for *systems* of hyperbolic partial differential equations. As in the previous section, we will initially only consider first-order hyperbolic PDE.

Constant Coefficient Systems We start by considering the constant coefficient systems,

$$\partial_t\phi + \sum_{j=1}^{d} A_j\partial_{x^j}\phi = f,$$

where $\phi = (\phi_1,\dots,\phi_d)^\top$, and each A_j is a matrix-valued function. One has the following definition,

Definition 3.4 (Symmetric Hyperbolic Operator (Constant Coefficient))**.** *A constant coefficient operator*

$$L = \partial_t + \sum_{j=1}^{d} A_j\partial_{x^j}$$

*on $\mathbb{R}^d$ is said to be **symmetric hyperbolic** if each A_j is hermitian symmetric.*

Requiring that the A_j be hermitian *symmetric* is very useful for computing energy estimates. We recall that if A is symmetric, then the matrix-valued operator $\exp(-itA)$ is *unitary*. So now, computing the Fourier transform of the PDE $L\phi = 0$, one obtains,

$$\hat\phi(t,\xi) = \exp\left(-it\sum_{j=1}^{d} A_j\xi_j\right)\hat\phi(0,\xi).$$

But then, using the unitariness of the the exponential operator, we recover that for all $t \ge 0$,

$$|\hat\phi(t,\xi)|^2 = |\hat\phi(0,\xi)|^2.$$

In particular, integrating over $d\xi$ and using Plancherel's theorem, one easily obtains conservation of the L^2-norm: $\|\phi(t)\|_{L^2(\mathbb{R}^d)} = \|\phi(0)\|_{L^2(\mathbb{R}^d)}$. Likewise, under appropriate assumptions on ϕ, one could also obtain conservation of the Sobolev norms.

As we saw before, these types of energy estimates segue easily into a discussion of (local) well-posedness, existence, and unique. Thus as an analogue to the existence and uniqueness results in the above subsections, we have the following result.

Proposition 3.10 (Existence and Uniqueness for Constant Coefficient Symmetric Hyperbolic PDE)**.** *Suppose that $L = \partial_t + \sum_{j=1}^{d} A_j \partial_j$ is a constant coefficient symmetric hyperbolic operator. For $g \in H^\infty(\mathbb{R}^d)$, there exists a unique solution $\phi \in C^\infty(\mathbb{R}; H^\infty(\mathbb{R}^d))$ to the initial value problem*

$$L\phi = 0, \qquad \phi|_{\{t=0\}} = g. \tag{22}$$

This solution is given by $\widehat{\phi}(t,\xi) = e^{-2\pi i t \sum A_j \xi_j} \hat{g}$[8]. Furthermore, for all t, s, $\|\phi(t)\|_{H^s(\mathbb{R}^d)} = \|\phi(0)\|_{H^s(\mathbb{R}^d)}$.

Proof. We first prove existence of a solution: consider the function $\hat{\phi}(t,\xi) = \exp(-2\pi i t \sum A_j \xi_j)\, \hat{g}$. Then one notes that

$$|\hat{\phi}(t,\xi)|^2 = |\hat{g}|^2 \implies \left\|\hat{\phi}(t)\right\|^2_{L^2(\mathbb{R}^d)} = \|\hat{g}\|^2_{L^2(\mathbb{R}^d)} = \|g\|^2_{L^2(\mathbb{R}^d)} < \infty,$$

where the last equality follows from Plancherel's theorem, and the final inequality follows from the fact that $g \in H^\infty(\mathbb{R}^d) \subset L^2(\mathbb{R}^d)$. Thus the inverse Fourier transform of $\hat{\phi}(t,\xi)$ is well-defined and also L^2. In particular, one has

$$\phi(t,x) = \int_{\mathbb{R}^d} e^{2\pi i (x\cdot\xi - t\sum A_j \xi_j)} \hat{g}(\xi)\, d\xi.$$

It is immediately clear that $\phi(t,\cdot) \in C^\infty$. Now we must show that $\phi(\cdot, x) \in H^\infty(\mathbb{R}^d)$. Let $s \in \mathbb{R}$ be an arbitrary fixed real number. Then,

$$|(1+|\xi|^2)^{s/2}\hat{\phi}(t,\xi)|^2 = (1+|\xi|^2)^s \cdot |\hat{g}|^2.$$

But since we assumed that $g \in H^\infty(\mathbb{R}^d)$, it follows that $(1+|\xi|^2)^{s/2}\hat{g}(\xi) \in L^2(\mathbb{R}^d)$. Thus, it follows that $\phi(\cdot, x) \in H^s(\mathbb{R}^d)$ for all $s \in \mathbb{R}$. Hence, $\phi \in C^\infty(\mathbb{R}; H^\infty(\mathbb{R}^d))$. To show that ϕ actually solves the PDE, it suffices to show that $\hat{\phi}$ solves the ODE obtained by taking the Fourier transform of the PDE. But we've already shown this to be the case. Therefore, the proof of existence is complete. Moreover, the above equality also establishes conservatinon the Sobolev norms for all t, s. All that remains is to prove uniqueness. Assume that ϕ_1, ϕ_2 are two solutions to the initial value problem. Then their difference $\phi = \phi_1 - \phi_2$ also solves the PDE $L\phi = 0$, but with the initial value identically zero. Applying the above results (in particular, the Sobolev norm conservation), it follows that ϕ is identically zero. Thus the proof concludes. □

Variable Coefficient Systems Now we shall consider variable coefficient systems, analagous to the variable coefficient equations studied in Section 3.4. We start with the following definition.

Definition 3.5 (Symmetric Hyperbolic Systems)**.** *In $\mathbb{R}^{1+d}$ introduce the coordinates $y = (y_0, \ldots, y_d)$, where $y_0 = t$, and for each $1 \le j \le d$, $y_j = x_j$. Then the partial differential operator,*

$$L(y,\partial) = \sum_{\mu=0}^{d} A_\mu(y) \frac{\partial}{\partial y_\mu} + B(y) \tag{23}$$

*is said to be **symmetric hyperbolic** if the following hold:*

1. *The coefficient matrices A_μ and B have uniformly bounded derivatives: for all α,*

$$\sup_{y\in\mathbb{R}^{1+d}} \left\|\partial_y^\alpha (A_\mu(y), B(y))\right\| < \infty;$$

2. *A_μ are hermitian symmetric;*
3. *A_0 is strictly positive in the sense that there exists $c > 0$ so that for all y, $A_0(y) \ge cI$.*

It follows that this system satisfies an L^2 estimate of the following form.

[8] Note the 2π constant; [7] uses a slightly different convention for the Fourier transform.

Proposition 3.11. *L^2 Estimates For every $s \in \mathbb{R}$, there is a constant $C(s, L)$ so that for all $t \geq 0$ and all $u \in C^1(\mathbb{R}; H^s(\mathbb{R}^d)) \cap C(\mathbb{R}; H^{s+1}(\mathbb{R}^d))$,*

$$\|u(t)\|_{H^s(\mathbb{R}^d)} \leq Ce^{Ct} \|u(0)\|_{H^s(\mathbb{R}^d)} + \int_0^t Ce^{C(t-\sigma)} \|(Lu)(\sigma)\|_{H^s(\mathbb{R}^d)} \, d\sigma,$$

where L is a symmetric hyperbolic system.

Proof. Let us consider a few cases.

1. First assume that $A_0 = I$, and define

$$G(t) = \sum_j A_j(t,x) \, \partial_j + B(t,x),$$

and define its formal adjoint differential operator G^* by

$$G^*(t) = -\sum_j A_j^*(t,x)\partial_j + B(t,x)^* - \sum_j (\partial_j A_j^*)$$

so that for all $\phi, \psi \in C_c^\infty(\mathbb{R}^d)$, $\langle G(t)\phi, \psi \rangle = \langle \phi, G^*(t)\psi \rangle$. In particular, one has

$$G(t) + G^*(t) = B(t,x) + B^*(t,x) - \sum_{j=1}^d \partial_j A_j,$$

where we used the fact that since the A_j are hermitian symmetric, $A_j^* = A_j$. Therefore, $G(t) + G^*(t)$ is multiplication by a uniformly bounded matrix, which means that as an operator on $L^2(\mathbb{R}^d)$, $\|G(t) + G^*(t)\| \leq 2C$ for some $C > 0$. Note that this means that $G(t)$ is **nearly anti-self-adjoint**. If we assume that $u \in C(\mathbb{R}; H^1(\mathbb{R}^d))$, $u_t \in C(\mathbb{R}; L^2(\mathbb{R}^d))$, and $f \coloneqq Lu$, then it is straightforward to see that $u'(t) + G(t)u(t) = f(t)$.

□

4 Local Theory for Nonlinear Wave Equations

4.1 Quasilinear Equations

In this subsection, we are most interested in the following quasilinear wave equations for $\phi : I \times \mathbb{R}^n \to \mathbb{R}$:

$$\begin{cases} \partial_\alpha(a^{\alpha\beta}(\phi)\partial_\beta\phi) = F(\phi, \partial\phi) \\ (\phi, \partial_t\phi) \restriction_{\{t=0\}} = (\phi_0, \phi_1) \in H^k(\mathbb{R}^n) \times H^{k-1}(\mathbb{R}^n), \end{cases} \tag{24}$$

where $k \in \mathbb{N}$ is specified later, $a^{\alpha\beta}$ is symmetric and satisfies,

$$\sum_{\alpha,\beta} |a^{\alpha\beta} - m^{\alpha\beta}| \le \frac{1}{10}, \qquad a(0) = 0, \qquad F(0,0) = 0, \tag{$*$}$$

and $a^{\alpha\beta} : \mathbb{R} \to \mathbb{R}$ and $F : \mathbb{R} \times \mathbb{R}^n \to \mathbb{R}$ are smooth functions. As we will see, unlike the linear wave equation, (24) is only *locally* (in time) well-posed. We will first state the main result of this section, then prove a couple of technical results that will then enable us to prove the theorem.

Theorem 4.1 (Local Existence Theorem)**.** *Fix a and F satisfying assumptions ($*$). Then the Cauchy problem (24) is locally well-posed in the following sense.*

1. *(Existence and Uniqueness of Local-in-time Solutions) There exists*
$$T = T(\|\phi_0\|_{H^{n+2}(\mathbb{R}^n)}, \|\phi_1\|_{H^{n+1}(\mathbb{R}^n)}) > 0$$
*such that there exists a **classical** solution ϕ to (24) with*
$$(\phi, \partial_t\phi) \in L^\infty([0,T]; H^{n+2}(\mathbb{R}^n)) \times L^\infty([0,T]; H^{n+1}(\mathbb{R}^n)).$$
Moreover, this solution is unique in the function space above.

2. *(Continuous Dependence on Initial Data) Let $\phi_0^{(i)}, \phi_1^{(i)}$ be sequences of functions such that $\phi_0^{(i)} \to \phi_0$ in $H^{n+2}(\mathbb{R}^n)$ and $\phi_1^{(i)} \to \phi_1$ in $H^{n+1}(\mathbb{R}^n)$ as $i \to \infty$. Then taking $T > 0$ sufficiently small, we have*
$$\left\|(\phi^{(i)} - \phi, \partial_t(\phi^{(i)} - \phi))\right\|_{L^\infty([0,T];H^s(\mathbb{R}^n)) \times L^\infty([0,T];H^{s-1}(\mathbb{R}^n))} \to 0$$
as $i \to \infty$ for every $1 \le s < n+2$.

Before we can prove this theorem, we need the following estimates.

Lemma 4.1. *There exists $B = B(A, n, F) > 0$ such that for $t \in [0,T]$, we have*

$$\sum_{0 \le |\alpha| \le n+1} \left\|\partial_x^\alpha F(\phi^{(i-1)}, \partial\phi^{(i-1)})\right\|_{L^2}(t) \le B.$$

Proof. To establish this theorem, we need to first expand the partial derivatives, and use the chain rule. First, since F was assumed to be smooth, by the Extreme Value Theorem, one has that for any $A > 0$,

$$\sum_{|\gamma| \le N} \sup_{|x|, \|p\| \le A} |\partial_{x,p}^\gamma(F)|(x,p) \le C_{A,N}.$$

Thus expanding the partial derivatives using the chain rule, one gets

$$\begin{aligned} &\sum_{0 \le |\alpha| \le n+1} \left\|\partial_x^\alpha F(\phi^{(i-1)}, \partial\phi^{(i-1)})\right\|_{L^2(\mathbb{R}^n)}(t) \\ &\quad \le C\Bigg(\sum_{0 \le |\alpha| \le n+1} \left\|\partial\partial_x^\alpha \phi^{(i-1)}\right\|_{L^2(\mathbb{R}^n)}(t) + \sum_{0 \le |\alpha_1| + |\alpha_2| \le n+1} \left\|\partial\partial_x^{\alpha_1}\phi^{(i-1)} \partial\partial_x^{\alpha_2}\phi^{(i-1)}\right\|_{L^2}(\mathbb{R}^n)(t) \\ &\qquad + \text{cubic terms} + \cdots + \text{terms of order } n+1 \Bigg). \end{aligned}$$

Thus we need to bound each of the terms on the right side of the estimate above. Consider first the quadratic term. It follows that either $|\alpha_1| \le \frac{n+1}{2}$ or $|\alpha_2| \le \frac{n+1}{2}$. So without loss of generality, assume that $|\alpha_1| \le \frac{n+1}{2}$. Then since $\frac{n+1}{2} + \frac{n}{2} \le n+1$, the Sobolev embedding theorem tells us that

$$\sum_{0\le|\alpha_1|\le\frac{n+1}{2}} \left\|\partial\partial_x^{\alpha_1}\phi^{(i-1)}\right\|_{L^\infty(\mathbb{R}^n)} \le C\cdot \sum_{0\le|\alpha|\le n+1} \left\|\partial\partial_x^{\alpha}\phi^{(i-1)}\right\|_{L^2(\mathbb{R}^n)}.$$

Thus,

$$\sum_{0\le|\alpha_1|+|\alpha_2|\le n+1} \left\|\partial\partial_x^{\alpha_1}\phi^{(i-1)}\partial\partial_x^{\alpha_2}\phi^{(i-1)}\right\|_{L^2(\mathbb{R}^n)}(t) \le C\cdot \sum_{0\le|\alpha|\le n+1} \left\|\partial\partial_x^{\alpha}\phi^{(i-1)}\right\|^2_{L^2(\mathbb{R}^n)}(t).$$

Thus if we treat the higher-order terms in the above expansion analogously, we derive the estimate,

$$\sum_{0\le|\alpha|\le n+1} \left\|\partial_x^\alpha F(\phi^{(i-1)}, \partial\phi^{(i-1)})\right\|_{L^2(\mathbb{R}^n)}(t)$$
$$C\left(1 + \left\|\phi^{(i-1)}\right\|_{L^2(\mathbb{R}^n)} + \sum_{0\le|\alpha|\le n+1}\left\|\partial\partial_x^\alpha\phi^{(i-1)}\right\|_{L^2(\mathbb{R}^n)}\right)^{n+1} \le B.$$

□

Lemma 4.2 (Bound on the Two Remaining Quantities)**.** *There exists some $B = B(A, n, F) > 0$ such that*

$$\int_0^T \sum_{|\gamma|+|\sigma|\le n+1}\sum_{\alpha,\beta}\left\|\partial\partial_x^\gamma\left(a^{\alpha\beta}(\phi^{(i-1)})\right)\partial\partial_x^\sigma\phi^{(i)}\right\|_{L^2(\mathbb{R}^n)}(t)\,dt$$
$$+\int_0^T \sum_{\substack{|\gamma|+|\sigma|\le n+1\\|\sigma|\le n}}\sum_{\alpha,\beta}\left\|\partial_x^\gamma\left(a^{\alpha\beta}(\phi^{(i-1)})\right)\partial\partial\partial_x^\sigma\phi^{(i)}\right\|_{L^2(\mathbb{R}^n)}(t)\,dt \le B.$$

Lemma 4.3 (Bound on Derivatives of a)**.**

$$\sum_{|\gamma|=1}\left\|\partial_x\left(a^{\alpha\beta}(\phi^{(i-1)})\right)\right\|_{L^\infty(\mathbb{R}^n)} \le B.$$

Proof. The proof of this theorem follows in two main steps; in the first step, we prove the existence of a solution. In the second, we prove continuous dependence on the initial data, and thus uniqueness of solutions. It is not immediately clear how to construct a solution to this general nonlinear system. But we recall from the previous section that we have a global-in-time existence and uniqueness solution for nonconstant coefficient linear wave equations. Thus if we can find a way to link the quasilinear system above to linear systems, then we can work from there. Luckily, we already have a tool that assists in this approach (namely, *Picard iteration*[9]).

(1) By density arguments, it suffices to assume that the initial data are Schwarz functions. Let $\phi^{(1)} \in C^\infty$ be an initial guess, and for $i \ge 2$, define $\phi^{(i)}$ as the unique smooth solution to the *linear* system

$$\begin{cases}\partial_\alpha(a^{\alpha\beta}(\phi^{(i-1)})\partial_\beta\phi^{(i)}) = F(\phi^{(i-1)}, \partial\phi^{(i-1)})\\ (\phi^{(i)}, \partial_t\phi^{(i)})\restriction_{\{t=0\}} = (\phi_0, \phi_1).\end{cases}$$

To prove existence for the quasilinear system, it suffices to show that for $T > 0$ (sufficiently small, and depending only on the initial data), the sequence is uniformly bounded in the space $L^\infty([0,T]; H^{n+2}(\mathbb{R}^n))\times L^\infty([0,T]; H^{n+1}(\mathbb{R}^n))$, and is Cauchy in some larger function space. Thus we tackle each step at a time.

[9]Recall that in ODE theory, Picard iteration is as follows: suppose one has the differential equation $y'(t) = f(t, y(t))$, where $y(t)$ is the unknown function, and $y(t_0) = y_0$. If we simply integrate both sides, then we can extract $y(t) = y_0 + \int_{t_0}^t f(s, y(s))\,ds$. However, the integral depends explicitly on the unknown function. Thus we start with an initial ansatz y_0, and then use the above equation to produce a better estimate for the solution. We then take this estimate and use the above equation to obtain a better estimate. We do this iteratively, obtaining better solutions (under specific conditions).

(a) (Uniformly Bounded) What we wish to show is that for all $i \geq 1$, there exists a $C = C(n, a, F_0, \phi_0, \phi_1) > 0$ such that

$$\left\|(\phi^{(i)}, \partial_t^{(i)}\phi)\right\|_{L^\infty([0,T];H^{n+2}(\mathbb{R}^n))\times L^\infty([0,T];H^{n+1}(\mathbb{R}^n))} \leq C.$$

To show uniform boundedness, it suffices to show that the estimate holding for some $i-1$ implies the result holds for i. Thus for some $i \geq 1$, we will assume there exists some $A > 0$ such that

$$\left\|(\phi^{(i-1)}, \partial_t\phi^{(i-1)})\right\|_{L^\infty([0,T];H^{n+2}(\mathbb{R}^n))\times L^\infty([0,T];H^{n+1}(\mathbb{R}^n))} \leq A.$$

The principle of bootstrap arguments tells us that we can also assume that

$$\left\|(\phi^{(i)}, \partial_t\phi^{(i)})\right\|_{L^\infty([0,T];H^{n+2}(\mathbb{R}^n))\times L^\infty([0,T];H^{n+1}(\mathbb{R}^n))} \leq 4A,$$

as long as the bound can be proved (which our theorem says is possible). Now we note that the bounds that we require look a lot like the estimates in (18),

$$\begin{aligned}
&\sup_{t\in[0,T]} \|(\phi, \partial_t\phi)\|_{H^k(\mathbb{R}^n)\times H^{k-1}(\mathbb{R}^n)}(t) \\
&\leq C(1+T)\exp\left(C\int_0^T \|\partial a\|_{L^\infty(\mathbb{R}^n)}(t)\,dt\right)\left(\|f\|_{H^k(\mathbb{R}^n)} + \|g\|_{H^{k-1}(\mathbb{R}^n)} + \int_0^T \|F\|_{H^{k-1}(\mathbb{R}^n)}(t)\,dt\right. \\
&\qquad \left. + C\int_0^T \sum_{|\alpha|+|\beta|\leq k-1} \left\|\partial\partial_x^\alpha a\partial\partial_x^\beta\phi\right\|_{L^2(\mathbb{R}^n)}(t) + \sum_{|\alpha|+|\beta|\leq k-1} \left\|\partial_x^\alpha a\partial\partial_x^\beta\phi\right\|_{L^2(\mathbb{R}^n)}(t)\,dt\right).
\end{aligned}$$

With these estimates, we can finally use our energy estimate (18):

$$\begin{aligned}
&\left\|(\phi^{(i)}, \partial_t\phi^{(i)})\right\|_{L^\infty([0,T];H^{n+2}(\mathbb{R}^n))\times L^\infty([0,T];H^{n+1}(\mathbb{R}^n))} \\
&\qquad \leq C\left(\|(\phi_0,\phi_1)\|_{H^{n+2}(\mathbb{R}^n\times H^{n+1}(\mathbb{R}^n))} + BT + CBT^2\right)\exp(CBT)
\end{aligned}$$

Let us assume wlog that $C \geq 2$. While the actual constant B can be very large, we can choose sufficiently small $T > 0$ (depending on A), so that $(BT+CBT^2)\exp(CBT) \leq \|(\phi_0,\phi_1)\|_{H^{n+2}(\mathbb{R}^n)\times H^{n+1}(\mathbb{R}^n)}$ and $\exp(CBT) \leq 2$. Thus with T chosen as above, we get the bound

$$\left\|(\phi^{(i)}, \partial_t\phi^{(i)})\right\|_{L^\infty([0,T];H^{n+2}(\mathbb{R}^n))} \leq 4C_0\|(\phi_0,\phi_1)\|_{H^{n+2}(\mathbb{R}^n)\times H^{n+1}(\mathbb{R}^n)}.$$

Let $A = 4C_0\|(\phi_0,\phi_1)\|_{H^{n+2}(\mathbb{R}^n)\times H^{n+1}(\mathbb{R}^n)}$. Thus we have shown that

$$\left\|(\phi^{(i-1)}, \partial_t\phi^{(i-1)})\right\|_{L^\infty([0,T];H^{n+2}(\mathbb{R}^n))} \leq A$$

implies

$$\left\|(\phi^{(i)}, \partial_t\phi^{(i)})\right\|_{L^\infty([0,T];H^{n+2}(\mathbb{R}^n))} \leq A,$$

provided that $T = T(A)$ is sufficiently small. Thus this portion of the proof concludes.

(b) Now we will show that the sequence is Cauchy in $L^\infty([0,T];H^1(\mathbb{R}^n)) \times L^\infty([0,T];L^2(\mathbb{R}^n))$. For every $i \geq 3$, we consider the difference $\phi^{(i)} - \phi^{(i-1)}$:

$$\begin{aligned}
\partial_\alpha(a^{\alpha\beta}(\phi^{(i-1)})\partial_\beta(\phi^{(i)} - \phi^{(i-1)})) &= \partial_\alpha\left(a^{\alpha\beta}(\phi^{(i-1)})\partial_\beta\phi^{(i)} - a^{\alpha\beta}(\phi^{(i-1)})\partial_\beta\phi^{(i-1)}\right) \\
&= F(\phi^{(i-1)}, \partial\phi^{(i-1)}) - \partial_\alpha\left(a^{\alpha\beta}(\phi^{(i-1)})\partial_\beta\phi^{(i-1)}\right) \\
&= F(\phi^{(i-1)}, \partial\phi^{(i-1)}) - \partial_\alpha((a^{\alpha\beta}(\phi^{(i-1)}) + a^{\alpha\beta}(\phi^{(i-2)}) - a^{\alpha\beta}(\phi^{(i-2)}))\partial_\beta\phi^{(i-1)}) \\
&= F(\phi^{(i)}, \partial\phi^{(i-1)}) - F(\phi^{(i-2)}, \partial\phi^{(i-2)}) \\
&\qquad - \partial_\alpha((a^{\alpha\beta}(\phi^{(i-1)}) - a^{\alpha\beta}(\phi^{(i-2)}))\partial_\beta\phi^{(i-1)}).
\end{aligned}$$

We know that a is continuously differentiable, and thus, satisfies the bound,

$$\sum_{\alpha,\beta}\sum_{|\gamma|\leq N}\sup_{|x|\leq A}|\partial_x^\gamma(a^{\alpha\beta})|(x)\leq C_{A,N}.$$

Since we also know that

$$\left\|(\phi^{(i)},\partial_t\phi^{(i)})\right\|_{L^\infty([0,T];H^{n+2}(\mathbb{R}^n))\times L^\infty([0,T];H^{n+1}(\mathbb{R}^n))}\leq C,$$

the Mean Value Theorem tells us that there exists some $C>0$ (depending on the initial data, but neither on i nor on T) such that

$$\left|\partial_\alpha((a^{\alpha\beta}(\phi^{(i-1)})-a^{\alpha\beta}(\phi^{(i-2)}))\partial_\beta\phi^{(i-1)})\right|\leq C|\partial(\phi^{(i-1)}-\phi^{(i-2)})|,$$

which implies

$$\left\|\partial_\alpha((a^{\alpha\beta}(\phi^{(i-1)})-a^{\alpha\beta}(\phi^{(i-2)}))\partial_\beta\phi^{(i-1)})\right\|_{L^2(\mathbb{R}^n)}(t)\leq C\|\partial(\phi^{(i-1)}-\phi^{(i-2)})\|_{L^2(\mathbb{R}^n)}(t),$$

for all $t\in[0,T]$. Likewise, one has that

$$\left\|F(\phi^{(i-1)},\partial\phi^{(i-1)})-F(\phi^{(i-2)},\partial\phi^{(i-2)})\right\|_{L^2(\mathbb{R}^n)}(t)\leq C\|\partial(\phi^{(i-1)}-\phi^{(i-2)})\|_{L^2(\mathbb{R}^n)}(t),$$

and $\left\|\partial(a^{\alpha\beta}(\phi^{(i-1)}))\right\|_{L^\infty(\mathbb{R}^n)}\leq C$. Thus applying (18), we get

$$\begin{aligned}&\sup_{t\in[0,T]}\|(\phi^{(i)}-\phi^{(i-1)},\partial_t\phi^{(i)}-\partial_t\phi^{(i-1)})\|_{H^1(\mathbb{R}^n)\times L^2(\mathbb{R}^n)}(t)\\&\leq CT\sup_{t\in[0,T]}\|(\phi^{(i-1)}-\phi^{(i-2)},\partial_t\phi^{(i-1)}-\partial_t\phi^{(i-2)})\|_{H^1(\mathbb{R}^n)\times L^2(\mathbb{R}^n)}(t)\leq C_1.\end{aligned}$$

Thus if we choose $T>0$ sufficiently small, then

$$\sup_{t\in[0,T]}\left\|\phi^{(i)}-\phi^{(i-1)}\right\|_{H^1(\mathbb{R}^n)}(t)\leq\frac{1}{2}\sup_{t\in[0,T]}\left\|\phi^{(i)}-\phi^{(i-2)}\right\|_{H^1(\mathbb{R}^n)}(t),$$

whence

$$\sup_{t\in[0,T]}\left\|\phi^{(i)}-\phi^{(i-1)}\right\|_{H^1(\mathbb{R}^n)}(t)\leq 2^{2-i}C_1,$$

so that the Cauchy property follows immediately.

(c) Now we will prove existence (uniqueness will follow from the proof of the second part of the theorem). Since we showed that the sequence is Cauchy in $L^\infty([0,T];H^1(\mathbb{R}^n))\times L^\infty([0,T];L^2(\mathbb{R}^n))$, there exists a limit $(\phi,\partial_t\phi)\in L^\infty([0,T];H^1(\mathbb{R}^n))\times L^\infty([0,T];L^2(\mathbb{R}^n))$. But by the uniform bound in the other function space, we conclude that $\phi,\partial_t\phi$ are both contained in the desired regularity class.

(2) Now we will show a proof of continuous dependence on the initial data. Pick $i\in\mathbb{N}$ sufficiently large, and consider the difference $\phi^{(i)}-\phi$. Then we compute

$$\begin{aligned}&\partial_\alpha(a^{\alpha\beta}(\phi^{(i)}-\phi))\\&\qquad=-\partial_\alpha((a^{\alpha\beta}(\phi^{(i)})-a^{\alpha\beta}(\phi))\partial_\beta\phi)+F(\phi^{(i)},\partial\phi^{(i)})-F(\phi,\partial\phi).\end{aligned}$$

Applying (18), one obtains

$$\begin{aligned}&\sup_{s\in[0,T]}\left\|(\phi^{(i)}-\phi,\partial_t\phi^{(i)}-\partial_t\phi)\right\|_{H^1(\mathbb{R}^n)\times L^2(\mathbb{R}^n)}(s)\\&\leq C\left(\left\|(\phi_0^{(i)}-\phi_0,\phi_1^{(i)}-\phi_1)\right\|_{H^1(\mathbb{R}^n)\times L^2(\mathbb{R}^n)}+\int_0^t\left\|\partial_\alpha((a^{\alpha\beta}(\phi^{(i-1)})-a^{\alpha\beta}(\phi))\partial_\beta\phi)\right\|_{L^2(\mathbb{R}^n)}(t)\,dt\right.\\&\qquad\left.+\int_0^t\left\|F(\phi^{(i)},\partial\phi^{(i)})-F(\phi,\partial\phi)\right\|_{L^2(\mathbb{R}^n)}(t)\,dt\right)\end{aligned}$$

Then the proof of (1) gives the following estimate,

$$\sup_{s\in[0,t]} \left\|(\phi^{(i)}-\phi, \partial_t\phi^{(i)}-\partial_t\phi)\right\|_{H^1(\mathbb{R}^n)\times L^2(\mathbb{R}^n)}(s)$$
$$\leq C\left(\left\|(\phi_0^{(i)}-\phi_0, \phi_1^{(i)}-\phi_1)\right\|_{H^1(\mathbb{R}^n\times L^2(\mathbb{R}^n))} + \int_0^t \left\|(\phi^{(i)}-\phi, \partial_t\phi^{(i)}-\partial_t\phi)\right\|_{H^1(\mathbb{R}^n)\times L^2(\mathbb{R}^n)}(t)\,dt\right).$$

Thus by Grönwall's inequality, one has that for some $C = C(T) > 0$,

$$\sup_{t\in[0,T]} \left\|(\phi^{(i)}-\phi_0, \phi_1^{(i)}-\phi_1)\right\|_{H^1(\mathbb{R}^n)\times L^2(\mathbb{R}^n)}(t) \leq C\left\|(\phi_0^{(i)}-\phi_0, \phi_1^{(i)}-\phi_1)\right\|_{H^1(\mathbb{R}^n\times L^2(\mathbb{R}^n))} \xrightarrow{i\to\infty} 0.$$

Then to get the general estimate for $1 \leq s < n+2$, we can simply use the bound

$$\|\eta\|_{H^s(\mathbb{R}^n)} \leq C\|\eta\|^{\theta_1}_{H^{s_1}(\mathbb{R}^n)}\|\eta\|^{\theta_2}_{H^{s_2}(\mathbb{R}^n)}$$

for some $C = C(s_1, s_2, s, n) > 0$, where $0 \leq s_1 \leq s \leq s_2$, $\theta_1 + \theta_2 = 1$, and $\theta_1 s_1 + \theta_2 s_2 = s$ (proven in the homework).

□

We note the following difference between the existence and uniqueness results we have already seen in the linear case, and Theorem (4.1): in the previous results, the solutions existed on time intervals $[0,T]$ for any arbitrary $T > 0$. However, the above theorem only gives existence of a *classical* solution on some finite interval $[0,T]$, where T is explicitly determined by the initial data.

For physical theories, we often desire *smooth* solutions. However, the unique solution constructed above is contained in $L^\infty([0,T]; H^{n+2}(\mathbb{R}^n)) \times L^\infty([0,T]; H^{n+1}(\mathbb{R}^n))$ for $T > 0$ sufficiently small. But in some cases, we can also *lose* derivatives and regularity. Thus we desire some type of safeguard against the loss of regularity. The following result establishes this *persistence of regularity.*

Theorem 4.2 (Persistence of Regularity)**.** *Given initial data $(\phi_0, \phi_1) \in H^{n+2}(\mathbb{R}^n) \times H^{n+1}(\mathbb{R}^n)$ to (24). Furthermore, let*

$$T_* \coloneqq \sup\{T > 0 : \exists!\ \textit{solution}\ \in L^\infty([0,T]; H^{n+2}(\mathbb{R}^n)) \times L^\infty([0,T]; H^{n+1}(\mathbb{R}^n))\}, \tag{25}$$

be the ***maximal time of existence****.*

(a) If $(\phi_0, \phi_1) \in H^m(\mathbb{R}^n) \times H^{m-1}(\mathbb{R}^n)$ for some $m \in \mathbb{N}$ with $m > n+2$, then the solution $(\phi, \partial_t\phi) \in L^\infty([0,T]; H^m(\mathbb{R}^n)) \times L^\infty([0,T]; H^{m-1}(\mathbb{R}^n))$ for every $T < T_*$.

(b) If $(\phi_0, \phi_1) \in \bigcap_{m=1}^\infty (H^m(\mathbb{R}^n) \times H^{m-1}(\mathbb{R}^n))$, then the solution is smooth in $[0, T_*) \times \mathbb{R}^n$.

However, we are interested in what happens as we approach the maximal time of existence. To study this, we first have the following technical lemma.

Lemma 4.4 (Schauder Estimates I)**.** *Let $s \geq 0$, and $F : \mathbb{R}^m \to \mathbb{R}$ be a smooth function with $F(0) = 0$. Then if $\phi : \mathbb{R}^n \to \mathbb{R}^m$ belongs to the space $\phi \in L^\infty(\mathbb{R}^n; \mathbb{R}^m) \cap H^s(\mathbb{R}^n; \mathbb{R}^m)$, there exists a constant $C = C(F, \|\phi\|_{L^\infty(\mathbb{R}^n;\mathbb{R}^m)}, s, n, m) > 0$ such that*

$$\|F(\phi)\|_{H^s(\mathbb{R}^n)} \leq C\|\phi\|_{H^s(\mathbb{R}^n;\mathbb{R}^m)}.$$

Lemma 4.5 (Schauder Estimates II)**.** *For $s > 0$, there exists $C = C(n, s) > 0$ such that*

$$\|f\cdot(\partial_{x_i} g)\|_{H^s(\mathbb{R}^n)} \leq C(\|f\|_{H^{s+1}(\mathbb{R}^n)}\|g\|_{L^\infty(\mathbb{R}^n)} + \|g\|_{H^s(\mathbb{R}^n)}\|f\|_{W^{1,\infty}(\mathbb{R}^n)}).$$

Theorem 4.3 (Breakdown Criteria)**.** *Given initial data $(\phi_0, \phi_1) \in H^{n+2}(\mathbb{R}^n) \times H^{n+1}(\mathbb{R}^n)$ to (24), let T_* be the maximal time of existence. If $T_* < \infty$, then all of the following hold:*

(i)

$$\liminf_{t\to T_*} \|(\phi, \partial_t\phi)\|_{H^{n+2}(\mathbb{R}^n)\times H^{n+1}(\mathbb{R}^n)}(t) \to \infty,$$

(ii)

$$\limsup_{t\to T_*}\left(\sum_{|a|\le\lfloor\frac{n+2}{2}\rfloor}\|\partial_x^\alpha\phi\|_{L^\infty(\mathbb{R}^n)}(t)+\sum_{|\alpha|\le\lfloor\frac{n}{2}\rfloor}\|\partial_t\partial_x^\alpha\phi\|_{L^\infty(\mathbb{R}^n)}(t)\right)\to 0$$

.

(iii)

$$\limsup_{t\to T_*}\sum_{|\alpha|\le 1}\|\partial^\alpha\phi\|_{L^\infty(\mathbb{R}^n)}(t)\to\infty$$

.

4.1.1 Global Regularity for Subcritical Equations

In this subsubsection, we study the following nonlinear equation

$$\begin{cases}\Box\,\phi = |\phi|^2\phi,\\ (\phi,\partial_t\phi)\restriction_{\{t=0\}} = (\phi_0,\phi_1),\end{cases}\tag{26}$$

where $\phi : I\times\mathbb{R}^3\to\mathbb{R}$, and $I\subseteq\mathbb{R}$ is an interval. While the energy function, (13), is not conserved for solutions ϕ to the above system, we have the following conserved quantity.

Proposition 4.1. *As long as the solution to (26) remains sufficiently regular, the following quantity is independent of time,*

$$E(t) \coloneqq \int_{\mathbb{R}^3}\frac{1}{2}(\partial_t\phi)^2+\frac{1}{2}\sum_1^3(\partial_i\phi)^2+\frac{1}{4}|\phi|^4\,dx.\tag{27}$$

Proof. As long as the solution remains *sufficiently* regular, differentiating yields,

$$\begin{aligned}\frac{d}{dt}E(t) &= \int_{\mathbb{R}^3}(\partial_t\phi)(\partial_{tt}^2\phi)\,dx+\sum_1^3\int_{\mathbb{R}^3}\partial_i(\partial_t\phi)(\partial_i\phi)\,dx+\int_{\mathbb{R}^3}\phi^3\,dx\\ &= \int_{\mathbb{R}^3}(\partial_t\phi)\cdot\left(\sum_1^3\partial_{ii}^2\phi-\phi^3\right)dx-\int_{\mathbb{R}^3}(\partial_t\phi)\cdot\sum_1^3(\partial_i^2\phi)+\int_{\mathbb{R}^3}\phi^3\,dx=0,\end{aligned}$$

where the second term in the second line follows from using integration by parts. □

It is important to remember that just because we have a conserved quantity, the solution ϕ need *not* be globally regular. But we have the following result.

Proposition 4.2 (Control over Solutions)**.** *There exists a constant $C>0$ such that for every $\phi:\mathbb{R}^3\to\mathbb{R}$, we have*

$$\|\phi\|_{L^6(\mathbb{R}^3)}\le C\,\|\phi\|_{\dot H^1(\mathbb{R}^3)}.$$

Proof. This is a consequence of the Sobolev embedding theorems. □

Theorem 4.4 (Global-in-Time Solutions)**.** *Assume $(\phi_0,\phi_1)\in(H^5(\mathbb{R}^3)\times H^4(\mathbb{R}^3))\cap(C^\infty(\mathbb{R}^3)\times C^\infty(\mathbb{R}^3))$. Then (26) has a global-in-time solution.*

Remark 4.1. *Notice that the local well-posedness result for nonlinear wave equations only guaranteed us local-in-time solutions to the general nonlinear wave equation, with no additional information about whether a global-in-time solution exists. The above theorem states that for the specific nonlinear equation (26), we, in fact, have a global-in-time solution.*

Proof. By the local existence theorem for nonlinear wave equations, we know that we have a smooth local-in-time solution to (26). Thus it suffices for us to show that this solution is global-in-time. Therefore, we must show that for every $T > 0$, the following bound is satisfied.

$$\sum_{|\alpha|\leq 2} \|\partial_x^\alpha \phi\|_{L^\infty([0,T]\times\mathbb{R}^3)} + \sum_{|\alpha|\leq 1} \|\partial_t^\alpha \partial_x^\alpha \phi\|_{L^\infty([0,T]\times\mathbb{R}^3)} \leq C(T).$$

But by the Sobolev embedding theorem ($H^{s_1}(\mathbb{R}^3) \subset H^{s_2}(\mathbb{R}^3)$ for all $s_1 \geq s_2$), it suffices to prove that

$$\|(\phi, \partial_t\phi)\|_{L^\infty([0,T];H^4(\mathbb{R}^3))\times L^\infty([0,T];H^3(\mathbb{R}^3))} \leq C(T).$$

We start by showing that, at minimum,

$$\|(\phi, \partial_t\phi)\|_{L^\infty([0,T];H^2(\mathbb{R}^3))\times L^\infty([0,T];H^1(\mathbb{R}^3))} \leq C(T).$$

We start by computing,

$$\begin{aligned}
&\|(\phi, \partial_t\phi)\|_{L^\infty([0,T];H^2(\mathbb{R}^3))\times L^\infty([0,T];H^1(\mathbb{R}^3))} \\
&\leq C(T)\cdot\left(\|(\phi_0,\phi_1)\|_{H^2(\mathbb{R}^3)\times H^1(\mathbb{R}^3)} + \int_0^T \|\phi\cdot\phi\cdot\partial\phi\|_{L^2(\mathbb{R}^3)}\, dt\right) \\
&\leq C(T)\cdot\left(\|\phi_0,\phi_1\|_{H^2(\mathbb{R}^3)\times H^3(\mathbb{R}^3)} + \int_0^T \|\phi\|_{L^6(\mathbb{R}^3)}(t)\,\|\phi\|_{L^6(\mathbb{R}^3)}(t)\,\|\partial\phi\|_{L^6(\mathbb{R}^3)}(t)\, dt\right) \\
&\leq C(T)\cdot\left(\|(\phi_0,\phi_1)\|_{H^2(\mathbb{R}^3)\times H^1(\mathbb{R}^3)} + \int_0^T (E(t))^2\,\|(\phi,\partial_t\phi)\|_{H^2(\mathbb{R}^3)\times H^1(\mathbb{R}^3)}(t)\, dt\right).
\end{aligned}$$

Since $E(t)$ is *a priori* controlled by the conservation law, we can just use Grönwall's inequality to establish the bound on the H^2 norms. Thus, one also has that $(\phi, \partial_t\phi)$ is bounded in $L^\infty([0,T]\times\mathbb{R}^3)$. Likewise, we can also establish the appropriate bound on the H^4 norm, thus establishing the proof. □

[!! Try to think about why it **suffices** to show the bounds that were shown above !!]

Scaling and Criticality In this section, we introduce the notion of *scaling*, and *criticality*. First suppose ϕ is a solution to (26). Then for $\lambda > 0$, we claim that $\phi_\lambda(t,x) = \lambda\phi(\lambda t, \lambda x)$ is also a solution:

$$\begin{aligned}
\partial_t^2\phi_\lambda &= \lambda^3(\partial_t^2\phi)(\lambda t,\lambda x).\\
\partial_x^2\phi_\lambda &= \lambda^3(\partial_x^2\phi)(\lambda t,\lambda x).\\
|\phi_\lambda|^2\phi_\lambda &= \phi_\lambda^3 = \lambda^3(|\phi|^2\phi)(\lambda t,\lambda x).
\end{aligned}$$

Thus since ϕ is a solution to (26), we have that ϕ_λ is also a solution. To get a sense of criticality, let us consider the $\dot{H}^1$ seminorm of ϕ_λ. We compute,

$$\|\phi_\lambda\|_{\dot{H}^1}(t) = \left(\int |\nabla\phi_\lambda|^2(t,x)\, dx\right)^{1/2} = \left(\int \lambda^4|(\nabla\phi)|^2(\lambda t,\lambda x)\, dx\right)^{1/2} = \left(\int \lambda|\nabla\phi|^2(\lambda t,\widetilde{x})\, d\widetilde{x}\right)^{1/2} = \lambda^{1/2}\|\phi\|_{\dot{H}^1}(\lambda t).$$

Thus in the limit $\lambda \to \infty$, we notice that the $\dot{H}^1$ norm also tends to infinity. Thus it becomes clear that conservation laws because much more important at the smaller scales. Thus we have the following classification of equations. An equation is said to be

- **subcritical** if the "conservation law becomes more useful at smaller scales",
- **supercritical** if the opposite is true,
- **critical** if the conservation law scales in the same way as the scaling of the equation.

The classification is useful since generally speaking,

- most subcritical equations have a global regular solution,

- critical equations vary a lot in terms of having global regular solutions/finite-time blow up,
- very little is known for supercritical equations.

For instance, for the given equation,

$$\Box\phi = |\phi|^{p-1}\phi. \tag{28}$$

in $\mathbb{R}\times\mathbb{R}^3$, one observes that the $\dot{H}^1$ norm is always controlled by the conservation law, while the equation itself is invariant under the the scaling

$$\phi_\lambda(t,x) = \lambda^{\frac{2}{p-1}}\phi(\lambda t, \lambda x).$$

Then, we have the following.

Proposition 4.3. *For the equation (28) in $\mathbb{R}\times\mathbb{R}^3$, the equation is **subcritical** if $p<5$, **critical** if $p=5$, and supercritical if $p>5$.*

Proof. A straightforward computation yields,

$$\|\phi_\lambda\|_{\dot{H}^1}(t) = \lambda^{\frac{5-p}{p-1}}\|\phi\|_{\dot{H}^1}(\lambda t).$$

Thus for $p<5$, the exponent of λ is positive so that the $\dot{H}^1$ norm tends to infinity as $\lambda\to\infty$ (meaning the equation is subcritical), is exactly 0 at $p=5$ (so that the equation is critical), and is negative for $p>5$ (so that the equation is supercritical). □

5 An Introduction to Einstein Vacuum Equations in Wave Coordinates

In this section, we look at the *Einstein vaccum equations*. In particular, this requires us to cover some initial definitions in geometry. Thus we start with some basic definitions from the theory of Lorentzian manifolds.

Definition 5.1 (Christoffel Symbols)**.** *Let (M, g) be a Lorentzian manifold, where g is a Lorentzian metric on M. Then we define the* ***Christoffel symbols*** *on M to be the smooth functions,*

$$\Gamma^{\mu}_{\alpha\beta} = \frac{1}{2} g^{\mu\nu}(g_{\alpha\nu,\beta} + g_{\beta\nu,\alpha} - g_{\alpha\beta,\nu}).$$

Roughly speaking, the Christoffel symbol $\Gamma^{\mu}_{\alpha\beta}$ denotes the αth component of the derivative of a local basis vector field E_β in the direction of E_α.

Definition 5.2 (Levi-Civita Connection)**.** *Let (M, g) be a Lorentzian manifold, where g is a Lorentzian metric on the smooth manifold M. Then there exists a unique connection, ∇, on M, called the* ***Levi-Civita connection*** *on M defined by*

$$\nabla_{X^\alpha \partial_{x^\alpha}}(Y^\beta \partial_{x^\beta}) = X^\alpha \partial_{x^\alpha} Y^\beta \partial_{x^\beta} + \Gamma^{\mu}_{\alpha\beta} X^\alpha Y^\beta \partial_{x^\mu},$$

that is ***compatible*** *with g (in the sense that $\nabla g = 0$) and is* ***torsion-free*** *(in the sense that $\nabla_X Y - \nabla_Y X - [X, Y] = 0$).*

Definition 5.3 (Riemann Curvature Tensor)**.** *Let (M, g) be a Lorentzian manifold, where g i a Lorentzian metric on M. Then we define the* ***Riemann curvature tensor*** *on M to be the $(1,3)$-tensor field given by*

$$R^{\gamma}{}_{\alpha\mu\nu}\partial_\gamma = \nabla_\mu \nabla_\nu \partial_\alpha - \nabla_\nu \nabla_\mu \partial_\alpha.$$

We can also think of the Riemann curvature tensor as a $(0,4)$-tensor field obtained by lowering an index:

$$R_{\gamma\alpha\mu\nu} = g_{\gamma\eta} R^{\eta}{}_{\alpha\mu\nu}.$$

The Riemann curvature tensor has the following properties (of which we omit the proofs).

Proposition 5.1 (Properties of the Riemann Curvature Tensor)**.** *Let (M, g) be a Lorentzian manifold, and let R denote the Riemann curvature tensor on M. Then R satisfies the following properties.*

(i) ((Anti)Symmetricity) $R_{\mu\nu\alpha\beta} = R_{\alpha\beta\mu\nu} = -R_{\nu\mu\alpha\beta} = -R_{\mu\nu\beta\alpha}$.
(ii) (Algebraic Bianchi Identity) $R_{\mu\nu\alpha\beta} + R_{\alpha\mu\nu\beta} + R_{\nu\alpha\mu\beta} = 0$.
(iii) (Differential Bianchi Identity) $\nabla_\sigma R_{\mu\nu\alpha\beta} + \nabla_\mu R_{\nu\sigma\alpha\beta} + \nabla_\nu R_{\sigma\mu\alpha\beta} = 0$.

We also have the *Ricci curvature tensor*, which is a symmetric $(0,2)$-tensor field.

Definition 5.4 (Ricci Curvature Tensor)**.** *Let (M, g) be a Lorentzian manifold, and let the* ***Ricci curvature tensor****, $\mathrm{Ric}(g)$, on M be the $(0,2)$-tensor field on M obtained by taking the trace over 2 indices of the Riemann curvature tensor as follows:*

$$\mathrm{Ric}(g)_{\mu\nu} = g^{\alpha\beta} R_{\mu\alpha\nu\beta}.$$

In the homework, we showed that in local coordinates, $\mathrm{Ric}(g)$ can be written in the form,

$$\mathrm{Ric}(g)_{\mu\nu} = -\frac{1}{2} g^{\alpha\beta} \partial^2_{\alpha\beta} g_{\mu\nu} - \frac{1}{2} g^{\alpha\beta} \partial^2_{\mu\nu} g_{\alpha\beta} + \frac{1}{2} g^{\alpha\beta} \partial^2_{\alpha\nu} g_{\beta\mu} + \frac{1}{2} g^{\alpha\beta} \partial^2_{\beta\mu} g_{\alpha\nu} + F_{\mu\nu}(g, \partial g),$$

where $F_{\mu\nu}(g, \partial g)$ is a function of g and its derivatives. In particular, if only the first term remains, then the PDE (i.e., the Einstein vacuum equations) corresponding to $\mathrm{Ric}(g)_{\mu\nu} = 0$ becomes the wave equation so that the local existence theorem for quasilinear systems can be applied. Indeed, one has the following result: under certain (i.e., *wave coordinate* conditions), the Einstein vacuum equations turn into a system of nonlinear wave equations. In particular, the Ricci curvature tensor can be expressed as the sum of a quasilinear wave operator and two terms that consist of the first derivatives of the metric. We formalize this result now.

Definition 5.5 (Wave Coordinate Condition)**.** *Let (M, g) be a Lorentzian manifold. A local coordinate system (x^μ) is said to satisfy the* ***wave coordinate condition*** *on M if the following is true,*

$$0 = \Box_g x^\alpha \coloneqq \frac{1}{\sqrt{-g}} \partial_\mu \partial_\mu (g^{\mu\nu} \sqrt{-g} \partial_\nu x^\alpha).$$

Proposition 5.2. *Let (M, g) be a Lorentzian manifold, and let $\mathrm{Ric}(g)$ be the Ricci curvature tensor on M. Then*

$$\mathrm{Ric}(g)_{\mu\nu} = \widetilde{\mathrm{Ric}}(g)_{\mu\nu} + \frac{1}{2}\partial_\mu \lambda_\nu + \frac{1}{2}\partial_\nu \lambda_\mu,$$

where $\widetilde{Ric}(g)_{\mu\nu}$ is the ***reduced Ricci curvature tensor*** *defined by*

$$\widetilde{Ric}(g)_{\mu\nu} = -\frac{1}{2} g^{\alpha\beta} g_{\mu\nu,\alpha\beta} + \frac{1}{2} g^{\beta\rho} g_{\sigma\rho,\mu} g_{\alpha\nu,\beta} + \frac{1}{2} g^{\alpha\sigma} g^{\beta\rho} g_{\sigma\rho,\nu} g_{\beta\mu,\alpha} - \frac{1}{2} g^{\alpha\sigma} g^{\beta\rho} g_{\sigma\rho,\mu} g_{\alpha\beta,\nu} + F_{\mu\nu}(g, \partial g).$$

Proof. Recall the matrix identities,

$$\partial_\alpha g^{\mu\nu} = -g^{\mu\beta} g^{\sigma\nu} \partial_\alpha g_{\sigma\beta} \qquad \text{and} \qquad \partial_\alpha (\log |\det(g)|) = g^{\mu\nu} \partial_\alpha g_{\mu\nu}.$$

[!! Complete Later !!] □

Thus if the wave condition holds, then the Einstein vacuum equations reduce to a nonlinear system of wave equations, for which we have the general local existence theorem from above. But we must ensure that the wave condition actually holds. We have the following results to that end.

Lemma 5.1 (Ricci Identity)**.** *Let (M, g) be a Lorentzian manifold, and let $\mathrm{Ric}(g)$ be the Ricci curvature on M. Then the following identity holds:*

$$\nabla^\mu (\mathrm{Ric}_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R) = 0.$$

Proof. From the differential Bianchi identity, we have

$$\begin{aligned} \nabla^\mu R_{\mu\alpha\nu\beta} &= g^{\mu\sigma} \nabla_\sigma R_{\mu\nu\alpha\beta} = g^{\mu\sigma} \nabla_\alpha R_{\mu\nu\beta\sigma} - g^{\mu\sigma} \nabla_\beta R_{\mu\nu\sigma\alpha} \\ &= \nabla_\alpha \mathrm{Ric}_{\beta\nu} - \nabla_\beta \mathrm{Ric}_{\alpha\nu}. \end{aligned}$$

Thus taking the trace over α, ν, we get the above identity. □

Proposition 5.3 (Wave Equation)**.** *Given a Lorentzian metric g such that the reduced Einstein vacuum equations are satisfied, i.e., $\mathrm{Ric}(g) = 0$, λ satisfies a wave equation*

$$\frac{1}{2} g^{\sigma\mu} \partial^2_{\sigma\mu} \lambda_\nu + c_\nu^{\alpha\beta} \partial_\alpha \lambda_\beta = 0,$$

where $c_\nu^{\alpha\beta}$ are smooth functions of g and its derivatives.

From this result, we have the following theorem of Choquet-Bruhat.

Theorem 5.1 (Local Well-Posedness of Einstein Vacuum Equations)**.** *Given initial data $(\hat{g}_{ij}, \hat{k}_{ij})$ on $\mathbb{R}^n$ satisfying the constraint equations,*

$$\begin{aligned} \hat{\nabla}_i \hat{k}^i_j - \hat{\nabla}_j \hat{k}^i_j &= 0, \\ \hat{R}(\hat{g}) + (\hat{k}^i_i)^2 - \hat{k}^i_j \hat{k}^j_i &= 0, \end{aligned} \tag{29}$$

and such that $\sum_{i,j} |\hat{g}_{ij} - \delta_{ij}| \le \frac{1}{20}$, there exists a metric g in $I \times \mathbb{R}^n$ which solves the Einstein vacuum equations such that the induced metric and the induced second fundamental form on $\{0\} \times \mathbb{R}^n$ coincide with $\hat{g}_{ij}$ and $\hat{k}_{ij}$, respectively.

Part III
Applications

In this part of the notes, we transition from the general theory of (non)linear wave equations and hyperbolic PDE to some physical applications. In particular, we will mostly apply the results from the previous part to understanding the mathematical theory of (in)compressible fluids.

6 Incompressible Fluids

In this section, we introduce the mathematical theory of incompressible fluids. [6].

6.1 Introduction to the Euler Equation

In this subsection, we introduce the **Euler equations** for ideal, incompressible fluids. In general, we will work in $\mathbb{R}^3$ unless stated otherwise. We start with a definition of *incompressible displacements*.

Definition 6.1 (Incompressible Displacements)**.** *Let $\Omega \subseteq \mathbb{R}^3$ be an open and bounded region with sufficiently regular boundary $\partial\Omega$. Further assume that Ω contains a fluid. An **incompressible displacement** of the fluid is a transformation $\sigma : \Omega \to \Omega$ such that*

(i) σ is invertible, and $\sigma(D) = D$;
(ii) $\sigma, \sigma^{-1} \in C^1(\Omega)$ (i.e., σ is a diffeomorphism on Ω),
(iii) σ preserves the Lebesgue measure (in the sense that $m(\sigma(A)) = m(A)$, where m is the 3-dimensional Lebesgue measure, and $A \subseteq \Omega$).

Remark 6.1. *Point (iii) makes incompressibility explicitly clear to see; intuitively, a subvolume A of fluid in Ω cannot be compressed or stretched (which could change the Lebesgue measure).*

In particular, it is clear to see that the set S of all incompressible displacements on D forms a group under function composition as the product. Using this as a guide, we define **incompressible motion** as follows.

Definition 6.2 (Incompressible Motion)**.** *Let $\Omega \subseteq \mathbb{R}^3$ be an open and bounded region with sufficiently regular boundary $\partial\Omega$. Further assume that Ω contains a fluid. An **incompressible motion** is a function $s, t \in \mathbb{R} \mapsto \Phi_{s,t} \in S$ such that*

(i) $\Phi_{s,t}(\Phi_{t,r}(x)) = \Phi_{s,r}(x)$;
(ii) $\Phi_{t,s}(\Phi_{s,t}(x)) = x$;
(iii) $\Phi_{s,t}$ is continuously differentiable in t and s.

6.1.1 Derivation of the Fluid Equations

Using the above results, we derive the PDEs satisfied by systems of *ideal*, incompressible fluids. We begin with the equation for the *density field*. Let $\Omega \subseteq \mathbb{R}^3$ be an open and bounded region with sufficiently regular boundary $\partial\Omega$. Further assume that Ω contains a fluid. Let $\rho = \rho(x,t) \in C^1(\Omega)$ denote the *density fluid* of the fluid (the mass of the fluid contained in the element of volume dx at time t is $\rho(x,t)\,dx$). Fix some subregion $V_0 \subset \Omega$, and let

$$V_t \coloneqq \{\Phi_t(x) : x \in V_0\}.$$

Note, we define $\Phi_t(x) \coloneqq \Phi_{t,0}(x)$. Also let $u(\Phi_t(x), t) = \frac{d}{dt}\Phi_t(x)$ denote the velocity field associated with this incompressible motion. Conservation of mass tells us that

$$\frac{d}{dt}\int_{V_t} \rho(x,t)\,dx = 0.$$

Thus making a "change of variables", we have that

$$0 = \frac{d}{dt}\int_{V_t} \rho(x,t)\,dx = \frac{d}{dt}\int_{V_0} \rho(\Phi_t(x),t)J_t(x)\,dx = \frac{d}{dt}\int_{V_0} \rho(\Phi_t(x),t)\,dx,$$

where $J_t(x)$ is the Jacobian of the transformation $x \mapsto \Phi_t(x)$. Since Φ_t is an incompressible motion, the Jacobian must identically be one. Passing the time derivative inside the integral (which we can do because of the assumed regularity of ρ and boundedness of V_0), we then obtain

$$0 = \frac{d}{dt}\int_{V_0} \rho(\Phi_t(x),t)\,dx = \int_{V_0} \frac{\partial}{\partial t}\rho(\Phi_t(x),t)\,dx = \int_{V_0} \partial_t\rho(\Phi_t(x),t) + u\cdot\nabla\rho(\Phi_t(x),t)\,dx.$$

Since V_0 was arbitrary, we obtain the PDE,

$$(\partial_t + u\cdot\nabla)\rho(\Phi_t(x),t) = 0,$$

which is identically the linear first-order transport equation; we call this equation the *continuity equation*. For the most part, however, we will assume the density to be constant ($\rho \equiv 1$), so that the above equation is not immediately useful. Instead, we have the more important variation of the *continuity equation*.

Proposition 6.1 (Continuity Equation for Incompressible Flows)**.** *Let $\Omega \subseteq \mathbb{R}^3$ be an open and bounded region containing a fluid. Let V_0 be a fixed subregion in Ω, and let $V_t = \{\Phi_t(x) : x \in V_0\}$, where $\Phi_t = \Phi_{t,0} \in S$ is an incompressible motion. Let $u(\Phi_t(x),t) = \frac{d}{dt}\Phi_t(x)$ denote the velocity field for the motion. Then the* ***continuity equation*** *is given by*

$$\operatorname{div}(u(x,t)) = \nabla\cdot u(x,t) \qquad (\forall x \in \Omega, t \in \mathbb{R}). \tag{30}$$

Hereafter in this subsubsection, we will assume that $u \in C^1(\Omega\times\mathbb{R})$ and $u(t,\cdot) \in C(\bar{\Omega})$. So far, we have only given a PDE for u, but have not yet assigned any additional constraints. Recall that for PDEs describing physical systems, boundary conditions tell us how the system interacts with the boundaries of its environment. Thus we are interested in giving u a boundary condition based on its interaction with $\partial\Omega$. By the definition of incompressible flow, we assume that no fluid particle can cross $\partial\Omega$. Thus if $v(x)$ denotes the velocity of the boundary $\partial\Omega$ at x, we must have $u(x,t)\cdot n = v(x)\cdot n$, where n is the outward unit normal to $\partial\Omega$. Thus once we have the velocity field $u(x,t)$, the trajectories $\Phi_t(x)$ can be uniquely determined.

6.1.2 Equations of Motion for an Incompressible Fluid

In the previous subsubsection, we studied how the density, and more importantly, the velocity fields behaved. In doing so, we wrote the continuity equation, which is a statement about mass conservation. If we assume the density is constant (as we shall), then this equation enables us to uniquely construct the incompressible motions for the fluid. In this subsubsection, we turn to the actual fluid. In particular, we will derive the *Euler equations* for the fluid. We make some simplifying assumptions for our model.

Definition 6.3 (Ideal Fluids)**.** *Let $\Omega \subseteq \mathbb{R}^3$ be an open and bounded region containing a fluid. We assume that the fluid is* ***ideal****, or* ***perfect****, in the sense that the only interaction between the fluid particles is that induced by our incompressibility condition (namely that two particles may not simultaneously occupy the same position). I.e., we will ignore any interactive forces (e.g., electromagnetic forces) between two distinct fluid particles.*

To derive the equations of motion, we will use the *Principle of Stationary Action*. The Lagrangian for the system is the kinetic energy,

$$\mathcal{L} = \frac{1}{2}\int_\Omega \left(\frac{d}{dt}\Phi_t(x)\right)^2 dx.$$

Thus, the action is given by

$$A(\Phi; t_1, t_2) \coloneqq \int_{t_1}^{t_2}\int_\Omega \left(\frac{d}{dt}\Phi_t(x)\right)^2 dx dt.$$

This defines a functional $\Phi \mapsto A(\Phi; t_1, t_2)$ on the space M of all incompressible motions. To determine a physical law, we will consider variations in this class M as follows: we require that the action be stationary for variations $\Phi \mapsto \Phi + \delta\Phi$ which are compatible with incompressiblity, and to satisfy $\delta\Phi_{t_1}(x) = \delta\Phi_{t_2}(x) = 0$ for all $x \in \Omega$. Moreover, we require the variation to satisfy the boundary conditions, $\frac{d}{dt}\Phi_t^\varepsilon(x) \cdot n = 0$, for all $x \in \partial\Omega$ (i.e., the velocity of Ω is assumed to be zero so that no fluid particle can cross the boundary). By Φ^ε, where $\varepsilon \in [0, \varepsilon_0]$, we denote a family of motions tangent to $\partial\Omega$ such that $\Phi^0 = \Phi$, $\Phi^\varepsilon_{t_1} = \Phi_{t_1}$, and $\Phi^\varepsilon_{t_2} = \Phi_{t_2}$. We want the action A to be stationary on Φ, which is to say that we want

$$\frac{d}{d\varepsilon} A(\Phi^\varepsilon; t_1, t_2)|_{\varepsilon=0} = 0.$$

Using the definition of A, it follows that

$$\begin{aligned} 0 = \frac{d}{d\varepsilon} A(\Phi^\varepsilon; t_1, t_2)|_{\varepsilon=0} &= \frac{d}{d\varepsilon} \int_{t_1}^{t_2} \int_\Omega \left(\frac{d}{dt}\Phi_t^\varepsilon(x)\right)^2 \, dxdt \Big|_{\varepsilon=0} \\ &= \int_{t_1}^{t_2} \int_\Omega 2\left(\frac{d}{dt}\Phi_t^\varepsilon(x)\right) \frac{d}{dt}\frac{d}{d\varepsilon}\Phi_t^\varepsilon(x)\Big|_{\varepsilon=0} \, dxdt \\ \implies \int_{t_1}^{t_2} \int_\Omega \frac{d}{dt}\Phi_t(x) \cdot \frac{d}{dt}\gamma_t(\Phi_t(x)) \, dxdt &= 0, \end{aligned}$$

where for $\varepsilon \in [0, \varepsilon_0]$, we define

$$\gamma_t^\varepsilon(\Phi_t(x)) = \frac{d}{d\varepsilon}\Phi_t^\varepsilon(x),$$

and denote $\gamma_t \coloneqq \gamma_t^0$. In other words, γ_t^ε is the vector field transversal to the motion that generates a flow parametrized by ε: $\Phi_t(x) \mapsto \Phi_t^\varepsilon(x)$. Then using integration by parts[10], we obtain the equation $D_t u = -\nabla p$, where $p : \mathbb{R} \times \Omega \to \mathbb{R}$ is some function, and $D_t = \partial_t + (u \cdot \nabla)$ is the *material* or *convective derivative*. Thus we have concluded the derivation of the incompressible Euler equations.

Proposition 6.2 (Incompressible Ideal Euler Equations)**.** *Let $\Omega \subset \mathbb{R}^3$ be an open and bounded region of $\mathbb{R}^3$ containing an incompressible fluid. Let u denote the velocity field of the fluid. Then u satisfies the* ***incompressible ideal Euler equations****,*

$$\begin{cases} \partial_t u + (u \cdot \nabla)u = -\nabla p, \\ \nabla \cdot u = 0, \\ u \cdot n = 0 \qquad (\text{on } \partial\Omega). \end{cases} \tag{31}$$

The equations above are a direct generaliation of the physical equations of motion for incompressible ideal fluids. In particular $D_t u$ can be thought of as the "acceleration" of the fluid, and the scalar function $p = p(x, t)$ as the *scalar pressure* (thus, the first equation in the system is a direct analog of Newton's law). The second equation is a statement of mass conservation; the third equation is a statement of the boundary conditions for our system of PDEs.

6.1.3 Vorticity and Stream Functions

In the previous subsection, we were looking at physical models that describe the motion of a fluid in a bounded open region Ω of $\mathbb{R}^3$. In this subsubsection, we dive deeper and study more specific properties of fluid motion. We start with the definition of *vorticity*.

Definition 6.4 (Vorticity Field)**.** *Let Ω be an open bounded region of $\mathbb{R}^3$ containing an incompressible ideal fluid, and let $u = u(t, x)$ be the velocity field of the fluid. Then we define the* ***vorticity field*** *of u as the vector field,*

$$\omega(t, x) = \operatorname{curl} u(t, x) = \nabla \times u(t, x), \tag{32}$$

where ∇ is taken with respect to the space variables only.

[10] [!! Fill in the details later !!]

Physically, the *vorticity* is a measure of how the fluid is rotating. We also define the following.

Definition 6.5 (Deformation Tensor)**.** *Let $\Omega \subset \mathbb{R}^3$ be an open and bounded region containing an incompressible ideal fluid, and let u denote the velocity field of the fluid. Let D be the matrix defined as*

$$D = \frac{1}{2}(\nabla u + (\nabla u)^{\perp}).$$

*Then D is said to be the **deformation tensor**.*

Proposition 6.3 (Local Motion of a Fluid)**.** *Let $\Omega \subset \mathbb{R}^3$ be an open and bounded region containing an incompressible ideal fluid, and let u denote the velocity field of the fluid. Then*

$$u(x+h,t) = u(x,t) + D \cdot h + \frac{1}{2}\omega(x,t) \times h + O(h^2), \tag{33}$$

where D is the deformation tensor.

I.e., locally, the velocity field of the fluid consists of three motions: (1) a translation motion, (2) a rotation with angular velocity $\frac{1}{2}\omega$, and (3) a deformation. One can see why D is called the **deformation tensor** by considering rigid motion, in which case, $D = 0$, and the angular velocity of the fluid is exactly $\Omega = \frac{1}{2}\omega$.

Constructing Solution to 2D Euler Equations from Vorticity Suppose that the vorticity $\omega = \omega(x) \in C(\Omega)$ is a given vector field, where $\Omega \subset \mathbb{R}^2$, and let $u(x)$ denote the velocity field of the fluid. Then u satisfies the system,

$$\begin{cases} \nabla \times u = \omega, \\ \nabla \cdot u = 0. \end{cases}$$

Thus we wish to know under what circumstances we can deduce u given ω. In general, even if a solution exists to the above system, it may not necessarily be unique. For instance, consider a solution u', and let $u = u' + \nabla\varphi$, where φ is a harmonic function. Then clearly u also satisfies the above system as does u'. Thus to force uniqueness, we must at least introduce some boundary conditions. Thus the motivation for the following section is to consider various cases for Ω and its consequences on solutions to the system above.

(1) ($\Omega \subset \mathbb{R}^2$**;** Ω **is simply connected and bounded**): Since Ω is simply connected and bounded, and u is divergence free, there exists[11] a function Ψ, called the **stream function**, such that $u = \nabla^{\perp}\Psi$, where $\nabla^{\perp} \coloneqq (\partial_{x_2}, -\partial_{x_1})$. Then we compute,

$$\Delta\Psi = \partial_{x_1}^2\Psi + \partial_{x_2}^2\Psi = \partial_{x_1}(-u_2) + \partial_{x_2}(u_1) = \partial_{x_2}(u_1) - \partial_{x_1}(u_2).$$

On the other hand, we compute, $\omega = \nabla \times u = \partial_{x_1}u_2 - \partial_{x_2}u_1$. Thus we obtain the Poisson equation, $\Delta\Psi = -\omega$. Since $u \cdot n = 0$ on ∂D, Ψ must be a constant on $\partial\Omega$. Thus without loss of generality, we may assume that $\Psi|_{\partial D} = 0$. Assuming ω has sufficient regularity, it then follows that there exists a unique solution Ψ satisfying $\Delta\Psi = -\omega$ and $\Psi|_{\partial\Omega} = 0$. Thus this gives us a solution for u. Now we establish uniqueness of this solution. Assume u' is another solution to the above system. Then $v = u - u'$ satisfies

$$\nabla \cdot v = 0 \qquad \text{and} \qquad \nabla \times v = 0.$$

Since Ω is simply connected, the curl-free condition on v implies that it is the gradient of a smooth function. Thus v satisfies $v = \nabla\varphi$ and $v \cdot n = 0$ on $\partial\Omega$. Taking the divergence of this relation, one then gets the condition $\Delta\varphi = 0$ and $\partial\varphi/\partial n = 0$ on ∂D, whence (by integration by parts), φ is a constant. Thus $v = 0$, from which $u = u'$.

(2) ($\Omega = \mathbb{R}^2$): In this case, we assume that $\Omega = \mathbb{R}^2$, which is simply connected, but not bounded. However, the initial analysis for the stream function remains the same; namely, we may introduce a stream function Ψ such that $u = \nabla^{\perp}\Psi$ and $\Delta\Psi = -\omega$, where ω is the vorticity of the fluid. Unlike in the

[11] via the Poincaré Lemma

previous example where we were simply able to exploit boundedness of Ω, integration by parts, and the theory of solutions to the Poisson equation, we must do some extra analysis in the unbounded case of $\mathbb{R}^2$. To do so, we will use the method of *Green functions*.

Let $G = G(x, x')$ be the fundamental solution of the Poisson equation,

$$\Delta_x G(x, x') = -\delta(x - x');$$

an explicit representation of G is given by [1],

$$G(x, x') = -\frac{1}{2\pi} \ln |x - x'|.$$

Thus, one has

$$\Psi(x) = \int G(x, x')\omega(x')\, dx'.$$

This then implies that

$$u(x) = \nabla^\perp \Psi(x) = \int K(x, x')\omega(x')\, dx',$$

where we define the velocity kernel,

$$K(x, x') = -\frac{1}{2\pi} \frac{(x - x')^\perp}{(x - x')^2}.$$

Intuitively, $K(x, x')$ denotes the velocity field (in x) generated by a point charge of intensity one fixed in x'. Now we are interested in uniqueness of solutions. Consider again v, which is the difference of two solutions. Then $v = \nabla\varphi$ for some harmonic function φ. Suppose we fix the asymptotic behavior $u(x) \to u_\infty \in \mathbb{R}^2$ as $|x| \to \infty$. Then the unique harmonic function $\nabla\varphi \to u_\infty$ has the general form $\varphi = u_\infty \cdot x + \text{const.}$ so that the solution to the problem is given by

$$u(x) = \nabla^\perp \Psi(x) + u_\infty.$$

6.1.4 Conservation Laws

As we did for the theory of linear wave equations, we start by assuming that solutions to the incompressible Euler equations exist, and then showing general properties of these solutions. Likewise, as in the previous part of these notes, we begin with an exposition on the *energy*. We start with the following definition

Definition 6.6 (Energy)**.** *Let $\Omega \subset \mathbb{R}^2$ be an open region containing an incompressible ideal fluid, and let $u \in C^1(\Omega \times [0, T])$ be the velocity field of the fluid. Then we define the* ***energy*** *of the fluid by*

$$E(t) \coloneqq \frac{1}{2} \int_\Omega u^2(t)\, dx. \tag{34}$$

Since the fluid is assumed ideal, we have the following energy conservation law.

Theorem 6.1 (Energy Conservation Law)**.** *Let $\Omega \subset \mathbb{R}^3$ be a bounded domain, and let u be a solution of the Euler equation with conservative external forces*

$$D_t u = -\nabla(p + V),$$

where $V = V(x, t)$ is a known function. Then $\dot{E}(t) = 0$.

Proof. Since Ω is a bounded domain, and u is continuous on $\Omega \times [0, T]$ (so that it is continuous on Ω), one has that u and $\partial_t u$ is integrable over Ω. Thus,

$$\dot{E}(t) = \int_\Omega u \cdot \partial_t u = -\int_\Omega \{u \cdot (u \cdot \nabla)u + u \cdot \nabla P\},$$

where $P \coloneqq p + v$. Next, since $\operatorname{div}(Pu) = u \cdot \nabla P + P \nabla \cdot u$, and u was assumed to be divergence-free,

$$\int_\Omega u \cdot \nabla P = \int_\Omega \operatorname{div}(Pu) = \int_{\partial\Omega} (Pu) \cdot n.$$

But since $u \cdot n = 0$ on $\partial\Omega$, it follows that the above integral is zero. Thus it suffices to show that the first term is also zero. We compute,

$$\begin{aligned}
\int_\Omega u \cdot (\nabla \cdot u) u &= \sum_i \int_\Omega u_i u \cdot \nabla u_i \\
&= \sum_i \int_\Omega \nabla \cdot \left(\frac{1}{2} u_i^2 u \right) - \frac{1}{2} u_i^2 (\nabla \cdot u) \\
&= \sum_i \int_{\partial\Omega} \frac{1}{2} u_i^2 u \cdot n - \sum_i \frac{1}{2} \int_\Omega u_i^2 (\nabla \cdot u) = 0,
\end{aligned}$$

where we used the fact that u is divergence-free, $u \cdot n = 0$ on $\partial\Omega$, and the divergence theorem. □

Now we introduce a special type of flow, called *stationary flows*.

Definition 6.7 (Stationary Flows)**.** *Let $\Omega \subset \mathbb{R}^3$ be a bounded domain containing a fluid, and let u be a solution to the Euler equations in Ω. Then u is said to be a **stationary** or **steady flow** if $u = u(x)$ is independent of time.*

For such flows, we have the following theorems of Bernoulli.

Theorem 6.2 (Bernoulli I)**.** *In an ideal fluid in stationary motion under the action of a conservative force with potential V independent of time, the quantity*

$$\mathcal{E} = \frac{1}{2} u^2 + (p + V) \tag{35}$$

is constant along the flow lines.

Theorem 6.3 (Bernoulli II)**.** *Consider an ideal fluid in a stationary irrotational motion in a domain D, under the action of a conservative force with potential V independent of time. Then the quantity,*

$$\mathcal{E} = \frac{1}{2} u^2 + (p + V) \tag{36}$$

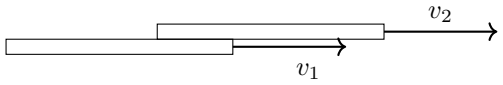
is constant.

While the two theorems *look* similar, the corresponding hypotheses are not identical. In the first Bernoulli theorem, we consider merely a stationary solution to the Euler equations. Under these hypotheses, while the quantity $\mathcal{E}$ is constant along *each* flow line, the quantity may vary from one stream to another. On the other hand, the second Bernoulli theorem informs us that if the stationary solution is also *irrotational*, then the quantity $\mathcal{E}$ is independent of the flow line. The two theorems imply the following consequences.

Corollary 6.1 (Bernoulli III)**.** *Consider an ideal fluid in stationary (irrotational) motion in a domain Ω, independent of any external forces. Then the velocity is smallest when the pressure is the greatest, and vice-versa.*

Corollary 6.2 (Bernoulli IV)**.** *Consider an ideal fluid in stationary (irrotational) motion in a domain Ω, independent of any external forces, where Ω is assumed to be a "pipeline". Then the velocity is greater (thus pressure is smaller) when the pipeline has a smaller section.*

Corollary 6.3 (Stevino)**.** *Consider a fluid at rest in the presence of a gravitational field $V = -gx_3$. Then $p = gx_3$.*



Now we shift to studying conservation laws dealing with the vorticity of fluids. The physical picture is as follows. Suppose that we have two layers of fluids with different velocities, stacked on top of each other.

Since the fluid was assumed to be ideal (i.e., no interaction between the fluid particles outside of the incompressiblity clause), it is not possible for the two layers to transfer momentum between each other, and thus must move independently. In particular, this prevents the fluid from producing rotation on its own, and ensures that the fluid has no means to dissipate energy through rotation/friction. The following theorem of Kelvin formalizes the result using the notion of *circulations*.

Theorem 6.4 (Kelvin)**.** *Let $\Omega \subset \mathbb{R}^3$ be a bounded region containing an ideal fluid with velocity field u. Let C_0 be a closed curve in Ω at $t = 0$, and let $C_t = \Phi_t(C_0)$. Define the* ***circulation*** *as the quantity,*

$$\Gamma(C_t) = \oint_{C_t} u(t)\, dl. \tag{37}$$

Then

$$\frac{d}{dt}\Gamma(C_t) = 0. \tag{38}$$

Proof. The proof of Kelvin's theorem will follow in two steps. First, we will prove that $\frac{d}{dt}\oint_{C_t} u \cdot dl = \oint_{C_t} D_t u \cdot dl$. Next, we will use Euler's equations in conjunction with the above result to establish the theorem.

(1) Let $[0,1] \ni s \mapsto x(s) \in C_0$ be a parametrization of the initial curve C_0. Then the map $[0,1] \ni s \mapsto \Phi_t(x(s))$ is a parametrization of C_t. Then we compute,

$$\begin{aligned}\frac{d}{dt}\oint_{C_t} u \cdot dl &= \frac{d}{dt}\int_0^1 u(\Phi_t(x(s),t),t) \cdot \frac{\partial}{\partial s}\Phi_t(x(s))\, ds \\ &= \int_0^1 D_t u(\Phi_t(x(s)),t) \cdot \frac{\partial}{\partial s}\Phi_t(x(s))\, ds + \int_0^1 u(\Phi_t(x(s)),t) \cdot \frac{\partial}{\partial t}\frac{\partial}{\partial s}\Phi_t(x(s))\, ds.\end{aligned}$$

The final integral can be evaluated as follows:

$$\begin{aligned}\int_0^1 u(\Phi_t(x,s),t) \cdot \frac{\partial}{\partial t}\frac{\partial}{\partial s}\Phi_t(x(s))\, ds &= \int_0^1 u(\Phi_t(x,s),t) \cdot \frac{\partial}{\partial s}\frac{\partial}{\partial t}\Phi_t(x(s))\, ds \\ &= \int_0^1 u(\Phi_t(x(s)),t) \cdot \frac{\partial}{\partial s} u(\Phi_t(x(s)),t)\, ds \\ &= \frac{1}{2}\int_0^1 \frac{\partial}{\partial s} u^2(\Phi_t(x(s)),t)\, ds \\ &= 0,\end{aligned}$$

where the second line follows from the definition of the velocity line in terms of Φ_t, and the final line follows from the fact that since C_0 is a closed curve, $x(0) = x(1)$. Thus, we have shown the first claim.

(2) Using the above result, since $D_t u = -\nabla(p + V)$, where V is the potential for a conservative external force, we have

$$\frac{d}{dt}\Gamma(C_t) = -\oint_{C_t} \nabla(p + V) \cdot dl = 0.$$

□

Corollary 6.4 (Constant Vorticity Flux)**.** *Let $\Sigma_t \subset \Omega$ be a surface that moves with the fluid whose base is a closed curve C_t, n the unit normal vector field to the surface, and ω the vorticity field. If $d\sigma$ denotes the surface area element on Σ_t, then*

$$\frac{d}{dt}\int_{\Sigma_t} \omega \cdot n\, d\sigma = 0.$$

We can say more about the vorticity. First consider the following definitions.

Definition 6.8 (Vorticity Line)**.** *A **vorticity line** is a curve in Ω that is tangent at every point to the vorticity field ω.*

Definition 6.9 (Vortex Tube)**.** *Let C be a closed curve in Ω that is not tangent to the vorticity field at any point. Then the set of all vorticity lines which cross C transversally in each of its points is called the **vortex tube**.*

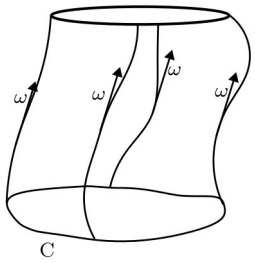


Figure 1: The lines in the above figure illustrate the vortex tube for C.

Theorem 6.5 (Helmholtz)**.** *Let C_1 and C_2 be two arbitrary cures encircling the same vorticity tube. Then*

$$\Gamma = \oint_{C_1} u \cdot dl = \oint_{C_2} u \cdot dl. \tag{39}$$

*Γ is called the **strength** of the vortex tube. By Kelvin's theorem, Γ is constant during the motion.*

Proof. Let C_1 and C_2 be two arbitrary closed curves, S the surface of the vortex tube enclosed by the two curves, and S_1, S_2 the surfaces whose borders are given by C_1 and C_2, respectively. Let $\Sigma = S \cup S_1 \cup S_2$. Since we assumed u was sufficiently regular, and ω is the curl of u, $\nabla \cdot \omega = 0$. Hence if V denotes the volume enclosed by Σ, then

$$\begin{aligned} 0 &= \int_V \nabla \cdot \omega \, dx = \int_\Sigma \omega \cdot n \, d\sigma \\ &= \int_{S_1 \cup S_2} \omega \cdot n \, d\sigma + \int_S \omega \cdot n \, d\sigma. \end{aligned}$$

Since by definition ω is tangent to S, the last integral identically vanishes. Applying consistent orientations on the boundary curves C_1 and C_2 of S_1 and S_2, respectively, and using Stokes' theorem, we obtain the desired conclusion. □

Conservation Laws when $\Omega = \mathbb{R}^3$ Now we consider conservation laws when $\Omega = \mathbb{R}^3$. We shall assume $u \in C^1 \cap L^\infty(\mathbb{R}^3 \times [0,T])$ is a solution to the Euler equations with *finite* energy (so that $u(x,t) \in L^2(\mathbb{R}^3)$ for all (x,t)). Moreover, we also assume that $\omega \in C^1 \cup L^2(\mathbb{R}^3 \times [0,T])$. Then we make the following definition.

Definition 6.10 (Helicity)**.** *For any $t \in [0,T]$, for an ideal fluid in $\mathbb{R}^3$ with the above properties, we define the **helicity** of the fluid to be the quantity*

$$A(t) = \int_{\mathbb{R}^3} u(x,t) \cdot \omega(x,t) \, dx. \tag{40}$$

Intuitively, the helicity measures the extent to which vortex lines "wrap" and "twist" around the vortex tube.

Theorem 6.6 (Helicity as an Invariant Factor)**.** *In an ideal fluid under the action of conservative forces*

$$\frac{d}{dt} A = 0.$$

In particular, we have additional invariant factors of ideal fluids under the action of conservative forces, which include

- **Net Vorticity:** $\Omega = \int_{\mathbb{R}^3} \omega \, dx$;
- **Momentum:** $2\int_{\mathbb{R}^3} x \times \omega \, dx$;
- **Angular Momentum:** $-2\int_{\mathbb{R}^3} |x|^2 \omega \, dx$.

6.1.5 Potential and Irrotational Flows

In this subsubsection, we investigate the properties of specific solutions to the Euler equations, namely irrotational ideals fluids. We start with the following definition.

Definition 6.11 (Potential Flow)**.** *A solution $u = u(x,t)$, to the Euler equations is called a* ***potential flow*** *if there exists some sufficiently regular function $\varphi(x,t)$ such that $u(x,t) = \nabla\varphi(x,t)$.*

Since under the right conditions of regularity, the curl of $u = \nabla\varphi$ is identically zero, u is irrotational. Thus every potential flow is irrotational. However, the converse might not necessarily be true for nonsimply connected domains. We begin with a qualitative analysis of irrotational flows.

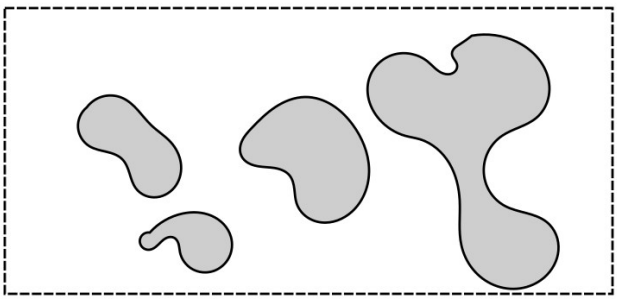
Proposition 6.4 (Irrotational Flows on Bounded, Simply Connected Domains)**.** *On a bounded, simply connnected domain Ω, all the irrotational flows reduce to the trivial one: $u = 0$.*

Proof. The proof of this statement is essentially straightforward. Since u is divergence free, and $u = \nabla\varphi$, one has that $\Delta\varphi = 0$. And since $u \cdot n = 0$ on $\partial\Omega$, $\partial u/\partial n = 0$ on $\partial\Omega$. Thus, $\varphi = \text{const.}$, whence $u \equiv 0$. Gometrically, this corresponds to the idea that it is impossible to extend to the interior of a bounded simply connected domain Ω, a vector field tangent to $\partial\Omega$ in a smooth, divergence-free, curl-free way. □

Therefore, if we wish to study *nontrivial* irrotational fields, we must either consider nonsimply connected or unbounded regions. In particular, we are interested in the notion of *external domains*.

Definition 6.12 (External Domains)**.** ***External domains*** *are defined as the complement of a finite union of simply connected bounded regions (e.g., obstacles).*

The consequences of such domains can significantly vary based on the dimension of the ambient space. E.g., consider the region depicted below. In $\mathbb{R}^2$, the external domain is not simply connected so that irrotational



flows are not necessarily potential flows. On the other hand, in $\mathbb{R}^3$, the external domain is simply connected. So any irrotational flow u is a potential flow: $u = \nabla\varphi$, where φ satisfies the boundary condition $\partial u/\partial n = 0$ on $\partial\Omega$. Thus if we specify the asymptotic behavior $\lim_{|x|\to\infty} u(x) = U_\infty$, the uniqueness theorem for Neumann problems gives us a unique potential flow up to a constant.

Now let us return back to the case of $\mathbb{R}^2$ by considering the external domain to be the space $\Omega = \{x \in \mathbb{R}^2 : x > 1\}$. Let φ be the potential function associated with the velocity field u. We know that as $|x| \to \infty$, the

obstruction becomes negligible so that $\partial\varphi/\partial x \sim 1$ as $|x| \to \infty$, which means that $\varphi \sim x = \rho\cos(\theta)$, where (ρ, θ) are polar coordinates on $\mathbb{R}^2$. Thus consider an initial ansatz of $\varphi(\rho, \theta) = A(\rho)\cos(\theta)$. Since φ must be harmonic, we have the following:

$$\frac{\partial^2}{\partial r^2}(A(\rho)\cos(\theta)) + \frac{1}{r}\frac{\partial}{\partial r}(A(\rho)\cos(\theta)) + \frac{1}{r^2}\frac{\partial^2}{\partial\theta^2}(A(\rho)\cos(\theta)) = 0$$
$$\implies A''(\rho)\cos(\theta) + \frac{1}{r}A'(\rho)\cos(\theta) - \frac{1}{r^2}A(\rho)\cos(\theta) = 0.$$

Thus we obtain the following ODE,

$$\rho^2 A''(\rho) + \rho A'(\rho) - A(\rho) = 0.$$

Solving this ODE, we obtain a solution of the form

$$\varphi(\rho, \theta) = \left(A\rho + \frac{B}{\rho}\right)\cos(\theta).$$

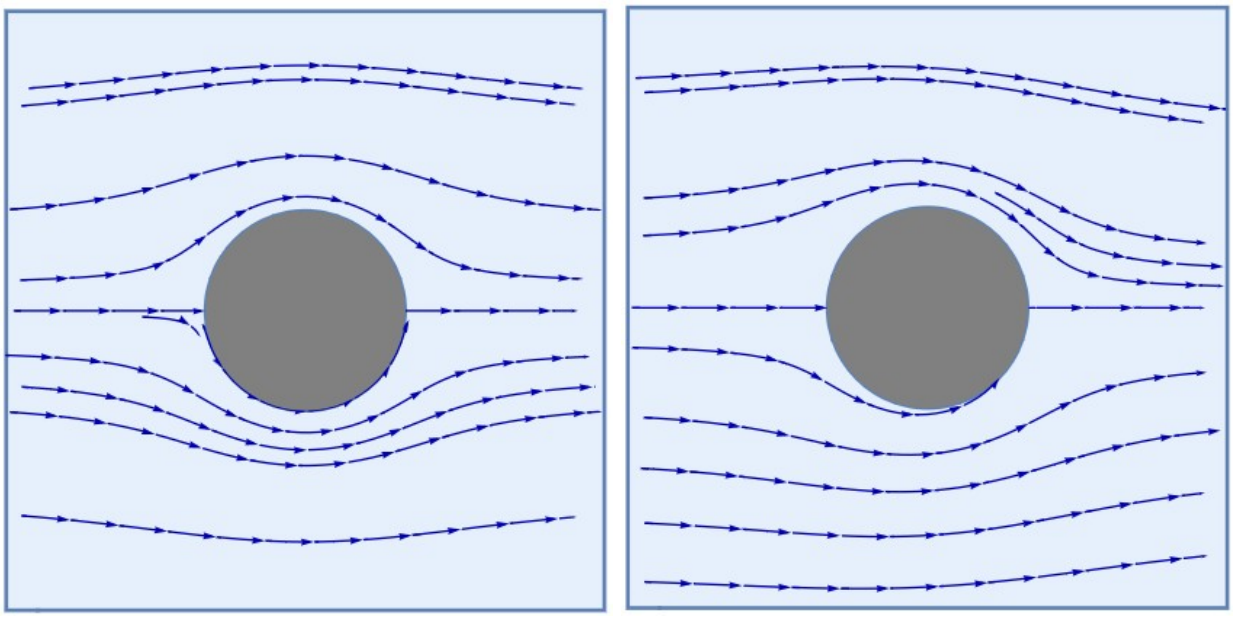
Because of the obstruction at the unit circle, we require $\partial\varphi/\partial\rho = 0$ when $\rho = 1$. We compute,

$$0 = \frac{\partial\varphi}{\partial\rho}\bigg|_{\rho=1} = \left(A - \frac{B}{\rho^2}\right)\cos(\theta)\bigg|_{\rho=1} \iff A - B = 0 \iff A = B.$$

Moreover, we want $\varphi \sim \rho\cos(\theta)$ as $\rho \to \infty$. Thus we get $A = 1$, whence $B = 1$. Therefore, the solution we obtain is

$$\varphi(\rho, \theta) = (\rho + \rho^{-1})\cos(\theta).$$

Figure (2) depicts the associated fluid flow around this obstruction.



(a) Fluid flow in the external domain without circulation.

(b) Fluid flow in the external domain, with circulation.

Figure 2: Fluid flows in external domain consisting of the space outide the unit circle in the plane.

Here, it is clear that the circulation around the obstacle identically vanishes.

Now we can also consider an irrotational flow with circulation α in the same external domain. The corresponding stream function is

$$\Psi(x) = -\frac{\alpha}{2\pi}\ln(|x|).$$

Thus taking the perpendicular gradient, we have the (constant-in-time) velocity field

$$u(x) = \frac{\alpha}{2\pi}\frac{x^{\perp}}{|x|^2}.$$

Then indeed, it is straightforward to check that the circulation around $\partial\Omega$ is α. Then we can consider the sum of the two flows,

$$\varphi(x,y) + \Psi(x,y) = \left(1 + \frac{1}{x^2+y^2}\right)x - \frac{\alpha}{2\pi}\ln(|x|).$$

Then we get the flow depicted in Figure 2.

6.2 Construction of Solutions to the Euler Equations

In this subsection, we study the theory of existence and uniqueness for the Euler equations. Thus far, we have deduced numerous qualitative and quantitative properties of solutions to the Euler equations, *if* they exist. But since the Euler equations were constructed with a specific physical model in mind, it behooves us to determine whether or not (unique) solutions exist (thus informing us if the model is reasonable). Unless otherwise specified, we will assume that our fluid moves in a three-dimensional torus $\Omega = [-\pi, \pi]^3$. First, we will provide some general analysis results/theory that will prove essential. Then we will prove one estimate for a solution to the Euler equations.

6.2.1 Some General Analysis Results

We start with some basic estimates for the Euler equations, inspired by functional analysis. In the space $C^\infty(\Omega)$ of smooth functions on Ω, we define the inner product

$$\langle f, g\rangle_m \coloneqq \sum_{0\le|\alpha|\le m} \int (D^\alpha f)\cdot(D^\alpha g)\, d,$$

where $\alpha = (\alpha_1, \alpha_2, \alpha_3)$ is a multiindex, and $|\alpha| = \alpha_1 + \alpha_2 + \alpha_3$. Thus this inner product defines $C^\infty(\Omega)$ as a **pre-Hilbert space**. However, crucially, $C^\infty(\Omega)$ is *not* complete under the norm induced by the inner product. Thus, completing the space $C^\infty(\Omega)$ under this norm, we obtain the *Sobolev spaces*,

$$H^m(\Omega) \coloneqq W^{m,2}(\Omega) = \{f \in L^2(\Omega) : \|f\|_{H^m(\Omega)} < \infty\},$$

where $\|f\|_{H^m(\Omega)} \coloneqq \langle f, g\rangle_m^{1/2}$. We can also extend this definition to vector fields in the natural way: given two vector fields $u = \{u_j\}_1^3$ and $v = \{v_j\}_1^3$, we define

$$\langle u, v\rangle_m \coloneqq \sum_{j=1}^3 \langle u_j, v_j\rangle_3 .$$

Thus by $\mathcal{H}^m$, we denote the Hilbert space consisting of all vector fields whose components are in H^m. Now we make note of the following inequalities from functional analysis.

Proposition 6.5 (Some Functional Analysis Estimates)**.** *For $f, g \in C^\infty(\Omega)$, the following estimates hold:*

(i) $\|f\|_{L^\infty} \le C_m \|f\|_{H^m}$ *if* $m \ge 2$ *(Sobolev Embedding Theorem)*
(ii) $\|fg\|_{H^m} \le C_m \|f\|_{H^m} \|g\|_{H^m}$ *if* $m \ge 2$
(iii) $\|fg\|_{L^1} \le C \|f\|_{H^2} \|g\|_{L^1}$.
(iv) $\|fg\|_{L^1} \le C \|f\|_{L^1} \|g\|_{L^1}$.

Much of this section will be spent trying to derive different (*a priori*) estimates for solutions to the Euler equations, and trying to understand their consequences. We begin with the following *a priori* estimate to demonstrate some of the techniques we will use. Assume, for now, that a solution u exists to the Euler equations. Then by the Euler equation, one has

$$\begin{aligned}\frac{1}{2}\frac{d}{dt}\langle D^\alpha u, D^\alpha u\rangle &= \frac{1}{2}\frac{d}{dt}\int_\Omega (D^\alpha u)\cdot(D^\alpha u)\, dx \\ &= \int_\Omega (D^\alpha u)\cdot D^\alpha(\partial_t u)\, dx \\ &= \int_\Omega (D^\alpha u)\cdot D^\alpha(-(u\cdot\nabla)u - \nabla p).\end{aligned}$$

Now recall that since u is divergence-free, $D^\alpha u$ is divergence-free. We recall the calculus identity, $\mathbf{F}\cdot\nabla f = \nabla\cdot(f\mathbf{F}) - f(\nabla\cdot\mathbf{F})$. Thus using this, one computes,

$$\begin{aligned}\int_\Omega (D^\alpha u)\cdot\nabla p\,dx &= \int_\Omega \nabla\cdot(pD^\alpha u)\,dx - \int_\Omega p(\nabla\cdot D^\alpha u)\,dx\\ &= \int_\Omega \nabla\cdot(pD^\alpha u)\,dx\\ &= \int_{\partial\Omega} pD^\alpha u\cdot n\,d\sigma = 0,\end{aligned}$$

where the second line follows from the divergence-free condition on $D^\alpha u$, and the final line follows from the fact that $\Omega = [-\pi,\pi]^3$ is periodic. Thus one has

$$\frac{1}{2}\frac{d}{dt}\langle D^\alpha u, D^\alpha u\rangle = -\langle D^\alpha u, D^\alpha(u\cdot\nabla)u\rangle.$$

Using the general Leibniz rule for differentiating with multiindices, one has that

$$\langle D^\alpha u, D^\alpha(u\cdot\nabla)u\rangle = \sum_\beta C_{\alpha,\beta}\langle D^\alpha u, [D^\beta u\cdot\nabla]D^{\alpha-\beta}u\rangle.$$

Here, the sum is made over all the terms with $\beta = \{\beta_i\}_{i=1}^3$ such that $0\le\beta_i\le\alpha_i$; the $C_{\alpha,\beta}$ are positive (combinatorial) constants. Since $\langle D^\alpha u, u\cdot\nabla D^\alpha u\rangle = 0$, the term with $\beta = 0$ identically vanishes in the sum above. Now we use the Cauchy-Schwarz inequality to write,

$$|\langle D^\alpha u, D^\alpha(u\cdot\nabla)u\rangle| \le \|D^\alpha u\|_{L^1}\cdot\left(\sum_{0<|\beta|\le\alpha} C_{\alpha,\beta}|(D^\beta u\cdot\nabla)D^{\alpha-\beta}u|\right)_{L^1}.$$

Now we will use the inequalities in Proposition 6.5 to bound the above terms.

- ($|\beta|\ge 3$) Using (iii), we have the bound
$$\left\|(D^\beta u\cdot\nabla)D^{\alpha-\beta}u\right\|_{L^1} \le C\|u\|_{H^{|\beta|}}|u|_{H^{|\alpha|-|\beta|+3}} \le C(|u|_{H^{|\alpha|}})^2.$$
- ($|\beta| = 1$) Using (iii), we have the bound,
$$\left\|(D^\beta u\cdot\nabla)D^{\alpha-\beta}\right\|_{L^1} \le C\|u\|_{H^3}\|u\|_{H^{|\alpha|}}.$$
- ($|\beta| = 2$) Using (iv), we have the same bound as above.

Thus using the three bounds found above, we conclude that

$$\frac{d}{dt}\|u\|_{H^m}^2 \le C_m\|u\|_{H^m}^3, \qquad (m\ge 3).$$

Thus by Grönwall's inequality, we conclude that

$$\|u\|_{H^m(\Omega)}(t) \le \frac{1}{1/\|u_0\|_{H^m} - \frac{1}{2}C_m t}.$$

Thus we make the following observation: the smaller the H^m norm of the initial data, the longer the solution u stays in $H^m(\Omega)$; the larger this norm is, the shorter the "time of existence". We will use this estimate when constructing solutions to the Euler equations.

6.2.2 Lagrangian Representation of the Vorticity

In this subsubsection, we will derive some explicit representation formulas for the time evolution of the vorticity field (which falls into line with our previous remark that knowing this field can help us determine the velocity field). Let $\omega_t := \omega(\cdot,t)$ and $u_t := u(\cdot,t)$. In terms of the vorticity, the Euler equations read

$$\begin{cases} D_t\omega = (\omega\cdot\nabla)u,\\ \nabla\cdot u = 0,\\ \omega = \nabla\times u.\end{cases} \tag{41}$$

In two dimensions, the construction of ω is straightforward. In two dimensions, the vorticity field can be thought of as a scalar function, $\omega = \omega_3$, and moreover, $(\omega \cdot \nabla)u = 0$. Thus, we have $D_t\omega = (\partial_t + u \cdot \nabla)\omega = 0$. I.e., the derivative along the particle paths vanish identically:

$$\frac{d}{dt}\omega_t(\Phi_t(x)) = 0.$$

Thus integrating with respect to time, we obtain

$$\omega_t(\Phi_t(x)) = \omega_0(x). \tag{42}$$

I.e., the vorticity in two dimensions is conserved along the particle paths. In particular, for all $p \in \mathbb{N} \cup \{\infty\}$, one has that $\|\omega_t\|_{L^p} = \|\omega_0\|_{L^p}$. Now we wish to generalize the above result to three-dimensions. We present the result first, then provide the proof.

Theorem 6.7 (Representation Formula for the Vorticity in 3D)**.** *Let $u = u(x,t)$, $u \in C^1(\Omega, [0,T])$ be a solution of the Euler equation, and let $\omega_t = \nabla \times u_t$. Then*

$$\omega_t(\Phi_t(x)) = \nabla\Phi_t(x) \cdot \omega_0(x). \tag{43}$$

Proof. We start with the ansatz $\omega_t(\Phi_t(x)) = A \cdot \omega_0(x)$, where $A = A(x,t)$ is some matrix field. Also assume that a smooth solution to the Euler equations exists. Taking the time derivative, we obtain

$$D_t\omega_t(\Phi_t(x)) = \partial_t A \cdot \Phi_0(x).$$

The left side can be simplified as follows using the Euler equations:

$$D_t\omega_t(\Phi_t(x)) = (\omega_t \cdot \nabla)u_t(\Phi_t(x)) = [(A \cdot \omega_0) \cdot \nabla]u_t(\Phi_t(x)).$$

Thus we have the equation,

$$[(A \cdot \omega_0) \cdot \nabla]u_t(\Phi_t(x)) = \partial_t A \cdot \omega_0(x).$$

Expanding in coordinates and using the arbitrariness of ω_0, we have the expression

$$\frac{\partial}{\partial t}A_{ik}(x,t) = \sum_j \partial_i u_j(\Phi_t(x),t)A_{jk}(x,t).$$

If u is sufficiently regular, then the above formula has a unique solution (e.g., via the standard existence and uniqueness theorem for ODEs: the Picard-Lindelöf theorem). But consider $\Phi_t(x)$. By definition,

$$\frac{\partial}{\partial t}\Phi_t(x) = u(\Phi_t(x),t) \implies \partial_i\Phi_t^j(x) = \sum_k \partial_k u_j(\Phi_t(x),t) \cdot \partial_i\Phi_t^k(x).$$

But then one observes that

$$(\nabla\Phi_t(x))_{ji} = \partial_i\Phi_t(x)^j.$$

Therefore, uniqueness of solutions to the ODE finally implies that $A = (\nabla\Phi_t)$. Thus the proof concludes. □

The geometric information contained in the solution is interesting. The vorticity field is transported along the path lines (via the term $(u \cdot \nabla)\omega$), while the field is deformed under the action of the matrix $\nabla\Phi_t$ (via the term $(\omega \cdot \nabla)u$, which is identically the identity in two-dimensions).

6.2.3 Global Existence and Uniqueness (2 Dimensions)

In this subsubsection, we will show that (classical/weak) solutions to the Euler equations in two dimensions actually exist. The setting will be as follows: we consider the Euler equations on an open regular bounded domain $\Omega \subset \mathbb{R}^2$, and we will study the equations

$$\begin{cases} D_t\omega = (\partial_t + u \cdot \nabla)\omega = 0, \\ \omega = \nabla \times u, \quad \nabla \cdot u = 0, \\ u \cdot n = 0, \qquad (\text{on } \partial\Omega) \\ u(x,0) = u_0(x). \end{cases}$$

Our strategy for this subsubsection is as follows. First, we will show that the equations above have *weak* solutions. Then we will consider some specific circumstances under which these weak solutions may be extended into classical solutions of the Euler equations. First, we define the notion of *weak solutions* to the Euler equations.

Definition 6.13 (Weak Solutions to the Euler Equations)**.** *Let $\Omega \subset \mathbb{R}^2$ be an open, bounded, regular domain in the two-plane. Then if by $\Phi_t(x)$ we denote the path lines starting at $x \in \Omega$; i.e., the solution of the ODE*

$$\begin{cases} \dfrac{d}{dt}\Phi_t(x) = u(\Phi_t(x), t) \\ \Phi_0(x) = x, \end{cases} \tag{$*_2$}$$

*for a fixed ω_0, a **weak solution** to the Euler equations is a triple (Φ_t, u_t, ω_t) such that Φ_t solves the above ODE, $\omega_t(x) = \omega_0(\Phi_{-t}(x))$, and*

$$u_t(x) = K_\Omega * \omega_t(x) = \int_\Omega K_\Omega(x,y)\omega(y,t)\, dy, \tag{$*_3$}$$

where $K_\Omega(x,y) = \nabla^\perp G_\Omega(x,y)$ and G_Ω is the fundamental solution of the Poisson equation with Dirichlet condition on $\partial\Omega$.

We have the following existence result.

Theorem 6.8 (Existence of Weak Solutions to the Euler Equations)**.** *Let $\omega_0 \in L^\infty(\Omega)$. There exists a unique triple (Φ_t, u_t, ω_t) satisfying ($*_2$) and ($*_3$), where $\omega \in L^\infty([0,T], L^\infty(\Omega))$.*

Proof. The argument will mostly be constructive. □

Part IV
Appendix

A Explicit Representation Formulas

In this section, we work through the derivations of explicit representation formulas for initial value problems of the *wave equation*,

$$\Box u \coloneqq u_{tt} - \Delta u = 0 \quad \text{(or more generally, } f\text{)}, \tag{44}$$

where the unknown is the function $u : \bar{U} \times [0,\infty) \to \mathbb{R}$, $u = u(x,t)$, with $x \in U^{\text{open}} \subseteq \mathbb{R}^n$ and $t > 0$, Δ is taken with respect to the spatial variables $x = (x_1, \dots, x_n)$, and $f : U \times [0,\infty) \to \mathbb{R}$ is a given function [1].

A.0.1 The Wave Equation in One Spatial Dimension – D'Alembert's Formula

The simplest example of the wave equation happens in the case $n = 1$. Consider the following initial value problem,

$$\begin{cases} u_{tt} - u_{xx} = 0 & \text{in } \mathbb{R} \times (0,\infty) \\ u = g, u_t = h & \text{on } \mathbb{R} \times \{t = 0\}. \end{cases} \tag{45}$$

We note that the wave equation can be written in a more suggestive manner:

$$0 = u_{tt} - u_{xx} = (\partial_t + \partial_x)(\partial_t - \partial_x)u =: (\partial_t + \partial_x)v, \tag{46}$$

where we set $v = v(x,t) = u_t - u_x$. Thus, v satisfies the transport equation, $v_t + v_x = 0$. Using the solution to the transport equation in one dimension, one obtains the solution,

$$v(x,t) = a(x-t),$$

where $a(x) \coloneqq v(x,0)$. This gives us the nonhomogeneous transport equation,

$$u_t - u_x = a(x-t),$$

for which the explicit representation formula was shown (see Appendix) to be

$$u(x,t) = \int_0^t a(x + (t-s) - s)\, ds + g(x+t) = \frac{1}{2}\int_{x-t}^{x+t} a(y)\, dy + g(x+t). \tag{47}$$

It remains for us to identify the function $a(x,t)$. Using the definition of $v(x,t)$, at $t = 0$, one has,

$$a(x) = v(x,0) = u_t(x,0) - u_x(x,0) = h(x) - g'(x).$$

Therefore, using the fundamental theorem of calculus and the above results, we conclude that

$$\boxed{u(x,t) = \frac{1}{2}\left[g(x+t) - g(x-t)\right] + \frac{1}{2}\int_{x-t}^{x+t} h(y)\, dy} \qquad (x \in \mathbb{R}^n, t \geq 0). \tag{48}$$

This explicit formula is called **d'Alembert's formula** for $n = 1$. Hence, we have shown that *if* the wave equation has a "sufficiently smooth solution u", then it must be given by d'Alembert's formula. The following theorem formalizes this result.

Theorem A.1 (Solution to the Wave Equation ($n = 1$))**.** *Assume $g \in C^2(\mathbb{R})$, $h \in C^1(\mathbb{R})$, and define u by d'Alembert's formula. Then*

(i) $u \in C^2(\mathbb{R} \times [0,\infty))$.

(ii) $u_{tt} - u_{xx} = 0$ *in* $\mathbb{R} \times (0,\infty)$

(iii) $\lim\limits_{\substack{(x,t)\to(x^0,0)\\ t>0}} u(x,t) = g(x^0)$ *and* $\lim\limits_{\substack{(x,t)\to(x^0,0)\\ t>0}} u_t(x,t) = h(x^0)$ *for each point* $x^0 \in \mathbb{R}$.

Proof.

(i) Define u by d'Alembert's formula. Then we compute the following:

$$u_x = \frac{1}{2}[g'(x+t) + g'(x-t)] + \frac{1}{2}[h(x+t) - h(x-t)].$$
$$u_t = \frac{1}{2}[g'(x+t) - g'(x-t)] + \frac{1}{2}[h(x+t) + h(x-t)],$$

whence

$$u_{xx} = \frac{1}{2}[g''(x+t) + g''(x-t)] + \frac{1}{2}[h'(x+t) - h'(x-t)].$$
$$u_{xt} = \frac{1}{2}[g''(x+t) - g''(x-t)] + \frac{1}{2}[h'(x+t) + h'(x-t)].$$
$$u_{tt} = \frac{1}{2}[g''(x+t) + g''(x-t)] + \frac{1}{2}[h'(x+t) - h'(x-t)].$$

Since g', g'' and h, h' are all continuous at every $x \in \mathbb{R}$ and $t \in [0, \infty)$, as are the functions $x + t$ and $x - t$, we conclude that each of the first and second-order partial derivatives of u are continuous on $\mathbb{R} \times [0, \infty)$. Thus, $u \in C^2(\mathbb{R} \times [0, \infty))$.

(ii) From the above calculations, it follows immediately that $u_{tt} - u_{xx} = 0$.

(iii) By (1), u is continuous at every $(x, t) \in \mathbb{R} \times [0, \infty)$, and thus continuous at every point of the form $(x^0, 0) \in \mathbb{R} \times \{0\}$. Thus by the limit definition of continuity, one has

$$\lim_{(x,t)\to(x^0,0)} u(x,t) = u(x^0, 0) =: g(x^0).$$

Likewise, by our remark above, $u_t(x, t) \in C^1(\mathbb{R} \times [0, \infty)$. Thus by the limit definition of continuity, one has

$$\lim_{(x,t)\to(x^0,0)} u_t(x,t) = u_t(x^0, 0) =: h(x^0).$$

□

A.0.2 The Wave Equation in One Spatial Dimension – Half-Line

Now we will consider the modified initial value problem,

$$\begin{cases} u_{tt} - u_{xx} = 0 & (\text{in } \mathbb{R}_+ \times (0, \infty) \\ u = g, u_t = h & (\text{in } \mathbb{R}_+ \times \{t = 0\}) \\ u = 0 & (\text{on } \{x = 0\} \times (0, \infty). \end{cases} \tag{49}$$

To solve this problem, we will use the *method of reflections*; i.e., we will extend the initial value problem to the entire real line by means of the odd reflections:

$$\widetilde{u}(x,t) = \begin{cases} u(x,t) & x \geq 0 \\ -u(-x,t) & x \leq 0. \end{cases}$$

$$\widetilde{g}(x) = \begin{cases} g(x), & x \geq 0 \\ -g(-x), & x \leq 0, \end{cases}$$

$$\widetilde{h}(x) = \begin{cases} h(x), & x \geq 0 \\ -h(-x), & x \leq 0. \end{cases}$$

It is straightforward to check that $\widetilde{u}, \widetilde{g}, \widetilde{h}$ satisfy the ordinary wave equation ($n = 1$) on the whole real line. Thus by d'Alembert's formula, we obtain the following solution:

$$\boxed{u(x,t) = \begin{cases} \frac{1}{2}[g(x+t) - g(x-t)] + \frac{1}{2}\int_{x-t}^{x+t} h(y)\,dy & (x \geq t \geq 0) \\ \frac{1}{2}[g(x+t) - g(t-x)] + \frac{1}{2}\int_{t-x}^{x+t} h(y)\,dy & (0 \leq x \leq t) \end{cases}} \tag{50}$$

A.0.3 The Wave Equation in $n \geq 2$ Spatial Dimension – Kirchoff's and Poisson's Formulas

In this subsection, we will develop explicit representation formulas for the wave equation for dimensions $n \geq 2$. To develop these formulas, we require the *method of spherical means*. Consider the following definitions.

Definition A.1 (Spherical Means)**.**

(i) Let $x \in \mathbb{R}^n$, $t > 0$, and $r > 0$. We define the average of $u(\cdot, t)$ over the sphere $\partial B(x,r)$ to be

$$U(x;r,t) = \fint_{\partial B(x,r)} u(y,t)\,dS(y). \tag{51}$$

(ii) We make the analogous definitions,

$$\begin{cases} G(x;r) & = \fint_{\partial B(x,r)} g(y)\,dS(y). \\ H(x;r) & = \fint_{\partial B(x,r)} h(y)\,dS(y). \end{cases} \tag{52}$$

For any fixed x, we will regard U, G, H as functions of r and t.

Lemma A.1 (Euler-Poisson-Darboux Equation)**.** *Fix $x \in \mathbb{R}^n$, $n, m \geq 2$, and let $u \in C^m(\mathbb{R}^n \times [0,\infty))$ satisfy the initial value problem,*

$$\begin{cases} u_{tt} - \Delta u = 0 & (\text{in } \mathbb{R}^n \times (0,\infty)) \\ u = g, u_t = h & (\text{in } \mathbb{R}^n \times \{t = 0\}). \end{cases}$$

Then $U \in C^m(\overline{\mathbb{R}}_+ \times [0,\infty))$, and U_{tt} satisfies the Euler-Poisson-Darboux initial value problem,

$$\begin{cases} U_{tt} - U_{rr} - \frac{n-1}{r} U_r = 0 & (\text{in } \mathbb{R}_+ \times (0,\infty)) \\ U = G, U_t = H & (\text{on } \mathbb{R}_+ \times \{t = 0\} \end{cases} \tag{53}$$

Proof. We begin by taking the partial derivative of the spherical mean U with respect to r. To do so, we rewrite

$$U(x;r,t) = \fint_{\partial B(x,r)} u(y,t)\,dS(y) = \frac{1}{n\alpha(n) r^{n-1}} \int_{\partial B(x,r)} u(y,t)\,dS(y) = \frac{1}{n\alpha(n)} \int_{\partial B(0,1)} u(x+rz,t)\,dS(z).$$

Thus,

$$\begin{aligned} U_r(x;r,t) &= \frac{1}{n\alpha(n)} \frac{\partial}{\partial r} \int_{\partial B(0,1)} u(x+rz,t)\,dS(z) \\ &= \frac{1}{n\alpha(n)} \int_{\partial B(0,1)} Du(x+rz,t) \cdot z\,dS(z) \\ &= \frac{r^{1-n}}{n\alpha(n)} \int_{\partial B(x,r)} Du(y,t) \cdot \frac{y-x}{r}\,dS(y) \\ &= \frac{r}{n\alpha(n) r^n} \int_{\partial B(x,r)} \frac{\partial u}{\partial \nu}\,dS(y) \\ &= \frac{r}{n\alpha(n) r^n} \int_{B(x,r)} \Delta u\,dy = \frac{r}{n} \fint_{B(x,r)} \Delta u\,dy, \end{aligned}$$

where passing the derivative inside the integral in the second line is justified since u is sufficiently smooth (meaning that Du exists a.e. and is integrable on $\partial B(0,1)$), and the final line follows from Green's formula. Now taking some large $R > 0$, and restricting to $0 < r \le R/2$, one has that since $B(x,r) \subset \overline{B(x,R/2)}$ and Δu is continuous (so that the extreme value theorem applies),

$$\|\Delta u\|_{L^\infty(B(x,r))} \le \|\Delta u\|_{L^\infty(B(x,R))} =: C < \infty.$$

Hence,

$$U_r(x;r,t) \le \frac{C}{n} r \implies \lim_{r\to 0^+} U_r(x;r,t) = 0.$$

Now we will compute the second partial derivative of U with respect to r. We compute,

$$\begin{aligned} U_{rr}(x;r,t) &= \frac{1-n}{n\alpha(n)r^n}\int_{B(x,r)} \Delta u\, dy + \frac{r^{1-n}}{n\alpha(n)}\frac{\partial}{\partial r}\int_{B(x,r)} \Delta u(y,t)\, dy \\ &= \frac{1-n}{n}\fint_{B(x,r)} \Delta u(y,t)\, dy + \frac{r^{1-n}}{n\alpha(n)}\int_{\partial B(x,r)} \Delta u(y,t)\, dS(y) \\ &= \fint_{\partial B(x,r)} \Delta u(y,t)\, dS(y) + \left(\frac{1}{n} - 1\right)\fint_{B(x,r)} \Delta u(y,t)\, dy, \end{aligned}$$

where the second line follows from the coarea formula (see Appendix). By applying similar reasoning to above, one computes that $\lim_{r\to 0^+} U_{rr}(x;r,t) = (1/n)\Delta u$. In this manner, repeatedly differentiating U, one sees that $U \in C^m(\overline{\mathbb{R}}_+ \times [0,\infty))$. Now differentiating,

$$\begin{aligned} (n-1)r^{n-2}U_r + r^{n-1}U_{rr} &= \frac{d}{dr}(r^{n-1}U_r) \\ &= \frac{d}{dr}\frac{1}{n\alpha(n)}\int_{B(x,r)} u_{tt}\, dy \\ &= \frac{1}{n\alpha(n)}\int_{\partial B(x,r)} u_{tt}\, dS(y) \\ &= r^{n-1}\fint_{\partial B(x,r)} u_{tt}\, dS(y) =: r^{n-1}U_{tt}, \end{aligned}$$

with the third line following from the coarea formula. □

Now we will derive the explicit representation formula for the 3-dimensional wave equation by reducing the system to the one-dimensional Euler-Poisson-Darboux equation and solving this system using the formula we derived for the one-dimensional half-line equation.

Kirchoff's Formula for the Wave Equation ($n = 3$) Suppose $u \in C^2(\mathbb{R}^3 \times [0,\infty))$ solves the initial-value problem

$$\begin{cases} u_{tt} - \Delta u = 0 & (\text{in } \mathbb{R}^3 \times [0,\infty)) \\ u = g,\ u_t = h & (\text{on } \mathbb{R}^3 \times \{t=0\}) \end{cases} \tag{54}$$

Define the functions, $\widetilde{U} = rU(x;r,t)$, $\widetilde{G}(r) = rG(x;r)$, and $\widetilde{H}(r) = rH(x;r)$. Then we observe the following:

(i)
$$\begin{aligned} \widetilde{U}_{tt} = rU_{tt} &= r\left[U_{rr} + \frac{2}{r}U_r\right] \\ &= rU_{rr} + 2U_r = (U + rU_r)_r = \widetilde{U}_{rr}. \end{aligned}$$

(ii) On $\mathbb{R}_+ \times \{t=0\}$, $U = G$ and $U_t = H$ so that $\widetilde{U} = \widetilde{G}$ and $\widetilde{U}_t = \widetilde{H}$.

(iii) For $r = 0$, $\partial B(x,0) = \varnothing$ so that $\widetilde{U} = 0$ on $\{r=0\} \times (0,\infty)$.

In other words, we have shown that $\widetilde{U}$ satisfies the following initial value problem,

$$\begin{cases} \widetilde{U}_{tt} - \widetilde{U}_{rr} = 0 & (\text{in } \mathbb{R}_+ \times (0,\infty)) \\ \widetilde{U} = \widetilde{G}, \widetilde{U}_t = \widetilde{H} & (\text{on } \mathbb{R}_+ \times \{t=0\}) \\ \widetilde{U} = 0 & (\text{on } \{r=0\} \times (0,\infty)) \end{cases}$$

Using the solution for the one-dimensional wave equation on the half-line, for $0 \le r \le t$, one has

$$\widetilde{U}(x;r,t) = \frac{1}{2}[\widetilde{G}(r+t) - \widetilde{G}(t-r)] + \frac{1}{2}\int_{t-r}^{r+t} \widetilde{H}(y)\, dy.$$

On the other hand, one has that

$$\begin{aligned} u(x,t) &= \lim_{r\to 0^+} \frac{\widetilde{U}(x;r,t)}{r} \\ &= \lim_{r\to 0^+} \left[\frac{\widetilde{G}(r+t) - \widetilde{G}(t-r)}{2r} + \frac{1}{2r}\int_{t-r}^{r+t} \widetilde{H}(y)\, dy \right] \\ &= \widetilde{G}'(t) + \widetilde{H}(t). \end{aligned}$$

Recall again that

$$\begin{aligned} \widetilde{G}'(t) &= \frac{\partial}{\partial t}\left(t \fint_{\partial B(x,t)} g(y)\, dS(y) \right) \\ &= \frac{\partial}{\partial t}\left(\frac{1}{n\alpha(n)t} \int_{\partial B(x,t)} g(y)\, dS(y) \right) \\ &= -\frac{1}{n\alpha(n)t^2}\int_{\partial B(x,t)} g(y)\, dS(y) + \frac{1}{n\alpha(n)t}\frac{\partial}{\partial t}\left(t^2 \int_{\partial B(0,1)} g(x+tz)\, dS(z) \right) \\ &= -\fint_{\partial B(x,t)} g(y)\, dS(y) + \frac{2}{n\alpha(n)}\int_{\partial B(0,1)} g(x+tz)\, dS(z) + \frac{1}{n\alpha(n)t}\int_{\partial B(x,t)} Dg(y)\cdot\left(\frac{y-x}{t}\right) dS(y) \\ &= \fint_{\partial B(x,t)} g(y)\, dS(y) + \fint_{\partial B(x,t)} Dg(y)\cdot(y-x)\, dS(y). \end{aligned} \tag{55}$$

Likewise, one has

$$\widetilde{H}(t) = tH(t) = \fint_{\partial B(x,t)} th(y)\, dS(y).$$

Therefore, we conclude that

$$\boxed{u(x,t) = \fint_{\partial B(x,t)} th(y) + g(y) + Dg(y)\cdot(y-x)\, dS(y)} \tag{56}$$

This is **Kirchoff's formula** for the wave equation in dimension 3.

Poisson's Formula for the Wave Equation ($n = 2$) Assume $u \in C^2(\mathbb{R}^2 \times [0,\infty))$ solves the IVP for $n = 2$. Unlike for the $n = 3$ case, we will consider the 2-dimensional problem and "view" it as a 3-dimensional problem. Define $\bar{u}(x_1,x_2,x_3,t) = u(x_1,x_2,t)$ so that u satisfies the IVP

$$\begin{cases} \bar{u}_{tt} - \Delta\bar{u} = 0 & (\text{in } \mathbb{R}^3 \times (0,\infty)) \\ \bar{u} = \bar{g}, \bar{u}_t = \bar{h} & (\text{on } \mathbb{R}^3 \times \{t=0\}), \end{cases}$$

where we define $\bar{g}$ and $\bar{h}$ analogously. For $x = (x_1,x_2) \in \mathbb{R}^2$, write $\bar{x} = (x_1,x_2,0) \in \mathbb{R}^3$. Then by Kirchoff's formula, one has

$$u(x,t) = \bar{u}(\bar{x},t) = \frac{\partial}{\partial t}\left(t\fint_{\partial\bar{B}(\bar{x},t)} \bar{g}\, d\bar{S} \right) + t\fint_{\partial\bar{B}(\bar{x},t)} \bar{h}\, d\bar{S},$$

where $\bar{B}(\bar{x},t)$ denotes the ball in $\mathbb{R}^3$ with center $\bar{x}$, radius $t>0$, and where $d\bar{S}$ denotes the 2-dimensional surface measure on $\partial\bar{B}(\bar{x},t)$. But we can write

$$\fint_{\partial\bar{B}(\bar{x},t)} \bar{g}\,d\bar{S} = \frac{1}{4\pi t^2}\int_{\partial\bar{B}(\bar{x},t)} \bar{g}\,d\bar{S} = \frac{2}{4\pi t^2}\int_{B(x,t)} g(y)(1+|D\gamma(y)|^2)^{1/2}\,dy,$$

where $\gamma(y) = (t^2-|y-x|^2)^{1/2}$ for $y\in B(x,t)$. The factor 2 comes from the two hemispheres of $\partial\bar{B}(\bar{x},t)$. Since $(1+|D\gamma|^2)^{1/2} = t(t^2-|y-x|^2)^{-1/2}$, one has

$$\begin{aligned}\fint_{\partial\bar{B}(\bar{x},t)} \bar{g}\,d\bar{S} &= \frac{1}{2\pi t}\int_{B(x,t)} \frac{g(y)}{(t^2-|y-x|^2)^{1/2}}\,dy \\ &= \frac{t}{2}\fint_{B(x,t)} \frac{g(y)}{(t^2-|y-x|^2)^{1/2}}\,dy.\end{aligned}$$

Using this and the coarea formula, one obtains

$$u(x,t) = \frac{1}{2}\frac{\partial}{\partial t}\left(t^2\fint_{B(x,t)} \frac{g(y)}{(t^2-|y-x|^2)^{1/2}}\,dy\right) + \frac{t^2}{2}\fint_{B(x,t)} \frac{h(y)}{(t^2-|y-x|^2)^{1/2}}\,dy.$$

Using the same reasoning as above to pass the derivative inside the integral, one has

$$\frac{\partial}{\partial t}\left(t^2\fint_{B(x,t)} \frac{g(y)}{(t^2-|y-x|^2)^{1/2}}\,dy\right) = t\fint_{B(x,t)} \frac{g(y)}{(t^2-|y-x|^2)^{1/2}}\,dy + t\fint_{B(x,t)} \frac{Dg(y)\cdot(y-x)}{(t^2-|y-x|^2)^{1/2}}\,dy.$$

Thus for $n=2$, we obtain **Poisson's Formula**,

$$\boxed{u(x,t) = \frac{1}{2}\fint_{B(x,t)} \frac{tg(y)+t^2h(y)+tDg(y)\cdot(y-x)}{(t^2-|y-x|^2)^{1/2}}\,dy \qquad (x\in\mathbb{R}^2, t>0)} \tag{57}$$

Notice that in solving the wave equation for $n=2$, we used the *method of descent*; i.e., we solved the problem for $n=3$ first, and then dropped down to $n=2$.

General Solutions for $n\geq 2$ Following similar extensive calculations as above (see Appendix for the work), we obtain the following solutions to the wave equation,

$$\boxed{u(x,t) = \begin{cases} \dfrac{1}{\gamma(n)}\left[\left(\dfrac{\partial}{\partial t}\right)\left(\dfrac{1}{t}\dfrac{\partial}{\partial t}\right)^{\frac{n-3}{2}}\left(t^{n-2}\fint_{\partial B(x,t)} g\,dS\right) + \left(\dfrac{1}{t}\dfrac{\partial}{\partial t}\right)^{\frac{n-3}{2}}\left(t^{n-2}\fint_{\partial B(x,t)} h\,dS\right)\right] & \begin{matrix} n\geq 3 \\ n \text{ odd}\end{matrix} \\ \dfrac{1}{\beta(n)}\left[\left(\dfrac{\partial}{\partial t}\right)\left(\dfrac{1}{t}\dfrac{\partial}{\partial t}\right)^{\frac{n-2}{2}}\left(t^n\fint_{B(x,t)} \dfrac{g(y)}{(t^2-|y-x|^2)^{1/2}}\,dy\right)\right. & \\ \qquad\left. + \left(\dfrac{1}{t}\dfrac{\partial}{\partial t}\right)^{\frac{n-2}{2}}\left(t^n\fint_{B(x,t)} \dfrac{h(y)}{(t^2-|y-x|^2)^{1/2}}\,dy\right)\right] & \begin{matrix} n\geq 2 \\ n \text{ even}\end{matrix} \end{cases}} \tag{58}$$

where in the first equation $\gamma_n = 1\cdot 3\cdot\cdots\cdot(n-2)$, and $\beta(n) = 2\cdot 4\cdot\cdots\cdot(n-2)\cdot n$. The above solutions highlight an interesting contrast between odd-dimensional, and even-dimensional solutions. Note that if n is odd and $n\geq 3$, the initial data at a point $x\in\mathbb{R}^n$ only affect the solution u along the boundary $\{(y,t)\mid t>0, |x-y|=t\}$ of the cone $C=\{(y,t)\mid t>0, |x-y|<t\}$. On the other hand, if n is given, the initial data can affect the solution within all of C. Thus, a disturbance originating at x propagates along a sharp wavefront in odd dimensions, but continues to have effects even after the leading edge of the wavefront passes in even dimensions. This phenomenon is **Huygen's principle**.

A.1 Solution to the Nonhomogeneous Wave Equation

Now we will use the solutions from the previous sections to address the nonhomogeneous wave equation,

$$\begin{cases} u_{tt} - \Delta u = f & (\text{in } \mathbb{R}^n \times (0,\infty)) \\ u = 0, u_t = 0 & (\text{on } \mathbb{R}^n \times \{t=0\}). \end{cases} \tag{59}$$

To solve this problem, we use Duhamel's principle[12] by setting $u = u(x,t;s)$ to be the solution of

$$\begin{cases} u_{tt}(\cdot;s) - \Delta u(\cdot,s) = 0 & (\text{in } \mathbb{R}^n \times (0,\infty)) \\ u(\cdot;s) = 0, u_t(\cdot;s) = f(\cdot,s) & (\text{on } \mathbb{R}^n \times \{t=0\}). \end{cases} \tag{60}$$

Thus, by Duhamel's principle,

$$u(x,t) \coloneqq \int_0^t u(x,t;s)\, ds \qquad (x \in \mathbb{R}^n, t \geq 0) \tag{61}$$

is a solution of the above nonhomogeneous wave equation. Consequently, the solution to the general inhomogeneous wave problem,

$$\begin{cases} u_{tt} - \Delta u = f & (\text{in } \mathbb{R}^n \times (0,\infty)) \\ u = g, u_t = h & (\text{on } \mathbb{R}^n \times \{t=0\}) \end{cases} \tag{62}$$

is equal to the sum of the solutions to the problems

$$\begin{cases} u_{tt} - \Delta u = 0 & (\text{in } \mathbb{R}^n \times (0,\infty)) \\ u = g, u_t = h & (\text{on } \mathbb{R}^n \times \{t=0\}) \end{cases} \quad \text{and} \quad \begin{cases} u_{tt} - \Delta u = f & (\text{in } \mathbb{R}^n \times (0,\infty)) \\ u = 0, u_t = 0 & (\text{on } \mathbb{R}^n \times \{t=0\}) \end{cases} \tag{63}$$

A.1.1 Example 1: Solution to Nonhomogeneous Problem for $n = 1$

By d'Alembert's formula, one has

$$u(x,t;s) = \frac{1}{2}\int_{x-t+s}^{x+t+s} f(y,s)\, dy.$$

Thus,

$$u(x,t) = \frac{1}{2}\int_0^t \int_{x-t+s}^{x+t+s} f(y,s)\, dyds = \frac{1}{2}\int_0^t \int_{x-s}^{x+s} f(y,t-s)\, dyds \qquad (x \in \mathbb{R}, t \geq 0).$$

A.1.2 Example 2: Solution to Nonhomogeneous Problem for $n = 3$

By Kirchoff's formula, one has

$$u(x,t;s) = (t-s)\fint_{\partial B(x,t-s)} f(y,s)\, dS.$$

This implies that

$$\begin{aligned} u(x,t) &= \int_0^t (t-s)\fint_{\partial B(x,t-s)} f(y,s)\, dSds \\ &= \int_0^t (t-s)\frac{1}{4\pi(t-s)^2}\int_{\partial B(x,t-s)} f(y,s)\, dSds \\ &= \frac{1}{4\pi}\int_0^t \int_{\partial B(x,t-s)} \frac{f(y,s)}{t-s}\, dSds \\ &= \frac{1}{4\pi}\int_0^t \int_{\partial B(x,r)} \frac{f(y,t-r)}{r}\, dSdr. \end{aligned}$$

[12] See https://en.wikipedia.org/wiki/Duhamel%27s_principle for an explanation.

Hence, we conclude that

$$u(x,t) = \frac{1}{4\pi}\int_{B(x,t)} \frac{f(y,t-|y-x|)}{|y-x|}\,dy \qquad (x \in \mathbb{R}^3, t \geq 0),$$

where the integrand on the right is called a *retarded potential*.

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